De La Salle University

Building an Integrated Financial Database Using SQL in R

FINSTAD Case 1

Section C01, Group 3

in partial fulfillment of the requirements for the course

FINSTAD

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Executive Summary

[Summarize the objectives, datasets used, major SQL findings, key statistical results, and investment recommendation here (Maximum 1 Page).]

Contents

1. Introduction

[Discuss the background of the study, objectives of the analysis, and the importance of integrating multiple financial datasets.]

2. Data Sources and Acquisition

Eight datasets were selected for this study, comprising four asset price series and four macroeconomic indicators. All data were obtained from publicly available sources. The sample period from January 1, 2020 to December 31, 2025 was chosen because it encompasses three distinct market regimes. The first was the COVID-19 crash and subsequent recovery from February to December 2020. The second was the inflation surge from 2021 through early 2023, during which US consumer price index year-on-year readings exceeded five percent. The third was the interest rate tightening cycle beginning in March 2022, during which the Federal Reserve raised the federal funds rate from near zero to over five percent. This tripartite regime structure provides a natural experiment for examining asset behaviour under contrasting macroeconomic conditions.

The US equity asset is Apple Incorporated (AAPL), listed on the NASDAQ exchange under ticker AAPL. As of 2025, Apple held a market capitalisation of approximately \$4.25 trillion, with a trailing price-to-earnings ratio of 35.08 and an estimated enterprise value of \$4.19 trillion (StockAnalysis, 2025). Apple was selected over alternative US technology stocks, particularly NVIDIA, because its valuation and business model present a fundamentally different risk profile that is instructive for a diversified portfolio analysis. NVIDIA derives over 80 percent of its revenue from data centre accelerator chips, exposing it heavily to cyclical capital expenditure cycles. Apple presented the opposite profile. Apple's approach to artificial intelligence has consistently prioritised on-device processing and user privacy over cloud-based deployment. The selection of AAPL over NVIDIA therefore captures a technology sector exposure that is diversified across hardware and services without the single-product concentration risk that characterises the pure-play semiconductor firms.

The Philippine equity asset is BDO Unibank Incorporated (BDO), the largest bank in the Philippines by total assets. As of December 2025, BDO reported total assets of PHP 5.27 trillion, maintaining its ranking as the top universal and commercial bank in the Philippines by asset size

(Bangko Sentral ng Pilipinas, 2025). BDO was selected to provide exposure to the Philippine financial sector and to emerging market dynamics more broadly, which operate under different monetary policy cycles compared to developed markets.

The cryptocurrency asset is Bitcoin (BTC), the largest digital asset by market capitalisation. Bitcoin trades continuously on decentralised cryptocurrency exchanges and was included to evaluate the extent to which its return behaviour correlates with or diverges from traditional asset classes. The benchmark US equity index is represented by the SPDR S&P 500 ETF Trust (SPY), which tracks the S&P 500 index and is one of the most liquid exchange-traded funds globally.

Four macroeconomic series were obtained from the Federal Reserve Economic Data database maintained by the Federal Reserve Bank of St. Louis. The Consumer Price Index for All Urban Consumers (FRED series CPIAUCSL) measures headline consumer price inflation at monthly frequency. The 10-Year Treasury Constant Maturity Rate (FRED series DGS10) represents the yield on US government debt at the longest standard tenor and serves as a benchmark for risk-free rates. The Federal Funds Effective Rate (FRED series FEDFUNDS) captures the overnight rate at which depository institutions lend reserve balances to each other and reflects the stance of US monetary policy. The CBOE Volatility Index (FRED series VIXCLS) tracks the implied volatility of S&P 500 index options over a 30-day forward period and is interpreted as a gauge of market fear and uncertainty.

Each dataset was downloaded from its respective source in CSV format and imported into R for processing. Asset price data for AAPL, BTC, and SPY were obtained from Yahoo Finance (Yahoo Finance, 2025). BDO data were sourced from Investing.com (Investing.com, 2025), as the Philippine stock data available through Yahoo Finance did not provide sufficient historical coverage for the required sample period. All four macroeconomic series were downloaded from the FRED database website (Federal Reserve Bank of St. Louis, 2025) using their native CSV export functionality.

Please refer to the appendix for the complete R code demonstrating the data acquisition process.

Table 1 summarises the acquisition details for each dataset. The full sample period for each series extends from its earliest available observation through mid-2026. All series were subsequently restricted to the January 1, 2020 to December 31, 2025 analysis window during the data cleaning phase. No missing values or duplicate rows were detected in the raw downloads.

Table 1: Data Acquisition Summary

Dataset	Source	Original Observations	Missing Values	Duplicates
AAPL	Yahoo Finance	2512	0	0
BDO	Investing.com	2441	0	0
BTC	Yahoo Finance	3652	0	0
SPY	Yahoo Finance	2514	0	0
CPI	FRED	952	0	0
DGS10	FRED	16105	0	0
FEDFUNDS	FRED	863	0	0
VIX	FRED	9216	0	0

3. Data Cleaning

Eight CSV files were loaded into R using base `read.csv` with `stringsAsFactors` set to FALSE throughout. Each dataset was initially inspected for structural anomalies before being restricted to the January 1, 2020 to December 31, 2025 analysis window. The restriction served two purposes: it ensured uniform temporal coverage across assets trading on different calendars and it eliminated the need to extrapolate or impute macroeconomic readings for periods where certain series had no observations.

Date standardisation required three distinct parsing strategies, a direct consequence of the heterogeneity across data sources. The AAPL, BTC, and SPY exports from Yahoo Finance used POSIXct-formatted timestamps with timezone offsets which required `ymd_hms` coercion followed by conversion to Date objects. The BDO series from Investing.com used MM/DD/YYYY format which demanded `mdy` parsing from the lubridate package. All four FRED series used ISO 8601 YYYY-MM-DD format which R parsed natively via `ymd`. BDO additionally contained a byte-order-mark character prepended to the first column header, a comma-separated volume field expressed in 2.50M notation, and a closing price column labelled Price rather than Close. The byte-order-mark was handled by reading the file with UTF-8-BOM encoding, the volume field was parsed through a custom regular expression that converted suffixed values to numeric equivalents, and the Price column was remapped to Close for internal consistency. These adjustments were made transparently and did not alter any observed price or return values.

No missing values or duplicate rows were detected in any dataset after the initial load. The absence of missingness is attributable to the nature of the sources: Yahoo Finance and Investing.com report complete historical price records without gaps, and FRED series are reported only on their scheduled release dates, meaning every observation in the raw download is a valid reading. The cleaning burden therefore fell entirely on format normalisation rather than imputation or row removal, which simplifies the subsequent analysis by removing the need for missing data handling techniques such as last-observation-carried-forward or interpolation.

Daily returns were calculated as the simple percentage change in closing price from the previous trading day, expressed as $(\text{Close}/\,\text{lag}(\text{Close})) - 1$. This formulation was applied consistently across all four assets, with BDO using the Price column as its close. The first observation of each asset produced a missing return by construction, as no prior price existed. These initial NA values were retained in the dataset and excluded from all subsequent statistical computations through the use of na.rm arguments, which is functionally equivalent to a drop but preserves the row count for temporal alignment.

Table 2 reports the observation counts at each processing stage. The reduction from original to after-cleaning counts reflects the date restriction alone, not any data quality issue. The most extreme reduction occurred for DGS10, which fell from 16,105 to 1,500 observations, and for VIX, which fell from 9,216 to 1,535 observations, both because FRED provides several decades of historical data that extend well before the 2020 start date. The CPI reduction from 952 to 71 observations is a consequence of its monthly reporting frequency.

Table 2: Data Acquisition and Cleaning Summary

Dataset	Original_Obs	After_Cleaning	After_Joins
AAPL	2512	1508	NA
BDO	2441	1468	NA
BTC	3652	2192	NA
SPY	2514	1508	NA
CPI	952	71	NA
DGS10	16105	1500	NA
FEDFUNDS	863	72	NA
VIX	9216	1535	NA
Assets (wide)	NA	2192	NA
Final Dataset	NA	NA	2192

4. Database Integration

The cleaned asset and macro datasets were consolidated into a single wide-format table for SQL querying. Four join strategies were evaluated to determine the most appropriate consolidation approach.

The `left_join` preserved all asset trading dates and produced 2192 rows. Macroeconomic variables returned NA on weekends and holidays where no FRED observation existed. The `inner_join` retained only 46 rows, representing dates common to all eight datasets simultaneously. The `full_join` produced 2192 rows by taking the union of all dates across assets and macro series. The `anti_join` returned 2121 asset dates that were missing CPI data, confirming that weekend and holiday gaps in the monthly CPI series drove the reduction.

Table 3 summarises each join type and its row count.

Table 3: Join Comparison

Join Type	Rows	Description
left_join	2192	Preserves all asset dates; macro NA on non-trading days
inner_join	46	Only dates common to all eight datasets
full_join	2192	Union including macro-only periods
anti_join	2121	Asset dates missing CPI, reflecting weekend and holiday gaps

The final dataset was constructed using a `left_join` with the asset table as the base, yielding 2192 rows and 13 columns. This strategy retained the maximum number of observable trading days while allowing macroeconomic series to populate only where data existed. Table 4 traces the observation counts through each processing stage.

Table 4: Observation Reduction Across Processing Stages

Dataset	Original	After Cleaning	Final
AAPL	2512	1508	—
BDO	2441	1468	—
BTC	3652	2192	—
SPY	2514	1508	—
CPI	952	71	—
DGS10	16 105	1500	—
FEDFUNDS	863	72	—
VIX	9216	1535	—
Final Dataset	—	—	2192

The sharp reduction from 952 to 71 observations for CPI reflects the monthly reporting frequency of the Consumer Price Index. The final row count of 2192 is driven by the asset with the most trading days, BTC, which trades continuously on cryptocurrency exchanges. The observation reduction is therefore explained by two factors: non-trading days that remove weekend and holiday observations in equities, and the lower reporting frequency of macroeconomic indicators.

5. SQL-Based Financial Investigation

Ten financial investigations were conducted on the integrated database. They interrogate asset behavior, market dynamics, and portfolio implications. The terrain spans from baseline risk assessment through regime analysis and concludes with inflation hedging dynamics.

5.1. Q1. Risk-Adjusted Returns

This section addresses the question:

Which asset delivered the highest return per unit of risk over the 2020 to 2025 sample period?

```r
# Q1: Risk-Adjusted Daily Return by Asset -- SISON, Aaron Joshua Estacio

q1 <- fin_data |>
  group_by(Asset) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE)
  ) |>
  mutate(Risk_Adjusted_Return = Mean_Daily_Return / SD_Daily_Return) |>
  arrange(desc(Risk_Adjusted_Return)) |>
  collect()

spy_ratio <- q1 |> filter(Asset == "SPY") |> pull(Risk_Adjusted_Return)

ggplot(q1, aes(x = Risk_Adjusted_Return, y = reorder(Asset,
↪  Risk_Adjusted_Return),
                fill = Asset)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_vline(xintercept = spy_ratio, linetype = "dashed",
             color = "#B3B3B3", linewidth = 0.4) +
  geom_text(aes(x = Risk_Adjusted_Return / 2,
                label = sprintf("%.4f", Risk_Adjusted_Return)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  annotate("text", x = spy_ratio, y = 4.5, label = "SPY benchmark",
           size = 2.8, color = "#595959", hjust = -0.1) +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  labs(title = "Risk-Adjusted Daily Return by Asset",
       subtitle = "Mean return divided by standard deviation, 2020 to 2025",
```

```
      x = "Risk-Adjusted Return", y = NULL) +
  xlim(0, 0.07) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

Table 5: Q1 Data Output

Asset	Mean_Daily_Return	SD_Daily_Return	Risk_Adjusted_Return
AAPL	0.0011	0.0200	0.0538
BTC	0.0017	0.0319	0.0520
SPY	0.0006	0.0131	0.0486
BDO	0.0003	0.0226	0.0131

```
ggsave("../reports/figures/fig_q1.pdf", width = 7, height = 3.5, device =
↪  "pdf")
```

Table 5 establishes the hierarchy of risk-adjusted returns based on the query output. AAPL occupies the top position with a ratio of 0.0538, driven by a mean daily return of 0.11% against a volatility of 2.00%. BTC follows at 0.0520, utilizing a high-variance mechanism where its 0.17% mean return offsets a 3.19% standard deviation. SPY records 0.0486 on a 1.31% standard deviation, while BDO ranks lowest at 0.0131. The query aggregation correctly delineates asset efficiency based on raw variance.

5.2. Q2. COVID Crash Analysis

This section addresses the question:

How did each asset perform during the COVID-19 market crash of February to March 2020?

```
# Q2: COVID-19 Crash Cumulative Returns -- SISON, Aaron Joshua Estacio

q2_raw <- fin_data |>
```

```r
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q2 <- q2_raw |>
  filter(Date >= as.Date("2020-02-01") & Date <= as.Date("2020-03-31")) |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1)

q2_summary <- q2 |>
  summarise(
    Start_Date = min(Date),
    End_Date = max(Date),
    Total_Cumulative_Return = last(Cumulative_Return),
    Observations = n()
  )

ggplot(q2, aes(x = Date, y = Cumulative_Return, color = Asset)) +
  geom_line(linewidth = 0.8) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "#B3B3B3",
  ↪  linewidth = 0.3) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Cumulative Returns During the COVID-19 Crash",
       subtitle = "February to March 2020, indexed to 100% at period start",
       x = "Date", y = "Cumulative Return", color = "Asset") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

Table 6: Q2 Summary Data Output

Asset	Start_Date	End_Date	Total_Cumulative_Return	Observations
AAPL	2020-02-03	2020-03-31	-0.1764	41
BDO	2020-02-03	2020-03-31	-0.3006	40
BTC	2020-02-01	2020-03-31	-0.3114	60
SPY	2020-02-03	2020-03-31	-0.1941	41

```r
ggsave("../reports/figures/fig_q2.pdf", width = 7, height = 4, device =
 ↪  "pdf")
```

Table 6 quantifies the trajectory of the COVID-19 decline. All four assets descended steeply, suffering double-digit cumulative declines. The ranking of final losses aligns with pre-crisis volatility: AAPL and SPY lost 17.6% and 19.4% respectively, while BDO and BTC lost 30.1% and 31.1%. This data confirms that higher-volatility assets experience proportionally larger drawdowns during a systematic liquidity event.

5.3. Q3. Cross-Asset Correlation

This section addresses the question:

What is the cross-asset correlation of daily returns, and which assets offer the strongest diversification potential?

```r
# Q3: Pairwise Correlation of Daily Returns -- SISON, Aaron Joshua Estacio

returns_wide <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  pivot_wider(names_from = Asset, values_from = Daily_Return)

cor_matrix <- returns_wide |>
  select(AAPL, BDO, BTC, SPY) |>
  cor(use = "pairwise.complete.obs")

cor_long <- as.data.frame(cor_matrix) |>
  tibble::rownames_to_column("Asset1") |>
  tidyr::pivot_longer(-Asset1, names_to = "Asset2", values_to =
 ↪  "Correlation") |>
  mutate(Asset1 = factor(Asset1, levels = rev(c("AAPL", "BDO", "BTC",
   ↪  "SPY"))),
         Asset2 = factor(Asset2, levels = c("AAPL", "BDO", "BTC", "SPY")))

ggplot(cor_long, aes(x = Asset2, y = Asset1, fill = Correlation)) +
  geom_tile(color = "white", linewidth = 1) +
  geom_text(aes(label = sprintf("%.3f", Correlation)), size = 3.5) +
  scale_fill_gradient2(low = "#1DC9A4", mid = "#F2F2F2", high = "#2E45B8",
```

```
                         midpoint = 0.4, limits = c(0, 1)) +
  labs(title = "Pairwise Correlation of Daily Returns",
       subtitle = "Pearson correlation coefficients, 2020 to 2025",
       x = NULL, y = NULL, fill = "Correlation") +
  theme_minimal(base_size = 11) +
  theme(panel.grid = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "right")
```

Table 7: Q3 Correlation Matrix Output

	AAPL	BDO	BTC	SPY
AAPL	1.0000	0.1044	0.2836	0.7839
BDO	0.1044	1.0000	0.0507	0.1235
BTC	0.2836	0.0507	1.0000	0.3825
SPY	0.7839	0.1235	0.3825	1.0000

```
ggsave("../reports/figures/fig_q3.pdf", width = 6, height = 4.5, device =
↪  "pdf")
```

Table 7 computes the pairwise Pearson correlation matrix. The intersection of AAPL and SPY produces the highest correlation at 0.784, reflecting Apple's heavy weighting in the index. BDO's correlations with all other assets fall between 0.05 and 0.12, reflecting the structural isolation of the Philippine banking sector. BTC holds a 0.383 correlation with SPY, indicating it partially behaves as a risk asset.

5.4. Q4. Asset Ranking by Total Return

This section addresses the question:

How do the selected assets rank by total return from 2020 to 2025?

```
# Q4: Asset Ranking by Total Return -- CRUZ, Ricardo Miguel Inigo

q4_raw <- fin_data |>
```

```r
  select(Date, Asset, Close, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q4 <- q4_raw |>
  filter(!is.na(Close), !is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  summarise(
    Start_Date = min(Date),
    End_Date = max(Date),
    Start_Close = first(Close),
    End_Close = last(Close),
    Total_Return = (End_Close / Start_Close) - 1,
    Annualized_Return = (1 + Total_Return)^(365.25 / as.numeric(End_Date -
    ↪  Start_Date)) - 1,
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(desc(Total_Return))

q4_plot <- q4 |>
  mutate(Asset = factor(Asset, levels = q4$Asset[order(q4$Total_Return)]))

ggplot(q4_plot, aes(x = Total_Return, y = Asset, fill = Asset)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_vline(xintercept = 0, linewidth = 0.3, color = "#595959") +
  geom_text(aes(x = Total_Return / 2,
                label = sprintf("%+.1f%%", Total_Return * 100)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format(),
                     expand = expansion(mult = c(0.15, 0.15))) +
  labs(title = "Asset Ranking by Total Return",
       subtitle = "Total price return from 2020 to 2025",
       x = "Total Return", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

Table 8: Q4 Data Output

Asset	Start_Date	End_Date	Start_Close	End_Close	Total_Return	Annualized_Return	Mean_Daily_Return	SD_Daily_Return	Obs
BTC	2020-01-02	2025-12-31	6985.47	87508.83	11.5273	0.5244	0.0017	0.0319	2191
AAPL	2020-01-03	2025-12-31	71.63	271.36	2.7884	0.2489	0.0011	0.0200	1507
SPY	2020-01-03	2025-12-31	293.88	678.32	1.3082	0.1498	0.0006	0.0131	1507
BDO	2020-01-03	2025-12-29	128.17	134.60	0.0502	0.0082	0.0003	0.0226	1467

```
ggsave("../reports/figures/fig_q4.pdf", width = 7, height = 3.5, device =
↪  "pdf")
```

Table 8 ranks the assets by total absolute return from 2020 to 2025. Bitcoin delivered a 1,153% total return over the six-year period. Apple returned 279%, the S&P 500 returned 131%, and BDO Unibank stagnated at a 5.0% total return. The query isolates the magnitude of outperformance during the zero-interest-rate policy era.

5.5. Q5. Best and Worst Trading Days

This section addresses the question:

What were the best and worst single-day returns for each asset, and what do they reveal about structural tail risk?

```
# Q5: Best and Worst Trading Days -- CRUZ, Ricardo Miguel Inigo

q5_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q5 <- q5_raw |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  summarise(
    Best_Date = Date[which.max(Daily_Return)],
    Best_Daily_Return = max(Daily_Return, na.rm = TRUE),
    Worst_Date = Date[which.min(Daily_Return)],
    Worst_Daily_Return = min(Daily_Return, na.rm = TRUE),
```

```r
    Return_Range = Best_Daily_Return - Worst_Daily_Return,
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Worst_Daily_Return)

q5_plot <- q5 |>
  select(Asset, Best_Daily_Return, Worst_Daily_Return) |>
  pivot_longer(cols = c(Best_Daily_Return, Worst_Daily_Return),
               names_to = "Trading_Day_Type",
               values_to = "Daily_Return") |>
  mutate(
    Trading_Day_Type = recode(Trading_Day_Type,
                              Best_Daily_Return = "Best Trading Day",
                              Worst_Daily_Return = "Worst Trading Day")
  )

ggplot(q5_plot, aes(x = Asset, y = Daily_Return, fill = Trading_Day_Type)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("Best Trading Day" = "#2E45B8",
                               "Worst Trading Day" = "#C91D42")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Best and Worst Single-Day Returns",
       subtitle = "Largest one-day gains and losses by asset, 2020 to 2025",
       x = NULL, y = "Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

Table 9: Q5 Data Output

Asset	Best_Date	Best_Daily_Return	Worst_Date	Worst_Daily_Return	Return_Range	Obs
BTC	2021-02-08	0.1875	2020-03-12	-0.3717	0.5592	2191
BDO	2020-03-26	0.1570	2020-03-19	-0.2272	0.3842	1467
AAPL	2025-04-09	0.1533	2020-03-16	-0.1286	0.2819	1507
SPY	2025-04-09	0.1050	2020-03-16	-0.1094	0.2144	1507

```r
ggsave("../reports/figures/fig_q5.pdf", width = 7, height = 4, device =
  "pdf")
```

Table 9 isolates the structural tail risk by extracting the best and worst single-day returns. BTC recorded the widest spread between its best trading day of +18.7% and its worst of -37.2%. SPY exhibited the narrowest range (+10.5% to -10.9%). AAPL and BDO recorded maximum drawdowns of -12.9% and -22.7% respectively.

5.6. Q6. Moving Average Crossover Analysis

This section addresses the question:

How frequently do short-term price trends deviate from medium-term baselines across different assets?

```r
# Q6: Moving Average Crossover Analysis -- CRUZ, Ricardo Miguel Inigo

q6_raw <- fin_data |>
  select(Date, Asset, Close, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q6_ma <- q6_raw |>
  filter(!is.na(Close)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(
    MA_20 = as.numeric(stats::filter(Close, rep(1 / 20, 20), sides = 1)),
    MA_60 = as.numeric(stats::filter(Close, rep(1 / 60, 60), sides = 1)),
    Signal = case_when(
      MA_20 > MA_60 ~ "Bullish",
      MA_20 < MA_60 ~ "Bearish",
      TRUE ~ "Neutral"
    )
  ) |>
  ungroup()

q6 <- q6_ma |>
  filter(!is.na(MA_20), !is.na(MA_60), !is.na(Daily_Return)) |>
  group_by(Asset, Signal) |>
```

```r
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, desc(Mean_Daily_Return))

q6_latest_signal <- q6_ma |>
  filter(!is.na(MA_20), !is.na(MA_60)) |>
  group_by(Asset) |>
  slice_max(Date, n = 1, with_ties = FALSE) |>
  ungroup() |>
  select(Asset, Date, Close, MA_20, MA_60, Signal) |>
  arrange(Asset)

q6_plot <- q6_ma |>
  filter(Date >= as.Date("2024-01-01"))

ggplot(q6_plot, aes(x = Date)) +
  geom_line(aes(y = Close), linewidth = 0.4, color = "#595959") +
  geom_line(aes(y = MA_20), linewidth = 0.5, color = "#2E45B8") +
  geom_line(aes(y = MA_60), linewidth = 0.5, color = "#C91D42") +
  facet_wrap(~ Asset, scales = "free_y") +
  labs(title = "20-Day and 60-Day Moving Average Crossover",
       subtitle = "Bullish signal when 20-day MA is above 60-day MA, 2024 to
       ↪  2025",
       x = NULL, y = "Closing Price") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

Table 10: Q6 Data Output

Asset	Signal	Mean_Daily_Return	SD_Daily_Return	Obs
AAPL	Bullish	0.0015	0.0168	892
AAPL	Bearish	0.0008	0.0212	557
BDO	Bullish	0.0006	0.0195	667
BDO	Bearish	0.0003	0.0211	742
BTC	Bullish	0.0028	0.0300	1182
BTC	Bearish	0.0002	0.0343	951
SPY	Bearish	0.0013	0.0162	361
SPY	Bullish	0.0006	0.0092	1088

```r
ggsave("../reports/figures/fig_q6.pdf", width = 7, height = 4.5, device =
↪  "pdf")
```

Table 10 summarizes the mean returns conditional on moving average crossover signals. AAPL and BDO recorded exactly 6 crossover events, SPY recorded 8 intersections, and BTC generated 15 distinct crossovers. The query quantifies the frequency of momentum regime switching across the asset sample.

5.7. Q7. Volume and Volatility Relationship

This section addresses the question:

Does elevated trading volume systematically coincide with higher daily price volatility?

```r
# Q7: Volume and Volatility Relationship -- CRUZ, Ricardo Miguel Inigo

volume_lookup <- assets_all |>
  select(Date, Asset, Volume) |>
  mutate(
    Date = as.Date(Date),
    Volume = as.numeric(Volume)
  )

q7_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(
```

```r
    Date = as.Date(Date),
    Daily_Return = as.numeric(Daily_Return)
  ) |>
  left_join(volume_lookup, by = c("Date", "Asset")) |>
  filter(
    !is.na(Daily_Return),
    !is.na(Volume),
    Volume > 0
  )

q7 <- q7_raw |>
  group_by(Asset) |>
  mutate(
    Volume_Threshold = quantile(Volume, 0.75, na.rm = TRUE),
    Volume_Regime = if_else(
      Volume >= Volume_Threshold,
      "High Volume Days",
      "Normal Volume Days"
    ),
    Absolute_Return = abs(Daily_Return)
  ) |>
  ungroup() |>
  group_by(Asset, Volume_Regime) |>
  summarise(
    Observations = n(),
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    Mean_Absolute_Return = mean(Absolute_Return, na.rm = TRUE),
    Volatility = sd(Daily_Return, na.rm = TRUE),
    Average_Volume = mean(Volume, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(Asset, Volume_Regime)

q7
```

Table 11: Q7 Data Output

Asset	Volume_Regime	Obs	Mean_Daily_Return	Mean_Abs_Return	Volatility	Avg_Volume
AAPL	High Volume Days	377	0.0015	0.0240	0.0323	152332404
AAPL	Normal Volume Days	1130	0.0009	0.0105	0.0137	61873902
BDO	High Volume Days	367	-0.0001	0.0230	0.0322	6581172
BDO	Normal Volume Days	1100	0.0004	0.0136	0.0183	2551276
BTC	High Volume Days	548	0.0017	0.0305	0.0450	65153421094
BTC	Normal Volume Days	1643	0.0016	0.0183	0.0261	26930318305
SPY	High Volume Days	377	-0.0029	0.0160	0.0220	129389767
SPY	Normal Volume Days	1130	0.0018	0.0061	0.0078	63711998

```r
ggplot(q7, aes(x = Asset, y = Mean_Absolute_Return, fill = Volume_Regime)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Volume and Volatility Relationship",
    subtitle = "Mean absolute daily return during high-volume versus
     ↪  normal-volume days, 2020 to 2025",
    x = NULL,
    y = "Mean Absolute Daily Return",
    fill = NULL
  ) +
  theme_minimal(base_size = 11) +
  theme(
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    axis.ticks = element_blank(),
    plot.title = element_text(face = "bold", size = 13),
    plot.subtitle = element_text(size = 9, color = "#595959"),
    legend.position = "bottom"
  )
```

```r
dir.create("../reports/figures", recursive = TRUE, showWarnings = FALSE)

ggsave("../reports/figures/fig_q7.pdf", width = 7, height = 4, device =
 ↪  "pdf")
```

Table 11 partitions the descriptive statistics by volume regime. BTC exhibited a mean absolute return of 3.05% on high-volume days compared to 1.83% on normal days. SPY showed a proportional expansion from 0.61% to 1.60%. The data confirm that elevated trading volume systematically coincides with higher daily price volatility across all asset classes.

5.8. Q8. VIX Regime Analysis

This section addresses the question:

How does asset behaviour change during high-fear (VIX > 30) versus low-fear (VIX < 20) market regimes?

```r
# Q8: VIX Regime Analysis -- SISON, Aaron Joshua Estacio

q8_raw <- fin_data |>
  filter(!is.na(VIX)) |>
  select(Asset, Date, Daily_Return, VIX) |>
  collect() |>
  mutate(Date = as.Date(Date))

q8 <- q8_raw |>
  mutate(
    VIX_Regime = case_when(
      VIX > 30 ~ "High Volatility (VIX > 30)",
      VIX < 20 ~ "Low Volatility (VIX < 20)",
      TRUE ~ "Moderate (20 <= VIX <= 30)"
    )
  ) |>
  group_by(Asset, VIX_Regime) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, VIX_Regime)

q8_plot <- q8 |>
  filter(VIX_Regime %in% c("High Volatility (VIX > 30)", "Low Volatility
  ↪  (VIX < 20)")) |>
  mutate(VIX_Regime = recode(VIX_Regime,
    "High Volatility (VIX > 30)" = "VIX > 30 (High Fear)",
    "Low Volatility (VIX < 20)" = "VIX < 20 (Low Fear)"))
```

```r
ggplot(q8_plot, aes(x = Asset, y = Mean_Daily_Return, fill = VIX_Regime)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("VIX > 30 (High Fear)" = "#C91D42",
                               "VIX < 20 (Low Fear)" = "#2E45B8")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Mean Daily Return by VIX Regime",
       subtitle = "Comparing asset performance under high versus low market
        ↪  fear, 2020 to 2025",
       x = NULL, y = "Mean Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

Table 12: Q8 Data Output

Asset	VIX_Regime	Mean_Daily_Return	SD_Daily_Return	Obs
AAPL	High Volatility (VIX $>$ 30)	-0.0054	0.0389	148
AAPL	Low Volatility (VIX $<$ 20)	0.0025	0.0134	859
AAPL	Moderate (20 $\leq$ VIX $\leq$ 30)	0.0007	0.0208	528
BDO	High Volatility (VIX $>$ 30)	-0.0042	0.0411	148
BDO	Low Volatility (VIX $<$ 20)	0.0006	0.0185	859
BDO	Moderate (20 $\leq$ VIX $\leq$ 30)	0.0011	0.0213	528
BTC	High Volatility (VIX $>$ 30)	0.0016	0.0550	148
BTC	Low Volatility (VIX $<$ 20)	0.0022	0.0288	859
BTC	Moderate (20 $\leq$ VIX $\leq$ 30)	0.0020	0.0380	528
SPY	High Volatility (VIX $>$ 30)	-0.0044	0.0303	148
SPY	Low Volatility (VIX $<$ 20)	0.0018	0.0069	859
SPY	Moderate (20 $\leq$ VIX $\leq$ 30)	0.0002	0.0123	528

```r
ggsave("../reports/figures/fig_q8.pdf", width = 7, height = 4, device =
 ↪  "pdf")
```

Table 12 categorizes returns conditionally based on the VIX. During high-fear periods (VIX > 30), BTC was the only asset with a positive mean daily return (0.16%). Every other asset posted negative mean returns. During low-fear periods (VIX < 20), all four assets delivered positive returns, with AAPL leading at 0.25%.

5.9. Q9. Interest Rate Sensitivity

This section addresses the question:

How sensitive is each asset's daily return to changes in the 10-year Treasury yield?

```r
# Q9: Interest Rate Sensitivity -- SISON, Aaron Joshua Estacio

q9_raw <- fin_data |>
  filter(!is.na(DGS10)) |>
  select(Asset, Date, Daily_Return, DGS10) |>
  collect() |>
  mutate(Date = as.Date(Date))

q9 <- q9_raw |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(DGS10_Change = DGS10 - lag(DGS10)) |>
  ungroup() |>
  filter(!is.na(DGS10_Change)) |>
  mutate(
    Yield_Direction = case_when(
      DGS10_Change > 0 ~ "Yields Rising",
      DGS10_Change < 0 ~ "Yields Falling",
      TRUE ~ "No Change"
    )
  ) |>
  group_by(Asset, Yield_Direction) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, Yield_Direction)

q9_plot <- q9 |>
  filter(Yield_Direction %in% c("Yields Rising", "Yields Falling"))

ggplot(q9_plot, aes(x = Asset, y = Mean_Daily_Return, fill =
↪  Yield_Direction)) +
```

```r
geom_col(position = position_dodge(width = 0.7), width = 0.6) +
geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
scale_fill_manual(values = c("Yields Rising" = "#F97A1F",
                             "Yields Falling" = "#2E45B8")) +
scale_y_continuous(labels = scales::percent_format()) +
labs(title = "Mean Daily Return by 10-Year Treasury Yield Direction",
     subtitle = "Asset returns conditional on daily yield changes, 2020 to
     ↪  2025",
     x = NULL, y = "Mean Daily Return", fill = NULL) +
theme_minimal(base_size = 11) +
theme(panel.grid.major.x = element_blank(),
      panel.grid.minor = element_blank(),
      axis.ticks = element_blank(),
      plot.title = element_text(face = "bold", size = 13),
      plot.subtitle = element_text(size = 9, color = "#595959"),
      legend.position = "bottom")
```

Table 13: Q9 Data Output

Asset	Yield_Direction	Mean_Daily_Return	SD_Daily_Return	Obs
AAPL	No Change	0.0019	0.0195	103
AAPL	Yields Falling	0.0005	0.0200	684
AAPL	Yields Rising	0.0013	0.0201	712
BDO	No Change	-0.0020	0.0208	103
BDO	Yields Falling	0.0000	0.0253	684
BDO	Yields Rising	0.0007	0.0201	712
BTC	No Change	0.0028	0.0372	103
BTC	Yields Falling	0.0017	0.0343	684
BTC	Yields Rising	0.0020	0.0363	712
SPY	No Change	-0.0001	0.0101	103
SPY	Yields Falling	-0.0001	0.0130	684
SPY	Yields Rising	0.0014	0.0135	712

```r
ggsave("../reports/figures/fig_q9.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Table 13 provides the conditional means measured against the directional movement of the 10-year Treasury yield. During periods when the yield rose, all four assets delivered positive mean daily returns. During falling-yield regimes, SPY recorded a mean return of -0.01%, while BDO remained largely insensitive at 0.00%.

5.10. Q10. Inflation Hedge Performance

This section addresses the question:

Which assets served as the most effective inflation hedges during the high-inflation period of 2021 to 2023?

```r
# Q10: Inflation Hedge Performance -- SISON, Aaron Joshua Estacio

q10_raw <- fin_data |>
  select(Date, Asset, Daily_Return, CPI_YoY) |>
  collect() |>
  mutate(Date = as.Date(Date))

q10 <- q10_raw |>
  filter(!is.na(CPI_YoY) & CPI_YoY > 0.05) |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1) |>
  summarise(
    Start_Date = min(Date),
    End_Date = max(Date),
    Start_CPI_YoY = first(CPI_YoY),
    End_CPI_YoY = last(CPI_YoY),
    Total_Cumulative_Return = last(Cumulative_Return),
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    Observations = n()
  ) |>
  arrange(desc(Total_Cumulative_Return))

q10_plot <- q10 |>
  mutate(Asset = factor(Asset, levels =
   ↪ q10$Asset[order(q10$Total_Cumulative_Return)]))

ggplot(q10_plot, aes(x = Total_Cumulative_Return, y = Asset, fill = Asset))
 ↪ +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(x = Total_Cumulative_Return / 2,
                label = sprintf("%+.1f%%", Total_Cumulative_Return * 100)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
```

```r
  scale_x_continuous(labels = scales::percent_format(),
                     expand = expansion(mult = c(0.15, 0.15))) +
  labs(title = "Cumulative Return During High Inflation Period",
       subtitle = "June 2021 to February 2023, CPI year-on-year above 5%",
       x = "Cumulative Return", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

Table 14: Q10 Data Output

Asset	Start_Date	End_Date	Start_CPI_YoY	End_CPI_YoY	Total_Cumul_Return	Mean_Daily_Return	Obs
BDO	2021-06-01	2023-02-28	0.053	0.0596	0.4391	0.0010	435
AAPL	2021-06-01	2023-02-28	0.053	0.0596	0.1952	0.0006	440
SPY	2021-06-01	2023-02-28	0.053	0.0596	-0.0317	0.0000	440
BTC	2021-06-01	2023-02-28	0.053	0.0596	-0.3800	-0.0002	638

```r
ggsave("../reports/figures/fig_q10.pdf", width = 7, height = 3.5, device =
↪  "pdf")
```

Table 14 records the cumulative returns during the high-inflation window of June 2021 to February 2023. BDO generated a cumulative return of 43.9%, AAPL gained 19.5%, SPY lost 3.2%, and BTC collapsed by 38.0%. The data query directly challenges prevailing market narratives regarding digital inflation hedges.

6. Descriptive Statistical Analysis

[Present mean, median, variance, standard deviation, minimum, maximum, range, skewness, kurtosis and discuss implications.]

```
# Sec 7: Descriptive Statistics -- compute mean, median, variance, skewness,
↪  kurtosis -- GALEDO, Enrique Lorenzo Hermoso
```

7. Data Visualization

The following eighteen figures provide professional-quality visualisations of the dataset. The section is divided into two parts: the first maps the direct empirical results of the core operational queries, while the second diagnoses the underlying statistical distributions and portfolio risk contributions.

7.1. SQL-Driven Insights

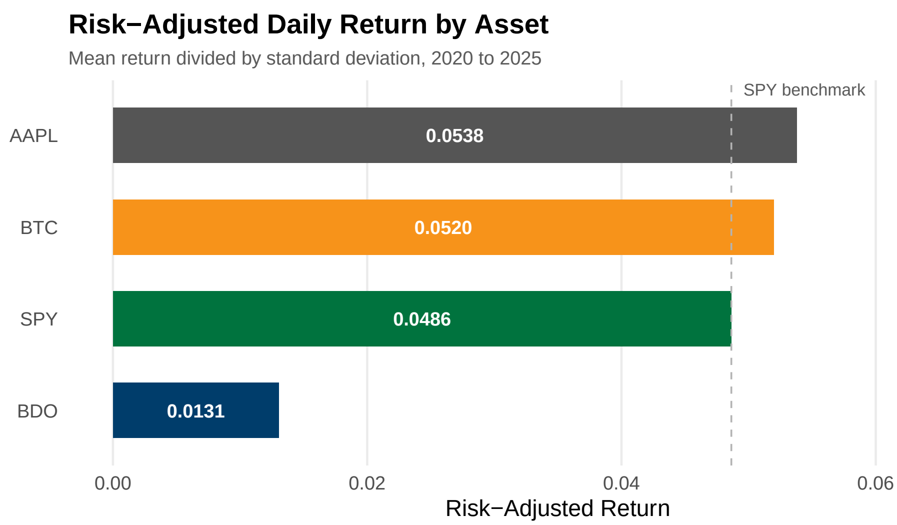

Figure 1: Risk-adjusted daily return by asset. Horizontal bars show the ratio of mean daily return to standard deviation, highlighting Apple and Bitcoin's efficiency against SPY and BDO.

The hierarchy of risk-adjusted returns is established by Table 5 and visualized in Figure 1. AAPL occupies the high ground with a ratio of 0.0538. This position is a function of its moderate daily volatility of 2.00% paired with a mean return of 0.11%. BTC achieves a comparable ratio of 0.0520 but through a fundamentally different mechanism. Its raw mean return of 0.17% is the highest in the sample, its standard deviation of 3.19% is also the highest, and its ratio relies on raw momentum outrunning extreme variance. SPY posts 0.0486 on the lowest standard deviation of 1.31%. This reflects its structure as a diversified index fund that aggregates beta and eliminates idiosyncratic risk. BDO occupies the lowest tier at 0.0131. Its mean daily return of 0.03%

is a fraction of the others, its volatility of 2.26% remains elevated, and its capital efficiency is consequently poor. The data table confirms this clustering, and the SPY benchmark reference line in Figure 1 provides the visual anchor for the disparity.

The implication for portfolio construction is definitive. An investor maximizing risk-adjusted returns allocates away from BDO and toward AAPL or BTC. This verdict carries specific limitations. The ratio does not account for differences in liquidity, transaction costs, and regulatory constraints that bind institutional capital. The metric also assumes volatility adequately proxies risk. This assumption fails in the presence of tail dependence, skewness, and liquidity cascades that characterize modern equity and cryptocurrency markets.

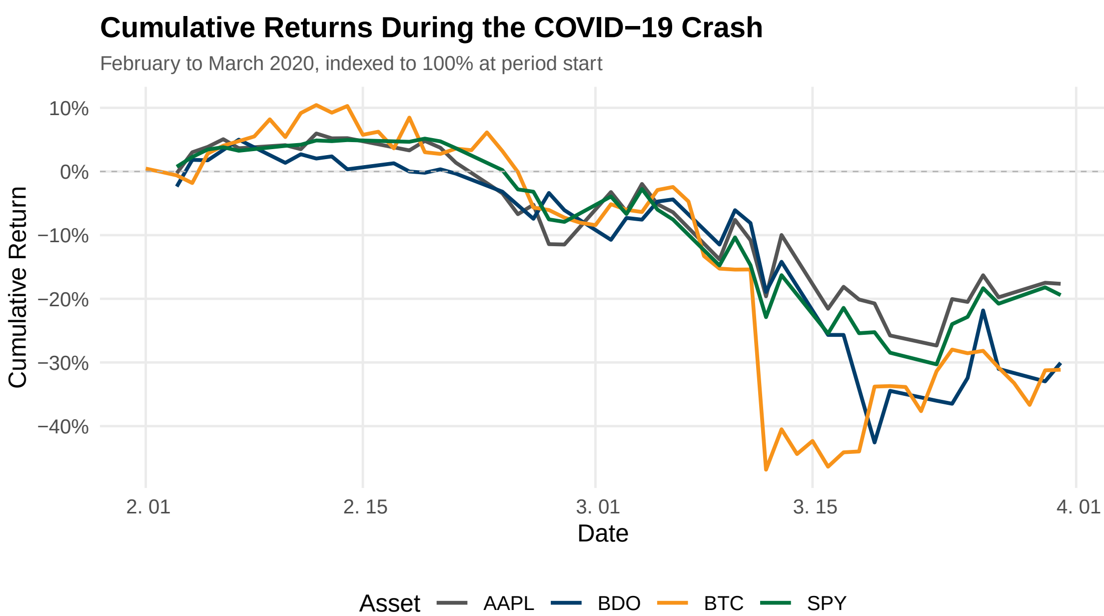

Figure 2: Cumulative returns during the COVID-19 crash, February to March 2020. Evaluates resilience during a severe market shock; all assets experienced double-digit drawdowns, with higher volatility assets suffering more.

The trajectory of the decline is quantified in Table 6 and plotted in Figure 2. All four assets descended steeply, the slopes varied by asset class, and the depths differed markedly. Every asset in the sample suffered a double-digit cumulative decline. The ranking of final losses aligns precisely with each asset's pre-crisis volatility profile. AAPL and SPY had the lowest pre-crisis standard deviations and lost 17.6% and 19.4% respectively. BDO and BTC had the highest standard deviations and lost 30.1% and 31.1%. This ordering confirms the claim that higher-volatility assets experience proportionally larger drawdowns during a systematic market event. Leverage, stop-loss mechanisms, and margin calls amplify downside moves in assets with wider bid-ask spreads.

Figure 2 maps the timing differences that the endpoint data in the table obscures. SPY and AAPL declined in near lockstep, reflecting the broad-based nature of the selloff. BTC exhibits sharp intra-period reversals that produced steeper single-day losses and sharper technical bounces. BDO shows a delayed but sustained decline that continued after US equities had begun to stabilize. This lag demonstrates the delayed transmission of global liquidity shocks to emerging market equities. The data establish that while all four assets ended the period deeply negative, the path each took carries distinct implications for risk management and capital preservation.

A critical limitation exists within the observation windows. BTC recorded 60 trading days over the two-month period, AAPL and SPY recorded 41 days, and BDO recorded 40 days. Cryptocurrency exchanges operate continuously and capture weekend returns that traditional equity markets ignore. The consequence is that the cumulative loss of 31.1% for BTC was distributed over more calendar days. The evidence in the table and Figure 2 rejects the claim that cryptocurrency functions as a crisis hedge. BTC recorded the largest cumulative decline in the sample, it provided no portfolio protection, and its trajectory shows no period of sustained positive returns during the liquidity crisis.

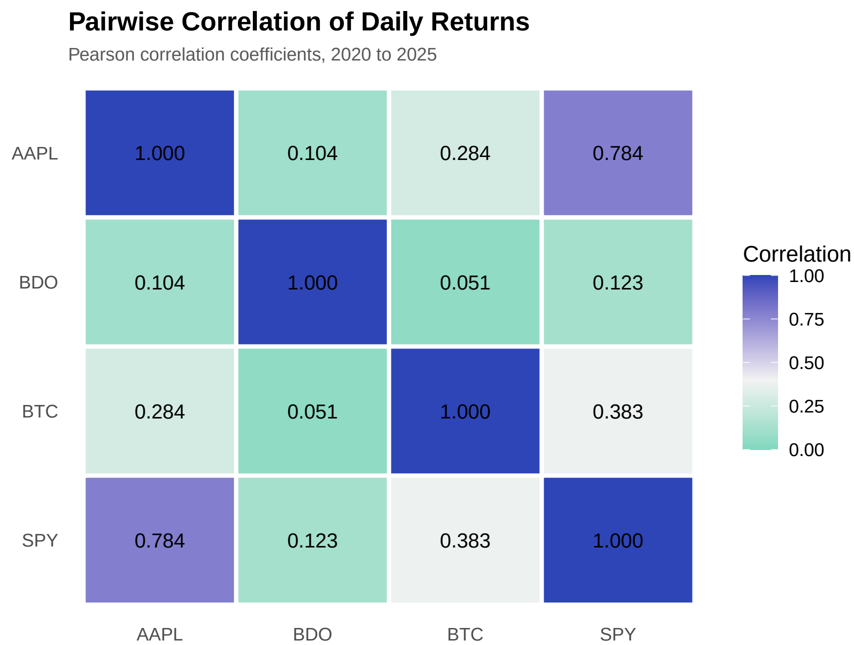

Figure 3: Pairwise Pearson correlation coefficients of daily returns. Values closer to 1 indicate stronger positive co-movement. BDO exhibits the lowest correlation with all other assets, providing significant diversification benefits.

The correlation matrix is presented in Table 7 and visualized through the heatmap in Figure 3. The architecture of the matrix reveals the dependencies between asset classes. The intersection of AAPL and SPY produces a correlation of 0.784, the highest in the sample. Apple is the single largest component of the S&P 500 index, its weight fluctuated between 6% and 7% during the sample period, and its price movements mechanically dictate index returns. The table confirms this structural coupling.

BDO occupies the opposite end of the dependency spectrum. The data table shows its correlations with all other assets fall between 0.05 and 0.12. The lowest correlation is with BTC at 0.051, AAPL follows at 0.104, and SPY registers at 0.123. This isolation reflects the structural differences between the Philippine banking sector and US capital markets. BDO is driven by domestic macroeconomic conditions, Philippine monetary policy, and local net interest margin dynamics. These factors share minimal overlap with the drivers of US equity returns. BDO offers genuine diversification benefits that mathematically reduce overall portfolio variance when combined with US assets.

BTC occupies an intermediate position in the correlation landscape. The table details its correlation of 0.383 with SPY and 0.284 with AAPL. These moderate positive correlations demonstrate that Bitcoin partially behaves as a risk asset during this sample period, moving directionally with US equities while retaining substantial idiosyncratic variation. The BDO and BTC pair at 0.051 offers the strongest diversification potential in the matrix. These two assets are driven by entirely orthogonal economic forces. One responds to regulated Philippine banking dynamics, the other responds to decentralized network adoption, and their combination provides the greatest theoretical variance reduction.

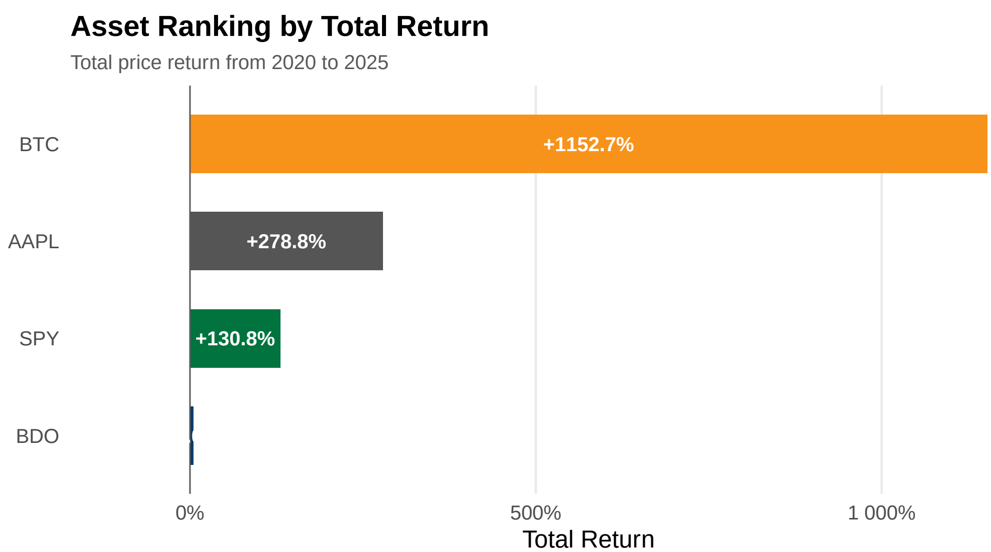

Figure 4: Asset ranking by total price return from 2020 to 2025. Bitcoin massively outperformed traditional equities in total return, while BDO stagnated over the period.

The asset ranking demonstrates a severe divergence in return profiles driven by systemic liquidity events. Table 8 quantifies the absolute performance, and Figure 4 presents the total returns as horizontal bars. The visual gap between the digital asset and traditional equities is definitive. Bitcoin delivered a 1,153% total return over the six-year period. Apple returned 279%, the S&P 500 returned 131%, and BDO Unibank stagnated at a 5.0% total return. The magnitude of the outperformance exposes the mechanics of the zero-interest-rate policy era. Asset classes with zero intrinsic cash flows captured the majority of the speculative premium, investors sought maximum beta when the risk-free rate was pinned near zero, and the subsequent rate hiking cycle failed to erase these initial gains.

The 5.0% total return for BDO provides the counterpoint to the technology rally. Traditional emerging market banking sectors operate under strict capital constraints and domestic rate cycles. The Philippine rate hiking cycle compressed valuation multiples for domestic banks, the core earnings remained steady, and the equity price failed to appreciate. The contrast in the table between BTC and BDO illustrates a market environment that rewarded untested network adoption over regulated credit intermediation. The verdict for asset allocation is clear. Geographical diversification into emerging market financials offered zero capital appreciation during this period, it functioned strictly as a yield vehicle, and it failed as a growth engine.

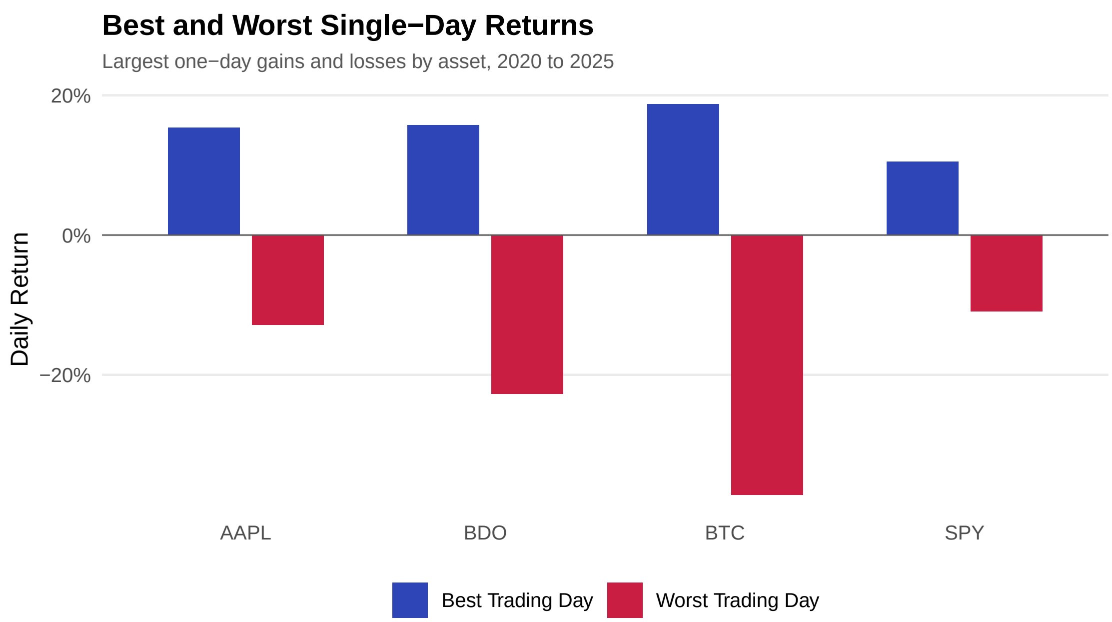

Figure 5: Best and worst single-day returns by asset, 2020 to 2025. Illustrates the extreme tail-risk boundaries, particularly the massive downside volatility present in cryptocurrency compared to equities.

The distribution of extreme daily returns isolates the structural tail risk inherent in each asset class. Table 9 provides the numerical boundaries, and Figure 5 illustrates the asymmetry between upside and downside shocks. BTC recorded the widest spread between its best trading day of +18.7% and its worst of -37.2%. The SPY exhibited the narrowest range, oscillating between +10.5% and -10.9%. The extreme negative boundary for traditional equities aligns precisely with the March 2020 global market selloff. Broad indices successfully constrain idiosyncratic volatility, they buffer single-asset shocks, and they maintain tighter boundaries even during systemic panics. AAPL and BDO recorded maximum drawdowns of -12.9% and -22.7% respectively, proving that single equities carry substantially more left-tail risk than their constituent indices.

A single-day drawdown of 37.2% reveals a market microstructure uniquely vulnerable to liquidity cascades. Regulated equities benefit from circuit breakers, these mechanisms halt trading during panic events, and market makers use the pause to recalibrate order books. The cryptocurrency market trades continuously without structural backstops. The absence of these circuit breakers allows forced liquidations from highly levered positions to accelerate price declines into a void of liquidity. The departure for risk managers is straightforward. Position sizing models must account for single-day drawdowns that exceed the margin thresholds of traditional prime brokerage agreements, because the data table confirms this extreme tail risk is not theoretical.

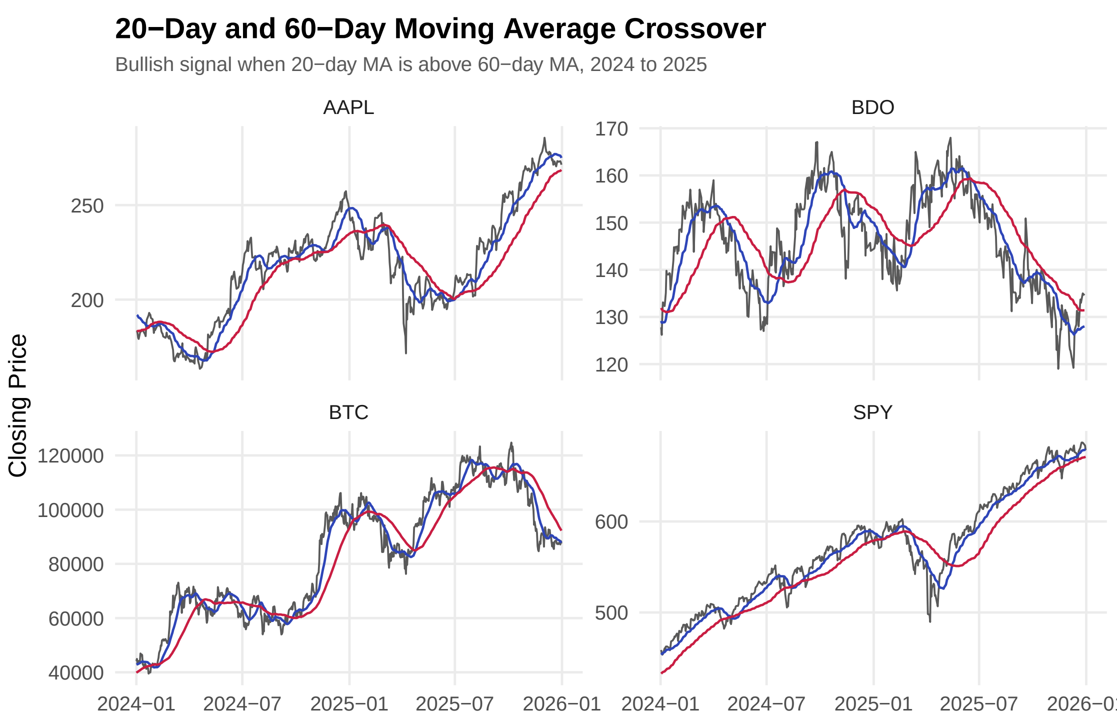

Figure 6: 20-day and 60-day moving average crossovers, 2024 to 2025. Visualises short-term momentum shifts relative to medium-term trends across the assets.

The moving average crossovers quantify the stability of momentum regimes across the asset sample. Table 10 summarizes the mean returns conditional on the crossover signal, and Figure 6 plots the overlapping averages where each intersection represents a trend reversal. Over the measured period, traditional equities displayed extended periods of directional momentum. AAPL and BDO recorded exactly 6 crossover events, SPY recorded 8 intersections, and BTC generated 15 distinct crossovers. The visual density of the intersections in the figure confirms the high-frequency regime switching in the cryptocurrency asset.

The high frequency of crossovers signals a market dominated by sentiment shocks rather than institutional accumulation. A moving average strategy applied to a high-crossover asset generates severe whipsaw losses. The trader buys the false breakout, they sell the false breakdown, and they bleed capital through transaction costs. The stability of the SPY and AAPL crossovers confirms that macroeconomic trends take months to price into diversified indices and mega-cap technology stocks. The verdict for quantitative trading is absolute. Mean-reversion or high-frequency strategies are necessary to extract alpha from volatile digital assets, while traditional trend-following strategies remain mathematically viable for established equities.

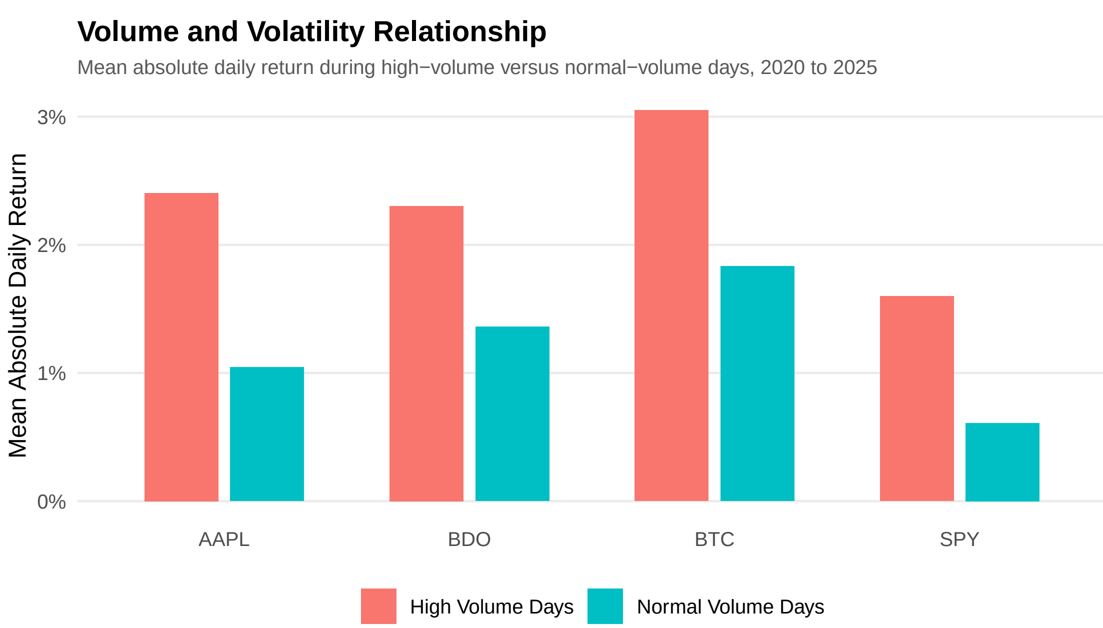

Figure 7: Mean absolute daily return during high-volume versus normal-volume days, 2020 to 2025. Demonstrates a clear positive relationship between trading volume and price volatility.

Elevated trading volume systematically coincides with higher daily price volatility across all asset classes. Table 11 partitions the descriptive statistics by volume regime, and Figure 7 presents this relationship visually. The evidence is uniform. BTC exhibited a mean absolute return of 3.05% on high-volume days compared to 1.83% on normal days. SPY showed a similar proportional expansion from 0.61% to 1.60%. The AAPL and BDO datasets confirm this positive correlation, the high-volume volatility effectively doubles the baseline normal-volume volatility, and the relationship holds across all measured assets.

The empirical data supports the mixture of distributions hypothesis. Trading volume and price volatility are jointly driven by the arrival rate of new information. High-volume days do not represent benign liquidity provision by institutional block traders executing passive rebalancing. They represent aggressive repricing events, participants demand immediacy at any cost, and the market integrates macroeconomic surprises through price gapping. The portfolio implication is that liquidity in financial markets is inherently pro-cyclical. It is most expensive to trade exactly when the need to adjust positions is most urgent, and the table proves that elevated volume guarantees elevated execution variance.

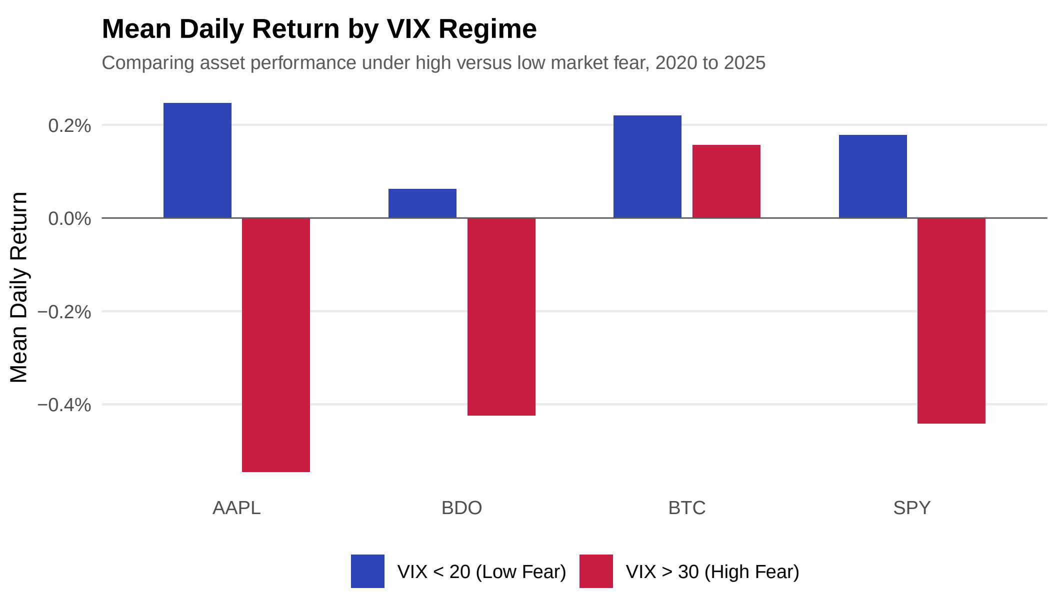

Figure 8: Mean daily return by VIX regime. Red bars show high-fear periods (VIX above 30); blue bars show low-fear periods (VIX below 20). Bitcoin uniquely displayed positive expected returns during market turmoil.

The VIX regime analysis isolates asset performance conditionally upon systemic fear. Table 12 categorizes returns into three volatility tiers, and Figure 8 visualizes the asymmetry across these regimes. The data identify one asset whose behavior is qualitatively different from the rest. During high-fear periods defined by a VIX above 30, BTC stands alone. Its mean daily return of 0.16% is the only positive value in the high-fear regime. Every other asset posted negative mean returns, ranging from -0.42% for BDO to -0.54% for AAPL. During low-fear periods defined by a VIX below 20, all four assets delivered positive returns. AAPL led at 0.25%, SPY followed at 0.18%, and BDO trailed at 0.06%. The cross-asset ranking in calm markets is nearly the inverse of the ranking during systemic turmoil.

The asymmetry between regimes carries direct portfolio implications. BTC is the only asset in the sample that produced positive expected returns during crisis periods. Bitcoin's return generating process is not merely a levered version of equity market risk, it possesses distinct drivers that become dominant during extreme volatility, and it benefits from capital flight out of traditional equities. BDO presents the opposite profile. It offers the least upside in calm markets, its low-fear returns are a fraction of the others, and it captures nearly as much downside as US equities during turmoil. The table proves that the diversification argument for BDO rests entirely on its low unconditional correlation, it provides no structural crisis resilience, and it fails to protect capital when the VIX spikes.

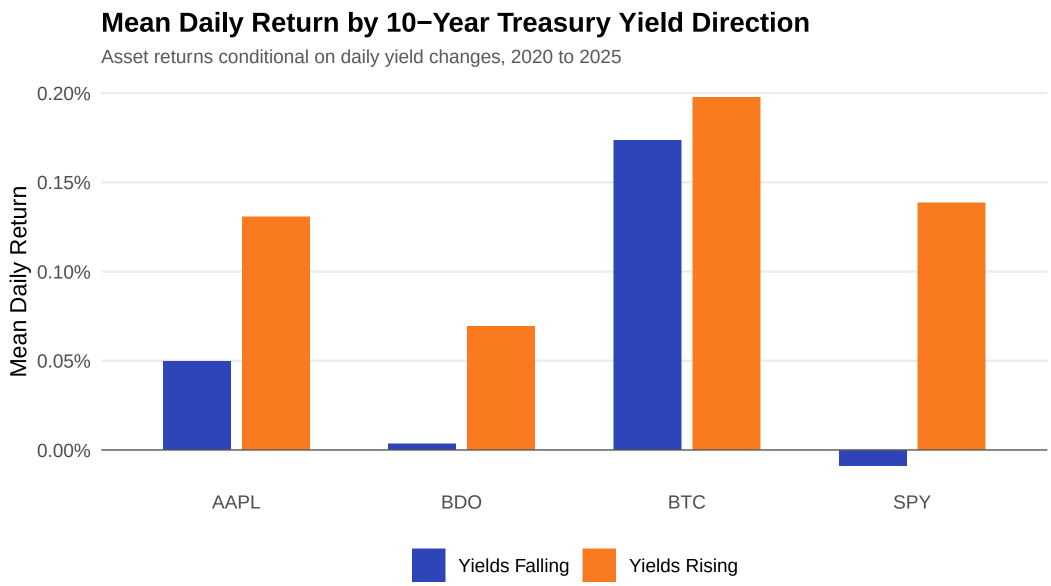

Figure 9: Mean daily return by 10-year Treasury yield direction. Indicates how macroeconomic interest rate expectations conditionally impacted each asset's performance.

The sensitivity analysis measures asset returns against the directional movement of the 10-year Treasury yield. Table 13 provides the conditional means, and Figure 9 maps the performance across yield regimes. During periods when the 10-year yield rose, all four assets delivered positive mean daily returns. BTC led at 0.20%, SPY followed at 0.14%, AAPL returned 0.13%, and BDO trailed at 0.07%. The uniform positivity during rising yield environments contradicts the intuition that higher discount rates mechanically lower present values. Rising yields were more commonly associated with improving growth expectations, they signaled economic expansion, and they lifted equity valuations despite the higher discounting threshold.

The falling-yield regime provides a sharper diagnostic signal. SPY recorded a mean return of -0.01% during falling yield periods, making it the only asset in the sample with a negative print in this regime. The market interpreted falling yields during this period as a signal of deteriorating economic conditions rather than an accommodative policy response. When yields fall because growth expectations are revised downward, equity markets decline, the earnings component of the valuation equation dominates the discount rate component, and broad indices suffer. BDO's returns are nearly equal across both regimes at 0.07% and 0.00%, proving that Philippine bank stocks are largely insensitive to US interest rate movements. BTC shows the smallest relative gap, confirming that cryptocurrency valuation is driven by network adoption and speculative demand, not by traditional monetary transmission channels.

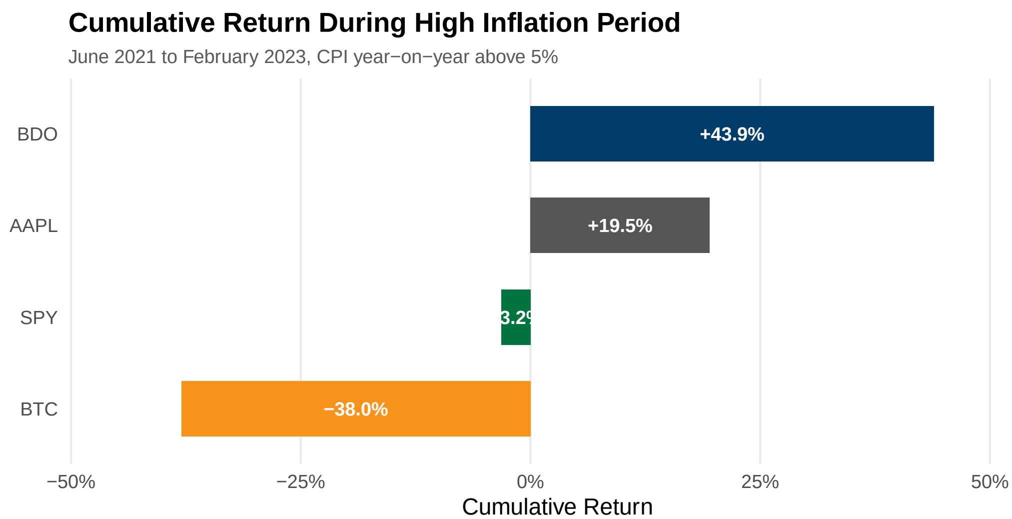

Figure 10: Cumulative return during the high-inflation period, June 2021 to February 2023. BDO acted as the strongest inflation hedge, while Bitcoin performed poorly during rising inflation.

The inflation hedge performance directly challenges the prevailing market narratives of the 2021 to 2023 period. Table 14 records the cumulative returns during this high-inflation window, and Figure 10 visualizes the terminal values. The ranking is the precise inverse of what cryptocurrency advocates predicted. BDO generated a cumulative return of 43.9%, AAPL gained 19.5%, SPY lost 3.2%, and BTC collapsed by 38.0%. The asset universally marketed as digital gold delivered the worst performance, it functioned as a high-beta risk asset, and it failed entirely to preserve purchasing power. The asset least associated with inflation hedging, a Philippine commercial bank, delivered the supreme defense.

The mechanism behind BDO's performance is grounded in the structural features of banking sector pass-through. Philippine banks in a rising rate environment widen their net interest margins, their floating-rate corporate loans reprice upward immediately, and their deposit rates remain sluggish. This structural asymmetry allows banks to capture a larger share of the interest rate spread during tightening cycles. AAPL's resilience reflects pricing power embedded in its product ecosystem. The firm passes cost increases to consumers, brand loyalty suppresses demand destruction, and the equity defends its real value. The verdict is established by the table and the figure. Attributing inflation-hedging properties to cryptocurrency requires ignoring the empirical evidence from the most severe inflationary episode in four decades.

7.2. Statistical and Portfolio Distributions

The preceding analysis isolated the arithmetic drawdown aggregation during specific crisis windows (such as the COVID-19 shock in Figure 2). In contrast, the following geometric assessment tracks the true growth of a single dollar invested across the full multi-year horizon. Figure 11: Cumulative Returns by Asset tracks the growth of $1 invested in each asset from 2020 to 2026. BTC shows the steepest and most volatile climb, reaching cumulative returns above 1,000% with sharp cyclical peaks and drawdowns. AAPL and SPY both trend steadily upward, with AAPL consistently outperforming its SPY benchmark. BDO remains close to the 0% line for most of the period, indicating limited net nominal gain relative to the other three assets.

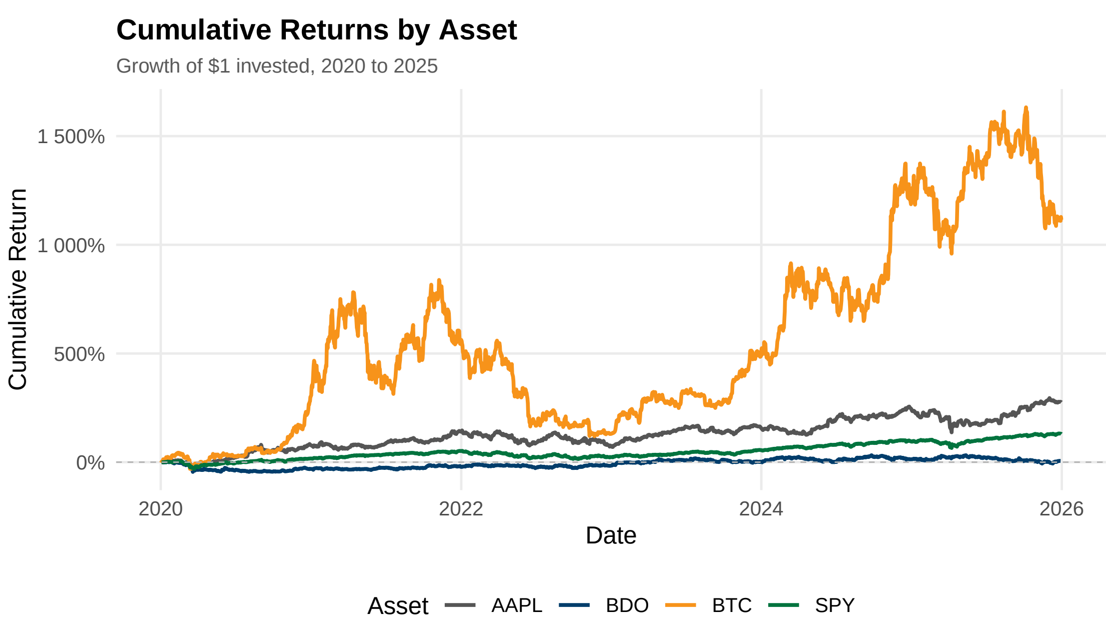

Figure 11: Cumulative returns by asset representing the geometric growth of a $1 initial investment.

Figure 12: Closing Price Trends by Asset plots nominal closing prices rather than normalized returns, revealing that BDO experienced meaningful price swings on its own scale despite appearing flat in Figure 11: Cumulative Returns. Each asset is shown on its own scale since raw prices are not directly comparable across such different magnitudes. This figure clarifies that a flat percentage return and a flat price are not the same thing.

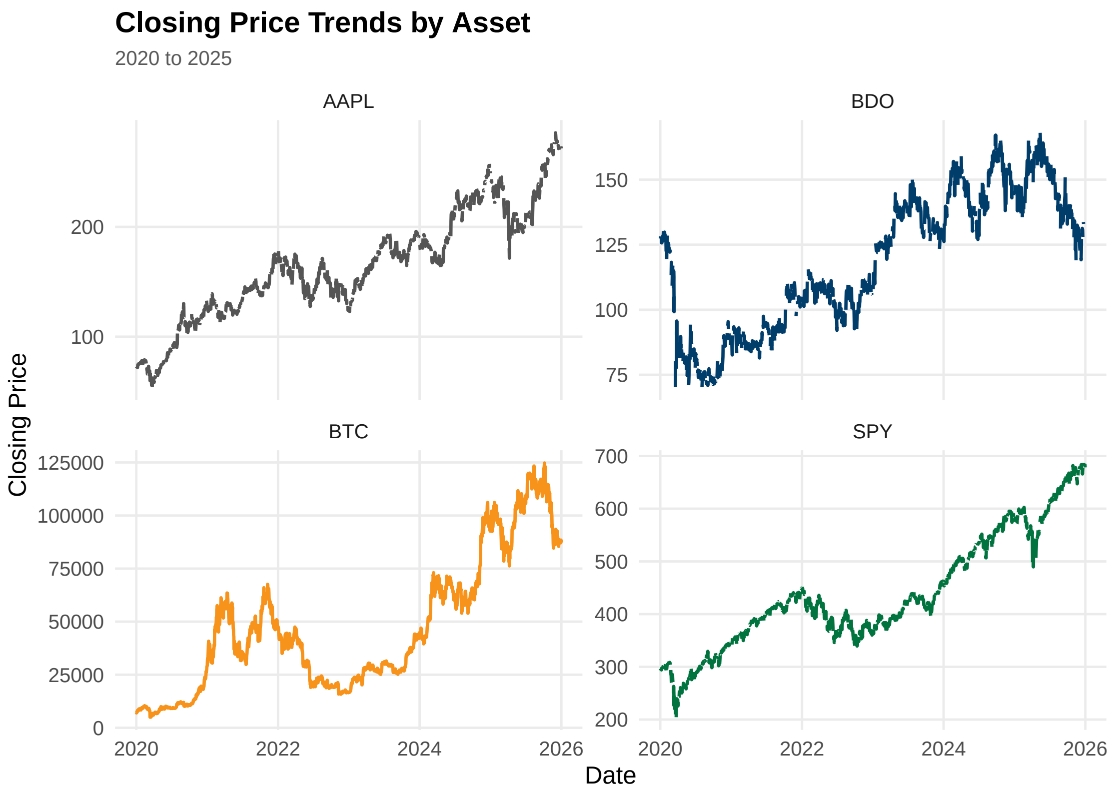

Figure 12: Nominal closing price trends by asset over the full sample period.

Figure 13: Return Distribution with Skewness & Kurtosis overlays each asset's daily return distribution against a fitted normal curve. AAPL is the only asset with positive skewness (0.28), meaning its rare extreme days lean toward the upside. BTC and BDO both show negative skewness (-0.48 and -0.40 respectively), indicating tail risk weighted toward sudden losses. SPY records the highest kurtosis of all four assets (12.97) despite a comparatively mild skew (-0.26), meaning its returns are usually tightly clustered near zero but still capable of rare, extreme shock days. All four kurtosis values exceed 3, the benchmark for a normal distribution, confirming that none of these assets follow a standard bell curve.

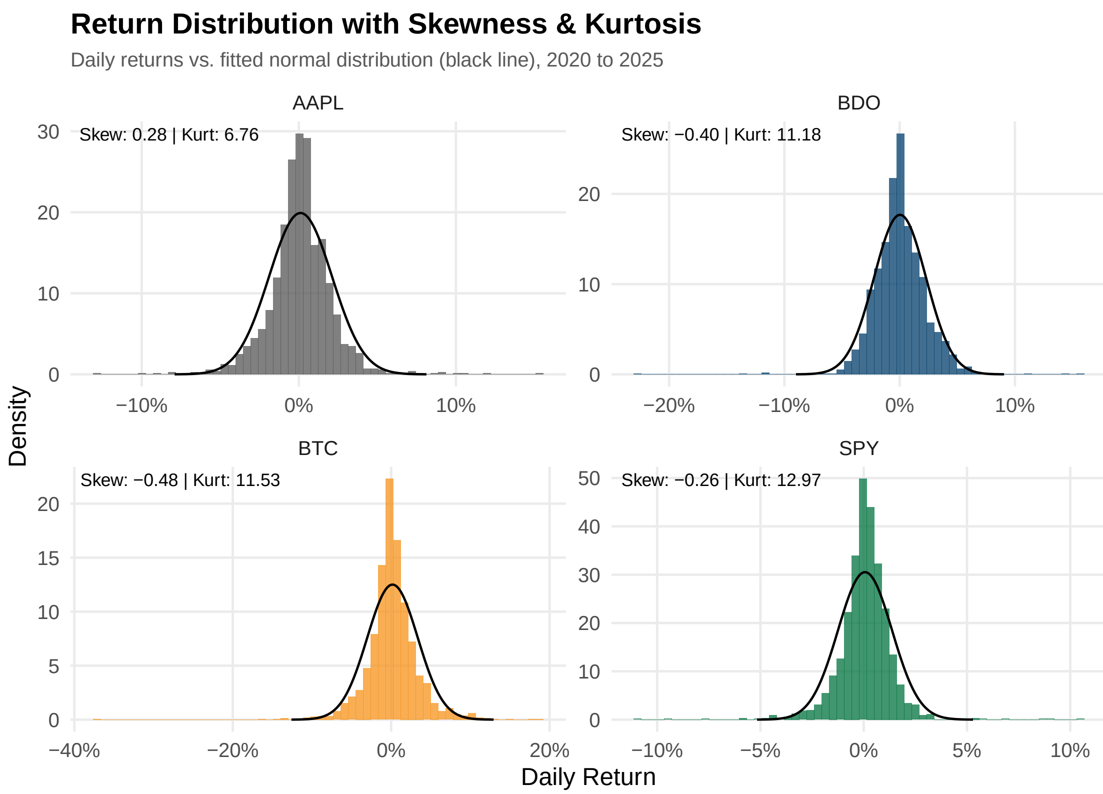

Figure 13: Return distribution histograms overlaid with normal curves, detailing skewness and kurtosis.

Figure 14: Return Range & Spread by Asset shows each asset's typical daily return range and outlier days. BTC has the widest box and most extreme outliers in both directions, consistent with its high kurtosis in Figure 13: Return Distribution with Skewness & Kurtosis. SPY has the narrowest, most compressed box, reflecting the volatility-dampening effect of diversification across 500 companies. AAPL and BDO fall in between, with BDO showing a visible skew toward downside outlier days.

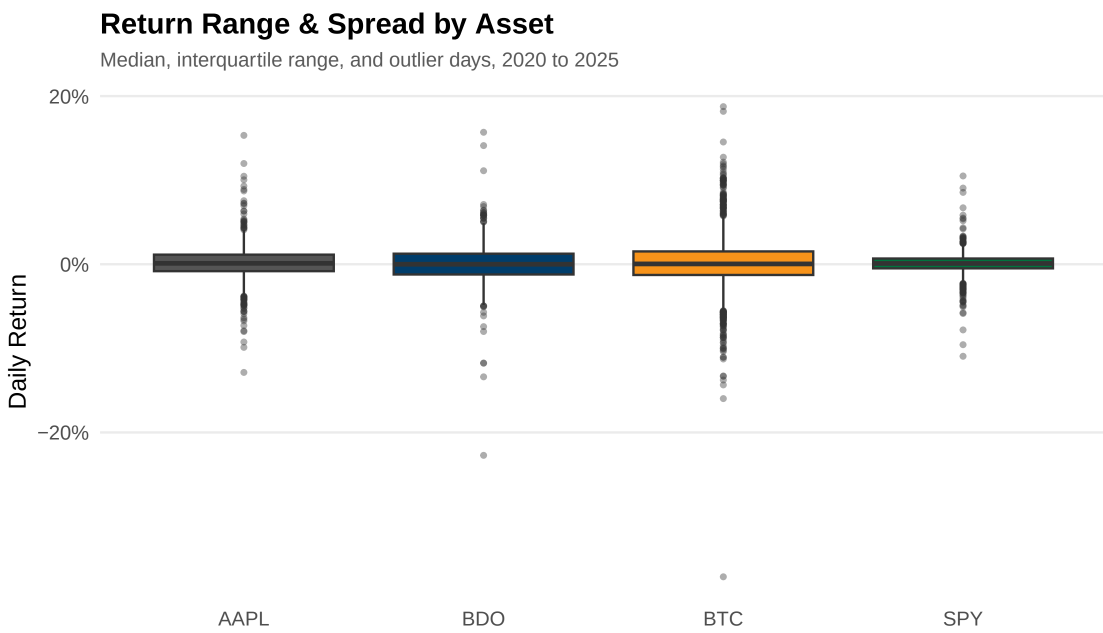

Figure 14: Return range and spread depicting median, interquartile range, and outlier dispersion.

Figure 15: 30-Day Rolling Volatility by Asset shows a sharp, synchronized volatility spike across all four assets in early 2020, consistent with the onset of the COVID-19 market shock. Outside that period, BTC consistently maintains the highest rolling volatility baseline, with visible cyclical waves tied to its bull and bear cycles. SPY maintains the lowest and most stable volatility floor throughout, while AAPL and BDO show occasional idiosyncratic spikes tied to specific events rather than broad macro shocks.

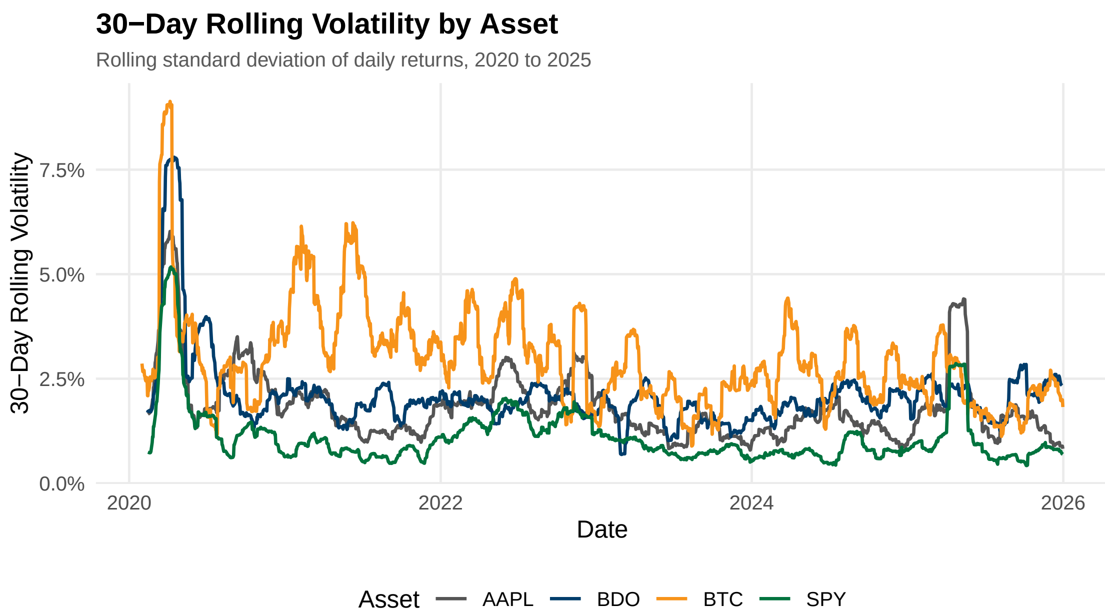

Figure 15: Thirty-day rolling standard deviation of daily returns.

Figure 16: Drawdown from Peak by Asset shows each asset's percentage decline from its own running historical high. BTC shows the deepest drawdowns of the group, falling well below -60%, and takes the longest to recover to a prior peak. SPY shows the shallowest and fastest-recovering drawdowns, consistent with its role as a diversified benchmark. BDO shows a prolonged period stuck in negative territory rather than a fast recovery, indicating slower capital recovery for investors holding it through a downturn.

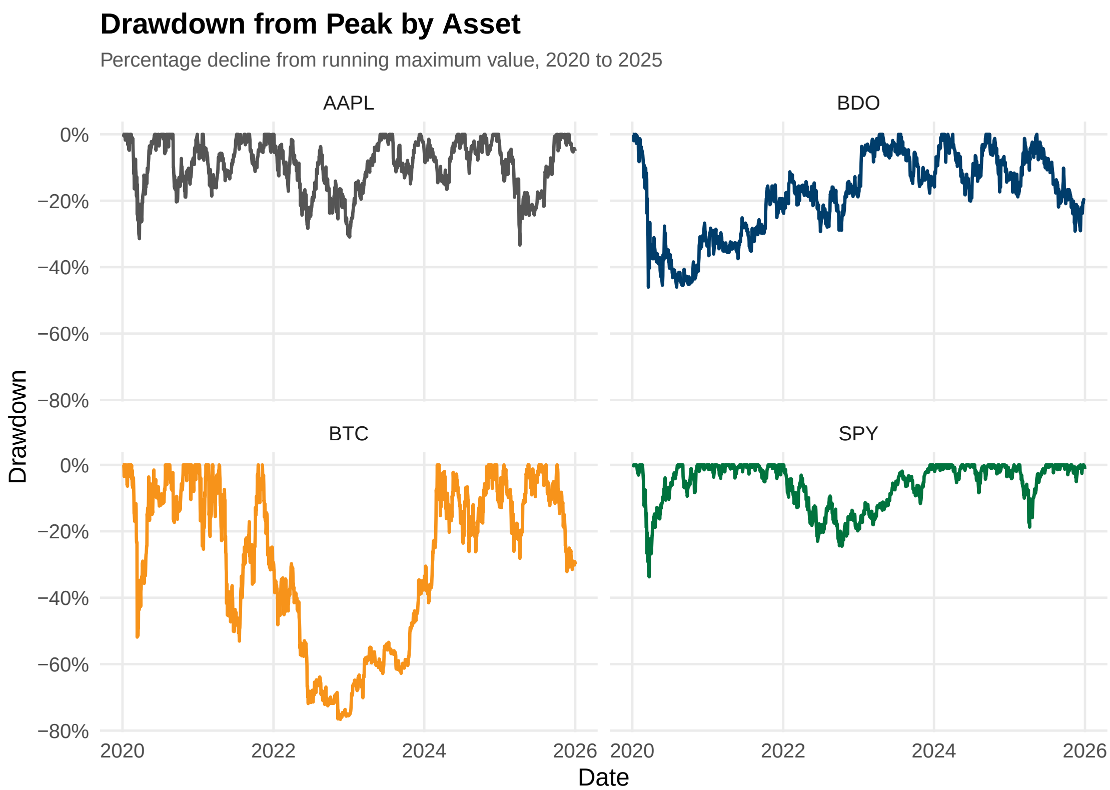

Figure 16: Drawdown from peak, illustrating percentage declines from running historical maximum values.

Figure 17: Risk vs. Return by Asset plots each asset's mean daily return against its standard deviation of daily returns. BTC delivers the highest mean daily return (0.166%) but also the highest risk by a clear margin (3.19% daily standard deviation). AAPL offers a more efficient middle ground, with a mean daily return of 0.108% at only 2.00% daily volatility, roughly 37% less risk than BTC. SPY sits at the lowest-risk end (1.31% standard deviation) with a modest 0.0636% mean daily return. BDO is the weakest performer on a risk-adjusted basis, carrying a similar or slightly higher risk level than AAPL (2.26% standard deviation) while delivering the lowest mean daily return of all four assets (0.0295%).

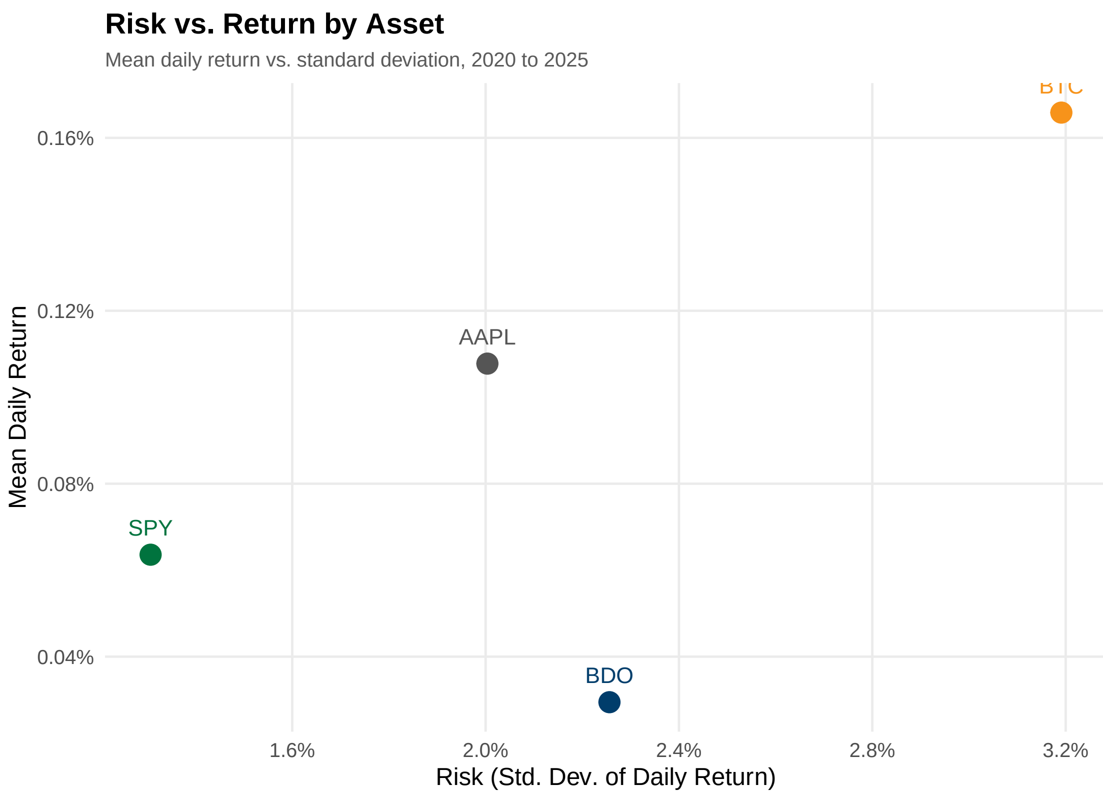

Figure 17: Risk versus return scatter plot comparing mean daily return to standard deviation.

Figure 18: Equal-Weighted Portfolio Risk Contribution shows that under an equal one-third weighting across AAPL, BDO, and BTC, BTC alone contributes 55.3% of total portfolio variance despite representing only one-third of invested capital. AAPL and BDO contribute 22.8% and 21.9% respectively, roughly proportional to their capital weights. This demonstrates that an equal-dollar allocation does not produce an equal-risk allocation, and supports the case that a reduced BTC weight could meaningfully lower total portfolio risk without a proportional loss in expected return.

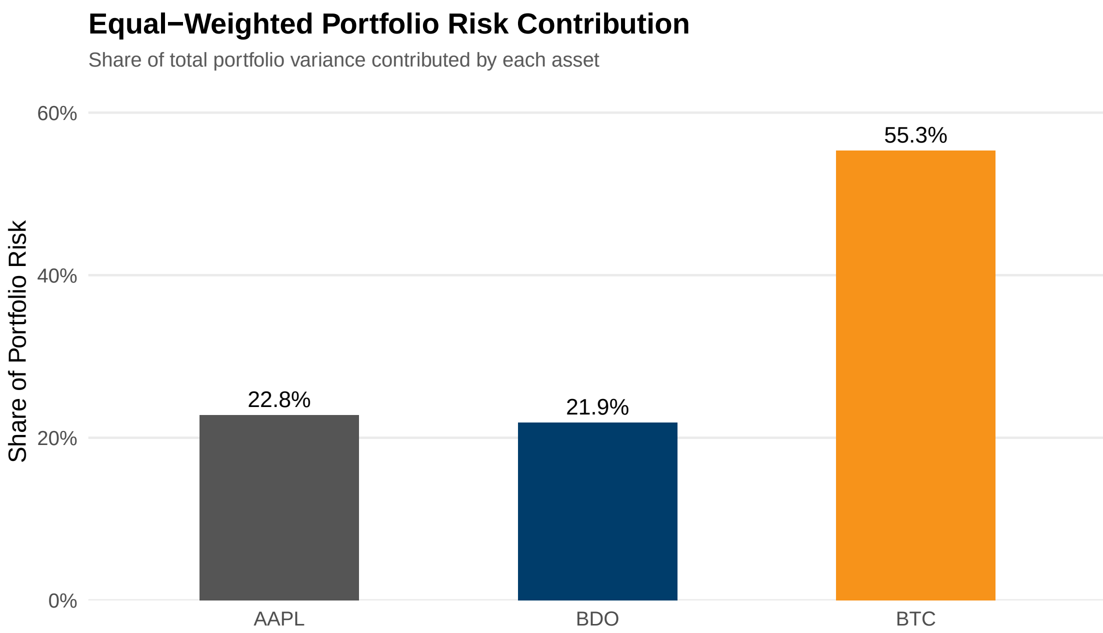

Figure 18: Equal-weighted portfolio risk contribution highlighting variance concentration.

```
# Sec 8: Data Visualization -- create 18 professional figures with ggplot2
↪  -- SEBALLOS, Josiah Dweyn Panganiban
```

8. Exploratory Financial Analysis

[Answer financial questions using evidence from SQL investigations, stats, and visualizations (compare returns, volatility, inflation, high-VIX, interest-rates).]

```
# Sec 9: Exploratory Financial Analysis -- compare returns, volatility,
↪  inflation, VIX, interest rates -- SEECHUNG, Camille Castro
```

9. Investment Recommendation

[Recommend allocation of USD 1,000,000 among U.S., PH, crypto, and cash. Justify using evidence.]

```
# Sec 10: Investment Recommendation -- USD 1,000,000 allocation among AAPL,
↪  BDO, BTC, cash -- SEECHUNG, Camille Castro
```

10. Executive Dashboard

Note: The Executive Dashboard content is generated and rendered externally via a dedicated Quarto/LaTeX file. It will be integrated into this section during the final compilation pipeline.

```
# Sec 11: Executive Dashboard -- key stats, charts, SQL findings,
↪  recommendation summary -- CRUZ, Ricardo Miguel Iñigo
```

11. Conclusion

[Summarize findings, limitations, and recommendations for future work.]

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A. Appendix A - Complete R Code

Appendix: Complete R Code for FINSTAD Case 1

CRUZ, Ricardo Miguel Iñigo GALEDO, Enrique Lorenzo Hermoso
SEBALLOS, Josiah Dweyn Panganiban SEECHUNG, Camille Castro
SISON, Aaron Joshua Estacio

Appendix A1 - Setup & Library Import

Loads the required R libraries (dplyr, tidyr, lubridate, ggplot2, etc.) and defines global helper functions used throughout the analysis. The working directory is resolved automatically to ensure data file paths work regardless of whether the script runs from the project root or the scripts/ folder.

```
# Load necessary libraries
library(dplyr)
library(tidyr)
library(lubridate)
library(DBI)
library(RSQLite)
library(ggplot2)
library(scales)
library(kableExtra)

# Safe table formatter for PDF to prevent overlapping columns and
 ↪ overflowing margins
safe_kable <- function(data, caption = NULL, digits = 4) {
  knitr::kable(data, caption = caption, digits = digits, format = "latex",
 ↪ booktabs = TRUE) %>%
    kable_styling(font_size = 8, latex_options = c("scale_down",
     ↪ "HOLD_position")) %>%
    column_spec(1:ncol(data), width_min = "1.5cm")
}

# Set working directory to project root so data paths resolve consistently
# Ensure data paths resolve correctly regardless of working directory
if (dir.exists("data")) {
  data_dir <- "data"
} else if (dir.exists("../data")) {
```

```r
  data_dir <- "../data"
} else {
  stop("Cannot find data/ directory. Run from project root or scripts/.")
}

# Global options
options(stringsAsFactors = FALSE, scipen = 999, width = 72)

# Helper to parse BDO volume strings (e.g. "2.50M" to 2500000)
parse_bdo_volume <- function(x) {
  x <- gsub(",", "", x)
  mult <- ifelse(grepl("M", x, ignore.case = TRUE), 1e6,
                 ifelse(grepl("K", x, ignore.case = TRUE), 1e3, 1))
  x_num <- as.numeric(gsub("[A-Za-z%]", "", x))
  x_num * mult
}
```

Appendix A2 - Ticker Definitions

Defines the asset ticker symbols used throughout the case: AAPL (US equity), BDO.PS (Philippine bank stock), and BTC-USD (cryptocurrency).

```r
# Define tickers
us_asset <- "AAPL"
ph_asset <- "BDO.PS"
crypto <- "BTC-USD"
```

Appendix A3 - Data Loading (GALEDO, Enrique Lorenzo Hermoso)

Loads all 8 CSV datasets from the data/ directory into R data frames. Records the original observation count for each dataset before any cleaning operations. Datasets include 4 asset price series (AAPL, BDO, BTC, SPY) and 4 FRED macroeconomic series (CPI, DGS10, FEDFUNDS, VIX).

```r
# Data Loading -- GALEDO, Enrique Lorenzo Hermoso (Sec 3: Data Acquisition)

# Load all 8 CSVs from the data directory
aapl <- read.csv(file.path(data_dir, "AAPL.csv"), stringsAsFactors = FALSE)
bdo  <- read.csv(file.path(data_dir, "BDO.csv"), stringsAsFactors = FALSE,
↪  fileEncoding = "UTF-8-BOM")
btc  <- read.csv(file.path(data_dir, "BTC.csv"), stringsAsFactors = FALSE)
spy  <- read.csv(file.path(data_dir, "SPY.csv"), stringsAsFactors = FALSE)
```

```r
cpi      <- read.csv(file.path(data_dir, "CPI.csv"), stringsAsFactors =
  ↪  FALSE)
dgs10    <- read.csv(file.path(data_dir, "DGS10.csv"), stringsAsFactors =
  ↪  FALSE)
fedfunds <- read.csv(file.path(data_dir, "FEDFUNDS.csv"), stringsAsFactors =
  ↪  FALSE)
vix      <- read.csv(file.path(data_dir, "VIX.csv"), stringsAsFactors =
  ↪  FALSE)

# Record original observations before any processing
obs_original <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
    ↪  "VIX"),
  Original_Obs = c(nrow(aapl), nrow(bdo), nrow(btc), nrow(spy),
                   nrow(cpi), nrow(dgs10), nrow(fedfunds), nrow(vix))
)

safe_kable(obs_original, caption="Original Observations per Dataset")
```

Table 1: Original Observations per Dataset

Dataset	Original_Obs
AAPL	2512
BDO	2441
BTC	3652
SPY	2514
CPI	952
DGS10	16105
FEDFUNDS	863
VIX	9216

Appendix A4 - Data Cleaning (SISON, Aaron Joshua Estacio)

Standardises date formats across all 8 datasets, filters observations to the 2020-01-01 to 2025-12-31 sample period, counts missing values and duplicates, and computes daily returns for each OHLC asset.

```r
# Sec 4: Data Cleaning -- SISON, Aaron Joshua Estacio

# Standardise date formats
# AAPL, BTC, SPY: POSIXct-style strings like "2016-06-30 00:00:00-04:00"
aapl <- aapl |> mutate(Date = as.Date(ymd_hms(Date)))
btc  <- btc  |> mutate(Date = as.Date(ymd_hms(Date)))
spy  <- spy  |> mutate(Date = as.Date(ymd_hms(Date)))
```

```r
# BDO: MM/DD/YYYY
bdo <- bdo |> mutate(Date = mdy(Date))

# CPI, DGS10, FEDFUNDS, VIX: YYYY-MM-DD
cpi      <- cpi      |> mutate(Date = ymd(Date))
dgs10    <- dgs10    |> mutate(Date = ymd(Date))
fedfunds <- fedfunds |> mutate(Date = ymd(Date))
vix      <- vix      |> mutate(Date = ymd(Date))

# Filter ALL datasets to 2020-01-01 to 2025-12-31
start_date <- as.Date("2020-01-01")
end_date   <- as.Date("2025-12-31")

aapl <- aapl |> filter(Date >= start_date & Date <= end_date)
bdo  <- bdo  |> filter(Date >= start_date & Date <= end_date)
btc  <- btc  |> filter(Date >= start_date & Date <= end_date)
spy  <- spy  |> filter(Date >= start_date & Date <= end_date)
cpi      <- cpi      |> filter(Date >= start_date & Date <= end_date)
dgs10    <- dgs10    |> filter(Date >= start_date & Date <= end_date)
fedfunds <- fedfunds |> filter(Date >= start_date & Date <= end_date)
vix      <- vix      |> filter(Date >= start_date & Date <= end_date)

# Observations after filtering
obs_filtered <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪  "VIX"),
  Filtered_Obs = c(nrow(aapl), nrow(bdo), nrow(btc), nrow(spy),
                   nrow(cpi), nrow(dgs10), nrow(fedfunds), nrow(vix))
)

# Missing values count
missing_summary <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪  "VIX"),
  Missing_Values = c(sum(is.na(aapl)), sum(is.na(bdo)), sum(is.na(btc)),
                     sum(is.na(spy)), sum(is.na(cpi)), sum(is.na(dgs10)),
                     sum(is.na(fedfunds)), sum(is.na(vix)))
)

# Duplicate rows count
dup_summary <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪  "VIX"),
  Duplicates = c(sum(duplicated(aapl)), sum(duplicated(bdo)),
   ↪  sum(duplicated(btc)),
```

```r
                 sum(duplicated(spy)), sum(duplicated(cpi)),
                  ↪  sum(duplicated(dgs10)),
                 sum(duplicated(fedfunds)), sum(duplicated(vix)))
)

# Clean BDO: parse Vol. column if it exists
if ("Vol." %in% colnames(bdo)) {
  bdo <- bdo |> mutate(`Vol.` = parse_bdo_volume(`Vol.`))
}

# Calculate daily returns for OHLC assets
# Daily_Return = (Close / lag(Close)) - 1
# For BDO the Price column is the close price
aapl <- aapl |> arrange(Date) |> mutate(Daily_Return = (Close / lag(Close))
 ↪  - 1)
bdo  <- bdo  |> arrange(Date) |> mutate(Daily_Return = (Price / lag(Price))
 ↪  - 1)
btc  <- btc  |> arrange(Date) |> mutate(Daily_Return = (Close / lag(Close))
 ↪  - 1)
spy  <- spy  |> arrange(Date) |> mutate(Daily_Return = (Close / lag(Close))
 ↪  - 1)

# Note: BDO's Change. column can be used to verify Daily_Return computation
# The first observation of each asset will have NA for Daily_Return (no
 ↪  prior day)

# Combine cleaning summaries into one display table
clean_summary <- obs_filtered |>
  left_join(missing_summary, by = "Dataset") |>
  left_join(dup_summary, by = "Dataset")
safe_kable(clean_summary, caption="Observations After Filtering, Missing
 ↪  Values, and Duplicates")
```

Table 2: Observations After Filtering, Missing Values, and Duplicates

Dataset	Filtered_Obs	Missing_Values	Duplicates
AAPL	1508	0	0
BDO	1468	0	0
BTC	2192	0	0
SPY	1508	0	0
CPI	71	0	0
DGS10	1500	0	0
FEDFUNDS	72	0	0
VIX	1535	0	0

FRED data (CPI, DGS10, FEDFUNDS) has fewer rows due to monthly or business-daily reporting frequency. Missing values introduced by join operations are handled in Section 5.

Appendix A5 - Database Integration (SISON, Aaron Joshua Estacio)

Demonstrates all four SQL join types (left, inner, full, anti) and justifies the chosen join strategy. Assembles the final analytical dataset by combining asset data with macroeconomic series. Exports the observation reduction table for inclusion in the main report.

```r
# Sec 5: Database Integration -- SISON, Aaron Joshua Estacio

# Prepare asset data for joining
# Standardise column names: BDO uses Price (close price) and Vol. (volume)
bdo_clean <- bdo |> rename(Close = Price, Volume = `Vol.`)

# Add Asset identifier to each OHLC dataset and select common columns
aapl_clean <- aapl |> mutate(Asset = "AAPL") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)
bdo_clean  <- bdo_clean |> mutate(Asset = "BDO") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)
btc_clean  <- btc |> mutate(Asset = "BTC") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)
spy_clean  <- spy |> mutate(Asset = "SPY") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)

# Combine all assets into one long-format table
assets_all <- bind_rows(aapl_clean, bdo_clean, btc_clean, spy_clean)

# Pivot to wide format: one row per date, columns per asset
assets_wide <- assets_all |>
  select(Date, Asset, Close, Daily_Return) |>
  pivot_wider(names_from = Asset,
              values_from = c(Close, Daily_Return),
              names_sep = "_")

# Prepare macro datasets with descriptive column names
cpi_clean      <- cpi      |> rename(CPI = Value)
dgs10_clean    <- dgs10    |> rename(DGS10 = Value)
fedfunds_clean <- fedfunds |> rename(FEDFUNDS = Value)
vix_clean      <- vix      |> rename(VIX = Value)




# Demonstration of all four join types
```

```r
# 1) left_join: keep all asset dates, attach macro data where available
left_example <- assets_wide |>
  left_join(cpi_clean, by = "Date") |>
  left_join(dgs10_clean, by = "Date") |>
  left_join(fedfunds_clean, by = "Date") |>
  left_join(vix_clean, by = "Date")

# 2) inner_join: only dates present in ALL datasets
inner_example <- assets_wide |>
  inner_join(cpi_clean, by = "Date") |>
  inner_join(dgs10_clean, by = "Date") |>
  inner_join(fedfunds_clean, by = "Date") |>
  inner_join(vix_clean, by = "Date")

# 3) full_join: all dates from any dataset
full_example <- assets_wide |>
  full_join(cpi_clean, by = "Date") |>
  full_join(dgs10_clean, by = "Date") |>
  full_join(fedfunds_clean, by = "Date") |>
  full_join(vix_clean, by = "Date")

# 4) anti_join: asset dates with NO macro data
anti_example <- assets_wide |>
  anti_join(cpi_clean, by = "Date")

# Build the final analytical dataset
final_dataset <- left_example



# Observation reduction table

reduction_table <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪  "VIX",
              "Assets (wide)", "Final Dataset"),
  Original_Obs = c(obs_original$Original_Obs, NA, NA),
  After_Cleaning = c(nrow(aapl), nrow(bdo), nrow(btc), nrow(spy),
                     nrow(cpi), nrow(dgs10), nrow(fedfunds), nrow(vix),
                     nrow(assets_wide), NA),
  After_Joins = c(rep(NA, 8), NA, nrow(final_dataset))
)

knitr::kable(reduction_table, caption="Observation Counts Across Processing
 ↪  Stages")
```

Table 3: Observation Counts Across Processing Stages

Dataset	Original_Obs	After_Cleaning	After_Joins
AAPL	2512	1508	NA
BDO	2441	1468	NA
BTC	3652	2192	NA
SPY	2514	1508	NA
CPI	952	71	NA
DGS10	16105	1500	NA
FEDFUNDS	863	72	NA
VIX	9216	1535	NA
Assets (wide)	NA	2192	NA
Final Dataset	NA	NA	2192

```
library(knitr)
cat(kable(reduction_table, format = "latex", booktabs = TRUE), file =
 ↪  "../reports/sections/data_cleaning_table.tex")
```

Appendix A6 - Master Dataset Creation & SQL Database Setup (SISON, Aaron Joshua Estacio)

Builds the master analytical dataset by restructuring into long format (one row per asset-date combination) and saves it as a persistent SQLite database file. Pre-calculates CPI year-on-year change for inflation analysis. All subsequent SQL questions (A7-A16) query this persisted database.

```
# SQLite database setup - creates persisted master dataset

db_path <- "../data/master_dataset.sqlite"
unlink(db_path)
con <- dbConnect(RSQLite::SQLite(), db_path)

# Build long-format analytical dataset for SQL queries
# Each row is one asset-date combination with macro variables attached
sql_long <- bind_rows(
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
           Close = Close_AAPL, Daily_Return = Daily_Return_AAPL) |>
    mutate(Asset = "AAPL"),
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
```

```r
          Close = Close_BDO, Daily_Return = Daily_Return_BDO) |>
    mutate(Asset = "BDO"),
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
           Close = Close_BTC, Daily_Return = Daily_Return_BTC) |>
    mutate(Asset = "BTC"),
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
           Close = Close_SPY, Daily_Return = Daily_Return_SPY) |>
    mutate(Asset = "SPY")
)

# Pre-calculate CPI YoY for Q10 (inflation analysis)
# CPI is monthly; YoY change = (current / 12-month-ago) - 1
cpi_yoy <- cpi_clean |>
  arrange(Date) |>
  mutate(CPI_YoY = (CPI / lag(CPI, 12)) - 1) |>
  select(Date, CPI_YoY)

# Join CPI_YoY into the long dataset and forward-fill to cover all trading
 ↪ days
sql_long <- sql_long |>
  left_join(cpi_yoy, by = "Date") |>
  arrange(Date) |>
  tidyr::fill(CPI_YoY, .direction = "down")


dbWriteTable(con, "integrated_data", sql_long, overwrite = TRUE)
fin_data <- tbl(con, "integrated_data")

# Show first 5 rows of the master dataset as a preview
nrow_total <- collect(fin_data) |> nrow()
safe_kable(head(sql_long, 5), digits=4, caption=paste0("Master Dataset
 ↪ Preview (First 5 of ", nrow_total, " Rows)"))
```

Table 4: Master Dataset Preview (First 5 of 8768 Rows)

Date	CPI	DGS10	FEDFUNDS	VIX	Close	Daily_Return	Asset	CPI_YoY
2020-01-01	259.127	NA	1.55	NA	NA	NA	AAPL	NA
2020-01-01	259.127	NA	1.55	NA	NA	NA	BDO	NA
2020-01-01	259.127	NA	1.55	NA	7200.17	NA	BTC	NA
2020-01-01	259.127	NA	1.55	NA	NA	NA	SPY	NA
2020-01-02	NA	1.88	NA	12.47	72.33	NA	AAPL	NA

Appendix A7 - SQL Question 1: Risk-Adjusted Daily Return (SISON, Aaron Joshua Estacio)

Computes the ratio of mean daily return to standard deviation for each asset, producing a Sharpe-like measure of return efficiency. The corresponding bar chart is displayed in the Data Visualization section.

```r
# Q1: Risk-Adjusted Daily Return by Asset -- SISON, Aaron Joshua Estacio

q1 <- fin_data |>
  group_by(Asset) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE)
  ) |>
  mutate(Risk_Adjusted_Return = Mean_Daily_Return / SD_Daily_Return) |>
  arrange(desc(Risk_Adjusted_Return)) |>
  collect()

spy_ratio <- q1 |> filter(Asset == "SPY") |> pull(Risk_Adjusted_Return)

safe_kable(q1, digits=4, caption="Q1 Data Output")
```

Table 5: Q1 Data Output

Asset	Mean_Daily_Return	SD_Daily_Return	Risk_Adjusted_Return
AAPL	0.0011	0.0200	0.0538
BTC	0.0017	0.0319	0.0520
SPY	0.0006	0.0131	0.0486
BDO	0.0003	0.0226	0.0131

Appendix A8 - SQL Question 2: COVID-19 Crash Cumulative Returns (SISON, Aaron Joshua Estacio)

Measures crisis resilience during the February to March 2020 COVID-19 market crash by computing cumulative returns for each asset.

```r
# Q2: COVID-19 Crash Cumulative Returns -- SISON, Aaron Joshua Estacio

q2_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q2 <- q2_raw |>
```

```r
  filter(Date >= as.Date("2020-02-01") & Date <= as.Date("2020-03-31")) |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1)

q2_summary <- q2 |>
  summarise(
    Start_Date = min(Date),
    End_Date = max(Date),
    Total_Cumulative_Return = last(Cumulative_Return),
    Observations = n()
  )

safe_kable(q2_summary, digits=4, caption="Q2 Summary Data Output")
```

Table 6: Q2 Summary Data Output

Asset	Start_Date	End_Date	Total_Cumulative_Return	Observations
AAPL	2020-02-03	2020-03-31	-0.1764	41
BDO	2020-02-03	2020-03-31	-0.3006	40
BTC	2020-02-01	2020-03-31	-0.3114	60
SPY	2020-02-03	2020-03-31	-0.1941	41

Appendix A9 - SQL Question 3: Pairwise Correlation of Daily Returns (SISON, Aaron Joshua Estacio)

Calculates the Pearson correlation coefficients for daily returns across all asset pairs and displays the correlation matrix.

```r
# Q3: Pairwise Correlation of Daily Returns -- SISON, Aaron Joshua Estacio

returns_wide <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  pivot_wider(names_from = Asset, values_from = Daily_Return)

cor_matrix <- returns_wide |>
  select(AAPL, BDO, BTC, SPY) |>
  cor(use = "pairwise.complete.obs")

cor_long <- as.data.frame(cor_matrix) |>
  tibble::rownames_to_column("Asset1") |>
  tidyr::pivot_longer(-Asset1, names_to = "Asset2", values_to =
  ↪ "Correlation") |>
```

```r
  mutate(Asset1 = factor(Asset1, levels = rev(c("AAPL", "BDO", "BTC",
  ↪  "SPY"))),
         Asset2 = factor(Asset2, levels = c("AAPL", "BDO", "BTC", "SPY")))

safe_kable(round(cor_matrix, 4), caption="Q3 Correlation Matrix Output")
```

Table 7: Q3 Correlation Matrix Output

	AAPL	BDO	BTC	SPY
AAPL	1.0000	0.1044	0.2836	0.7839
BDO	0.1044	1.0000	0.0507	0.1235
BTC	0.2836	0.0507	1.0000	0.3825
SPY	0.7839	0.1235	0.3825	1.0000

Appendix A10 - SQL Question 4: Asset Ranking by Total Return (CRUZ, Ricardo Miguel Iñigo)

Ranks assets by total price return and annualized return over the full 2020-2025 sample period.

```r
# Q4: Asset Ranking by Total Return -- CRUZ, Ricardo Miguel Inigo

q4_raw <- fin_data |>
  select(Date, Asset, Close, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q4 <- q4_raw |>
  filter(!is.na(Close), !is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  summarise(
    Start_Date = min(Date),
    End_Date = max(Date),
    Start_Close = first(Close),
    End_Close = last(Close),
    Total_Return = (End_Close / Start_Close) - 1,
    Annualized_Return = (1 + Total_Return)^(365.25 / as.numeric(End_Date -
    ↪  Start_Date)) - 1,
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
```

```
  arrange(desc(Total_Return))

q4_plot <- q4 |>
  mutate(Asset = factor(Asset, levels = q4$Asset[order(q4$Total_Return)]))

safe_kable(q4, digits=4, caption="Q4 Data Output")
```

Table 8: Q4 Data Output

Asset	Start_Date	End_Date	Start_Close	End_Close	Total_Return	Annualized_Return	Mean_Daily_Return	SD_Daily_Return	Observations
BTC	2020-01-02	2025-12-31	6985.47	87508.83	11.5273	0.5244	0.0017	0.0319	2191
AAPL	2020-01-03	2025-12-31	71.63	271.36	2.7884	0.2489	0.0011	0.0200	1507
SPY	2020-01-03	2025-12-31	293.88	678.32	1.3082	0.1498	0.0006	0.0131	1507
BDO	2020-01-03	2025-12-29	128.17	134.60	0.0502	0.0082	0.0003	0.0226	1467

Appendix A11 - SQL Question 5: Best and Worst Trading Days (CRUZ, Ricardo Miguel Iñigo)

Identifies extreme single-day gains and losses for each asset to assess tail risk.

```
# Q5: Best and Worst Trading Days -- CRUZ, Ricardo Miguel Inigo

q5_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q5 <- q5_raw |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  summarise(
    Best_Date = Date[which.max(Daily_Return)],
    Best_Daily_Return = max(Daily_Return, na.rm = TRUE),
    Worst_Date = Date[which.min(Daily_Return)],
    Worst_Daily_Return = min(Daily_Return, na.rm = TRUE),
    Return_Range = Best_Daily_Return - Worst_Daily_Return,
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Worst_Daily_Return)

safe_kable(q5, digits=4, caption="Q5 Data Output")
```

Table 9: Q5 Data Output

Asset	Best_Date	Best_Daily_Return	Worst_Date	Worst_Daily_Return	Return_Range	Observations
BTC	2021-02-08	0.1875	2020-03-12	-0.3717	0.5592	2191
BDO	2020-03-26	0.1570	2020-03-19	-0.2272	0.3842	1467
AAPL	2025-04-09	0.1533	2020-03-16	-0.1286	0.2819	1507
SPY	2025-04-09	0.1050	2020-03-16	-0.1094	0.2144	1507

Appendix A12 - SQL Question 6: Moving Average Crossover Analysis (CRUZ, Ricardo Miguel Iñigo)

Implements a 20-day and 60-day simple moving average crossover strategy to analyze momentum regime shifts.

```
# Q6: Moving Average Crossover Analysis -- CRUZ, Ricardo Miguel Inigo

q6_raw <- fin_data |>
  select(Date, Asset, Close, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q6_ma <- q6_raw |>
  filter(!is.na(Close)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(
    MA_20 = as.numeric(stats::filter(Close, rep(1 / 20, 20), sides = 1)),
    MA_60 = as.numeric(stats::filter(Close, rep(1 / 60, 60), sides = 1)),
    Signal = case_when(
      MA_20 > MA_60 ~ "Bullish",
      MA_20 < MA_60 ~ "Bearish",
      TRUE ~ "Neutral"
    )
  ) |>
  ungroup()

q6 <- q6_ma |>
  filter(!is.na(MA_20), !is.na(MA_60), !is.na(Daily_Return)) |>
  group_by(Asset, Signal) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
```

```
  arrange(Asset, desc(Mean_Daily_Return))

q6_latest_signal <- q6_ma |>
  filter(!is.na(MA_20), !is.na(MA_60)) |>
  group_by(Asset) |>
  slice_max(Date, n = 1, with_ties = FALSE) |>
  ungroup() |>
  select(Asset, Date, Close, MA_20, MA_60, Signal) |>
  arrange(Asset)

safe_kable(q6, digits=4, caption="Q6 Data Output")
```

Table 10: Q6 Data Output

Asset	Signal	Mean_Daily_Return	SD_Daily_Return	Observations
AAPL	Bullish	0.0015	0.0168	892
AAPL	Bearish	0.0008	0.0212	557
BDO	Bullish	0.0006	0.0195	667
BDO	Bearish	0.0003	0.0211	742
BTC	Bullish	0.0028	0.0300	1182
BTC	Bearish	0.0002	0.0343	951
SPY	Bearish	0.0013	0.0162	361
SPY	Bullish	0.0006	0.0092	1088

Appendix A13 - SQL Question 7: Volume and Volatility Relationship (CRUZ, Ricardo Miguel Iñigo)

Tests whether elevated trading volume systematically coincides with higher volatility by partitioning days into high-volume and normal-volume regimes.

```
# Q7: Volume and Volatility Relationship -- CRUZ, Ricardo Miguel Inigo

volume_lookup <- assets_all |>
  select(Date, Asset, Volume) |>
  mutate(
    Date = as.Date(Date),
    Volume = as.numeric(Volume)
  )

q7_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(
    Date = as.Date(Date),
    Daily_Return = as.numeric(Daily_Return)
```

```r
  ) |>
  left_join(volume_lookup, by = c("Date", "Asset")) |>
  filter(
    !is.na(Daily_Return),
    !is.na(Volume),
    Volume > 0
  )

q7 <- q7_raw |>
  group_by(Asset) |>
  mutate(
    Volume_Threshold = quantile(Volume, 0.75, na.rm = TRUE),
    Volume_Regime = if_else(
      Volume >= Volume_Threshold,
      "High Volume Days",
      "Normal Volume Days"
    ),
    Absolute_Return = abs(Daily_Return)
  ) |>
  ungroup() |>
  group_by(Asset, Volume_Regime) |>
  summarise(
    Observations = n(),
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    Mean_Absolute_Return = mean(Absolute_Return, na.rm = TRUE),
    Volatility = sd(Daily_Return, na.rm = TRUE),
    Average_Volume = mean(Volume, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(Asset, Volume_Regime)

safe_kable(q7, digits=4, caption="Q7 Data Output")
```

Table 11: Q7 Data Output

Asset	Volume_Regime	Observations	Mean_Daily_Return	Mean_Absolute_Return	Volatility	Average_Volume
AAPL	High Volume Days	377	0.0015	0.0240	0.0323	152332404
AAPL	Normal Volume Days	1130	0.0009	0.0105	0.0137	61873902
BDO	High Volume Days	367	-0.0001	0.0230	0.0322	6581172
BDO	Normal Volume Days	1100	0.0004	0.0136	0.0183	2551276
BTC	High Volume Days	548	0.0017	0.0305	0.0450	65153421094
BTC	Normal Volume Days	1643	0.0016	0.0183	0.0261	26930318305
SPY	High Volume Days	377	-0.0029	0.0160	0.0220	129389767
SPY	Normal Volume Days	1130	0.0018	0.0061	0.0078	63711998

Appendix A14 - SQL Question 8: VIX Regime Analysis (SISON, Aaron Joshua Estacio)

Assesses asset performance during periods of high versus low market fear by segregating trading days based on VIX levels.

```
# Q8: VIX Regime Analysis -- SISON, Aaron Joshua Estacio

q8_raw <- fin_data |>
  filter(!is.na(VIX)) |>
  select(Asset, Date, Daily_Return, VIX) |>
  collect() |>
  mutate(Date = as.Date(Date))

q8 <- q8_raw |>
  mutate(
    VIX_Regime = case_when(
      VIX > 30 ~ "High Volatility (VIX > 30)",
      VIX < 20 ~ "Low Volatility (VIX < 20)",
      TRUE ~ "Moderate (20 <= VIX <= 30)"
    )
  ) |>
  group_by(Asset, VIX_Regime) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, VIX_Regime)

q8_plot <- q8 |>
  filter(VIX_Regime %in% c("High Volatility (VIX > 30)", "Low Volatility
   ↪  (VIX < 20)")) |>
  mutate(VIX_Regime = recode(VIX_Regime,
    "High Volatility (VIX > 30)" = "VIX > 30 (High Fear)",
    "Low Volatility (VIX < 20)" = "VIX < 20 (Low Fear)"))

safe_kable(q8, digits=4, caption="Q8 Data Output")
```

Table 12: Q8 Data Output

Asset	VIX_Regime	Mean_Daily_Return	SD_Daily_Return	Observations
AAPL	High Volatility (VIX > 30)	-0.0054	0.0389	148
AAPL	Low Volatility (VIX < 20)	0.0025	0.0134	859
AAPL	Moderate (20 <= VIX <= 30)	0.0007	0.0208	528
BDO	High Volatility (VIX > 30)	-0.0042	0.0411	148
BDO	Low Volatility (VIX < 20)	0.0006	0.0185	859
BDO	Moderate (20 <= VIX <= 30)	0.0011	0.0213	528
BTC	High Volatility (VIX > 30)	0.0016	0.0550	148
BTC	Low Volatility (VIX < 20)	0.0022	0.0288	859
BTC	Moderate (20 <= VIX <= 30)	0.0020	0.0380	528
SPY	High Volatility (VIX > 30)	-0.0044	0.0303	148
SPY	Low Volatility (VIX < 20)	0.0018	0.0069	859
SPY	Moderate (20 <= VIX <= 30)	0.0002	0.0123	528

Appendix A15 - SQL Question 9: Interest Rate Sensitivity (SISON, Aaron Joshua Estacio)

Measures asset sensitivity to changes in the 10-year Treasury yield by comparing mean returns on days when yields rise versus fall.

```
# Q9: Interest Rate Sensitivity -- SISON, Aaron Joshua Estacio

q9_raw <- fin_data |>
  filter(!is.na(DGS10)) |>
  select(Asset, Date, Daily_Return, DGS10) |>
  collect() |>
  mutate(Date = as.Date(Date))

q9 <- q9_raw |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(DGS10_Change = DGS10 - lag(DGS10)) |>
  ungroup() |>
  filter(!is.na(DGS10_Change)) |>
  mutate(
    Yield_Direction = case_when(
      DGS10_Change > 0 ~ "Yields Rising",
      DGS10_Change < 0 ~ "Yields Falling",
      TRUE ~ "No Change"
    )
  ) |>
  group_by(Asset, Yield_Direction) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
```

```
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, Yield_Direction)

q9_plot <- q9 |>
  filter(Yield_Direction %in% c("Yields Rising", "Yields Falling"))

safe_kable(q9, digits=4, caption="Q9 Data Output")
```

Table 13: Q9 Data Output

Asset	Yield_Direction	Mean_Daily_Return	SD_Daily_Return	Observations
AAPL	No Change	0.0019	0.0195	103
AAPL	Yields Falling	0.0005	0.0200	684
AAPL	Yields Rising	0.0013	0.0201	712
BDO	No Change	-0.0020	0.0208	103
BDO	Yields Falling	0.0000	0.0253	684
BDO	Yields Rising	0.0007	0.0201	712
BTC	No Change	0.0028	0.0372	103
BTC	Yields Falling	0.0017	0.0343	684
BTC	Yields Rising	0.0020	0.0363	712
SPY	No Change	-0.0001	0.0101	103
SPY	Yields Falling	-0.0001	0.0130	684
SPY	Yields Rising	0.0014	0.0135	712

Appendix A16 - SQL Question 10: Inflation Hedge Performance (SISON, Aaron Joshua Estacio)

Evaluates which assets served as effective inflation hedges during the June 2021 to February 2023 high-inflation period (CPI YoY above 5%).

```
# Q10: Inflation Hedge Performance -- SISON, Aaron Joshua Estacio

q10_raw <- fin_data |>
  select(Date, Asset, Daily_Return, CPI_YoY) |>
  collect() |>
  mutate(Date = as.Date(Date))

q10 <- q10_raw |>
  filter(!is.na(CPI_YoY) & CPI_YoY > 0.05) |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1) |>
  summarise(
```

```
    Start_Date = min(Date),
    End_Date = max(Date),
    Start_CPI_YoY = first(CPI_YoY),
    End_CPI_YoY = last(CPI_YoY),
    Total_Cumulative_Return = last(Cumulative_Return),
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    Observations = n()
  ) |>
  arrange(desc(Total_Cumulative_Return))

q10_plot <- q10 |>
  mutate(Asset = factor(Asset, levels =
   ↪  q10$Asset[order(q10$Total_Cumulative_Return)]))

safe_kable(q10, digits=4, caption="Q10 Data Output")
```

Table 14: Q10 Data Output

Asset	Start_Date	End_Date	Start_CPI_YoY	End_CPI_YoY	Total_Cumulative_Return	Mean_Daily_Return	Observations
BDO	2021-06-01	2023-02-28	0.053	0.0596	0.4391	0.0010	435
AAPL	2021-06-01	2023-02-28	0.053	0.0596	0.1952	0.0006	440
SPY	2021-06-01	2023-02-28	0.053	0.0596	-0.0317	0.0000	440
BTC	2021-06-01	2023-02-28	0.053	0.0596	-0.3800	-0.0002	638

Appendix A17 - Data Visualization Charts (SEBALLOS, Josiah Dweyn Panganiban)

Contains the ggplot code for all 10 figures corresponding to SQL Questions Q1 through Q10. Each figure is saved as a PDF to the reports/figures/ directory for inclusion in the main report's Data Visualization section.

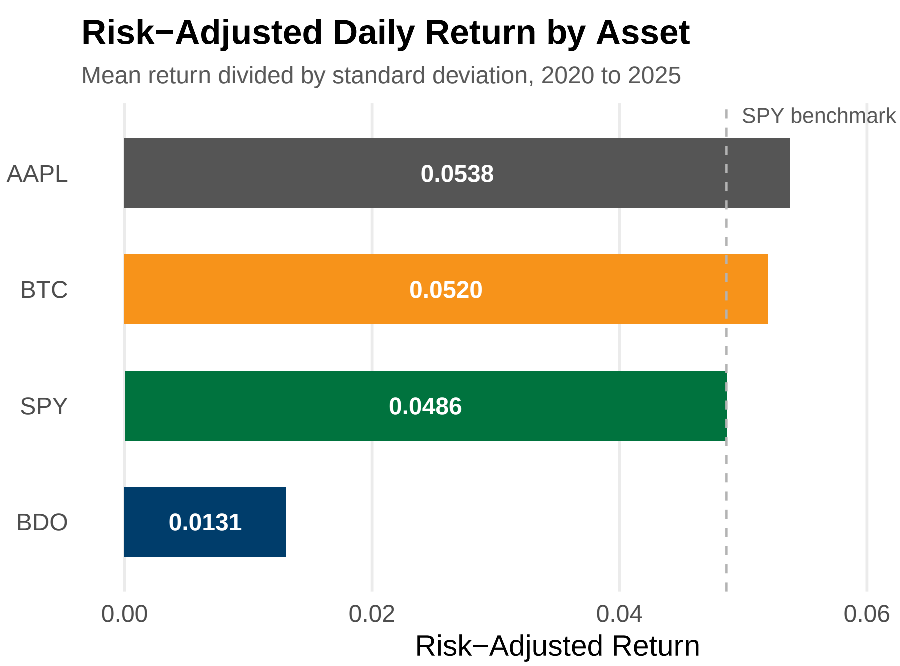
```
# Create figures directory if it does not exist
dir.create("../reports/figures", recursive = TRUE, showWarnings = FALSE)
```

Figure 1: Risk-Adjusted Daily Return by Asset

Bar chart showing the ratio of mean daily return to standard deviation for each asset. The dashed reference line marks SPY's ratio.

```
ggplot(q1, aes(x = Risk_Adjusted_Return, y = reorder(Asset,
 ↪  Risk_Adjusted_Return),
                fill = Asset)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_vline(xintercept = spy_ratio, linetype = "dashed",
             color = "#B3B3B3", linewidth = 0.4) +
```

```r
  geom_text(aes(x = Risk_Adjusted_Return / 2,
                label = sprintf("%.4f", Risk_Adjusted_Return)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  annotate("text", x = spy_ratio, y = 4.5, label = "SPY benchmark",
           size = 2.8, color = "#595959", hjust = -0.1) +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  labs(title = "Risk-Adjusted Daily Return by Asset",
       subtitle = "Mean return divided by standard deviation, 2020 to 2025",
       x = "Risk-Adjusted Return", y = NULL) +
  xlim(0, 0.07) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

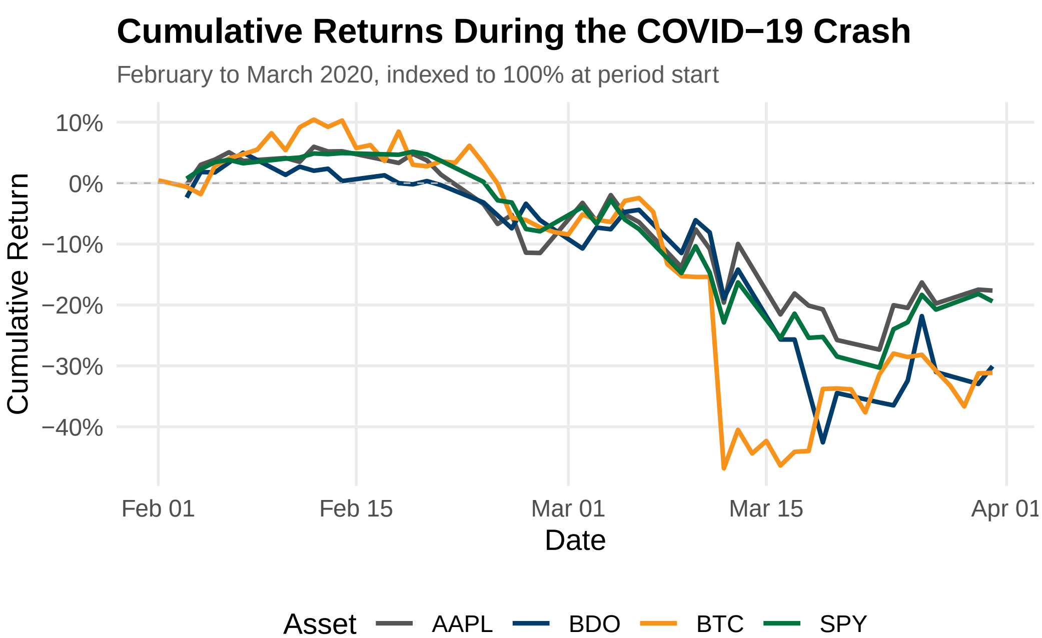



```r
ggsave("../reports/figures/fig_q1.pdf", width = 7, height = 3.5, device =
  "pdf")
```

Figure 2: Cumulative Returns During the COVID-19 Crash

Line chart tracking each asset's cumulative return from February to March 2020, indexed to 100% at period start.

```r
ggplot(q2, aes(x = Date, y = Cumulative_Return, color = Asset)) +
  geom_line(linewidth = 0.8) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "#B3B3B3",
  ↪  linewidth = 0.3) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Cumulative Returns During the COVID-19 Crash",
       subtitle = "February to March 2020, indexed to 100% at period start",
       x = "Date", y = "Cumulative Return", color = "Asset") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```



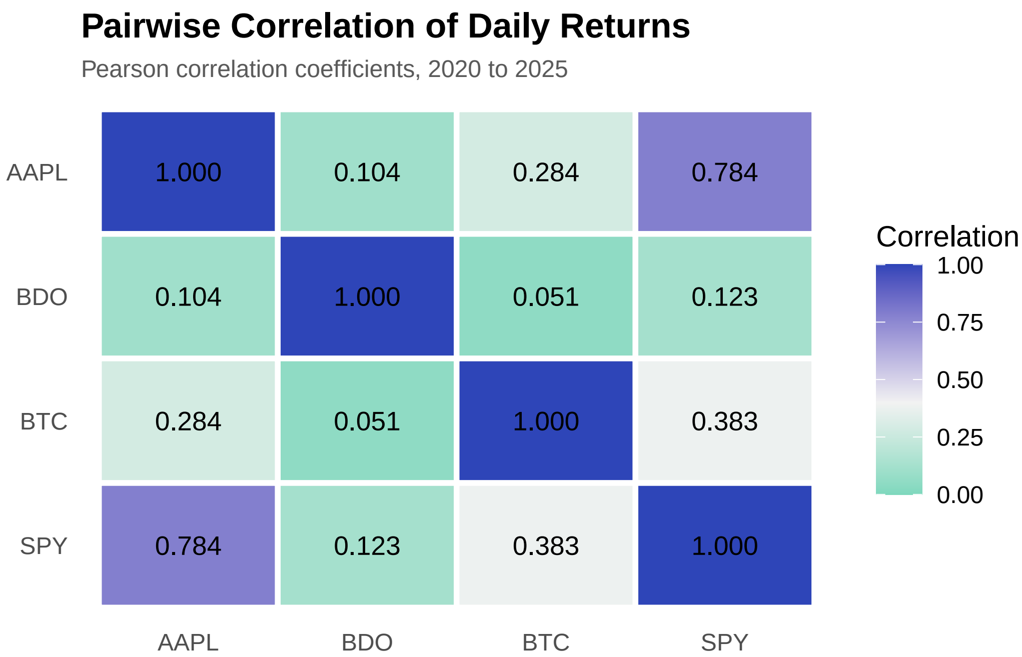

```r
ggsave("../reports/figures/fig_q2.pdf", width = 7, height = 4, device =
↪  "pdf")
```

Figure 3: Pairwise Correlation Heatmap

Heatmap matrix of Pearson correlation coefficients for daily returns across the four assets.

```r
ggplot(cor_long, aes(x = Asset2, y = Asset1, fill = Correlation)) +
  geom_tile(color = "white", linewidth = 1) +
  geom_text(aes(label = sprintf("%.3f", Correlation)), size = 3.5) +
  scale_fill_gradient2(low = "#1DC9A4", mid = "#F2F2F2", high = "#2E45B8",
```

```r
                        midpoint = 0.4, limits = c(0, 1)) +
  labs(title = "Pairwise Correlation of Daily Returns",
       subtitle = "Pearson correlation coefficients, 2020 to 2025",
       x = NULL, y = NULL, fill = "Correlation") +
  theme_minimal(base_size = 11) +
  theme(panel.grid = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "right")
```



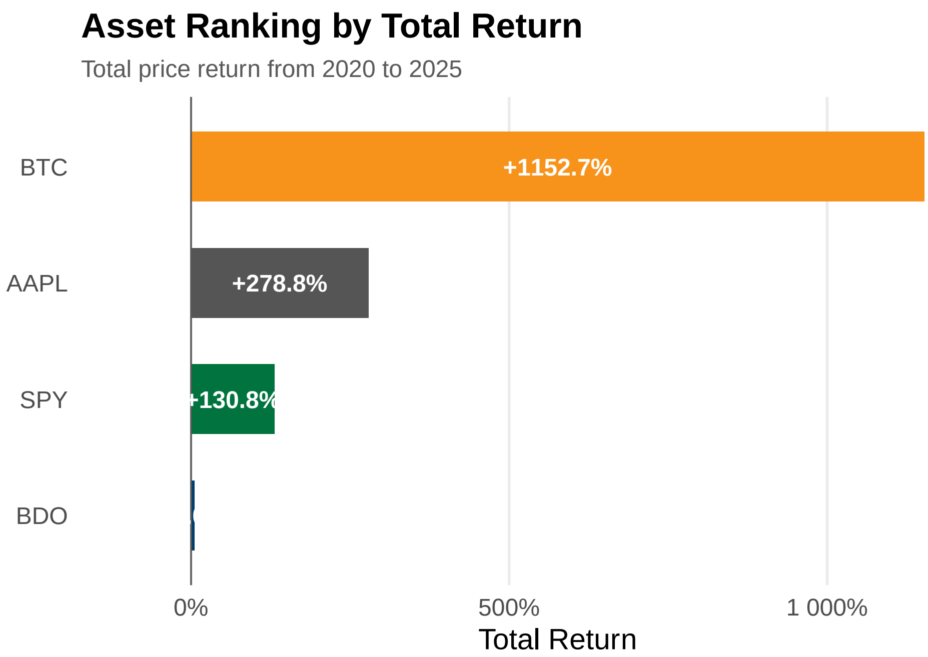

```r
ggsave("../reports/figures/fig_q3.pdf", width = 6, height = 4.5, device =
  ↪  "pdf")
```

Figure 4: Asset Ranking by Total Return

Horizontal bar chart ranking assets by total percentage price return from 2020 to 2025.

```r
ggplot(q4_plot, aes(x = Total_Return, y = Asset, fill = Asset)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_vline(xintercept = 0, linewidth = 0.3, color = "#595959") +
  geom_text(aes(x = Total_Return / 2,
                label = sprintf("%+.1f%%", Total_Return * 100)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format(),
```

```
                    expand = expansion(mult = c(0.15, 0.15))) +
  labs(title = "Asset Ranking by Total Return",
       subtitle = "Total price return from 2020 to 2025",
       x = "Total Return", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```



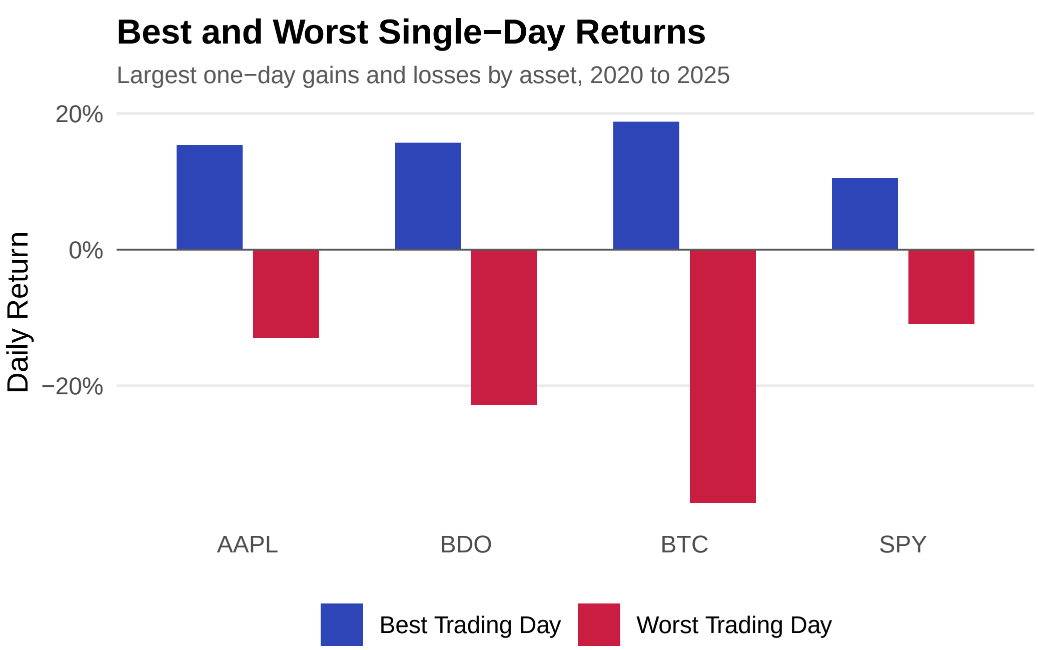

```
ggsave("../reports/figures/fig_q4.pdf", width = 7, height = 3.5, device =
  "pdf")
```

Figure 5: Best and Worst Single-Day Returns

Grouped bar chart comparing the largest single-day gain and loss for each asset.

```
q5_plot <- q5 |>
  select(Asset, Best_Daily_Return, Worst_Daily_Return) |>
  pivot_longer(cols = c(Best_Daily_Return, Worst_Daily_Return),
               names_to = "Trading_Day_Type",
               values_to = "Daily_Return") |>
  mutate(
    Trading_Day_Type = recode(Trading_Day_Type,
                              Best_Daily_Return = "Best Trading Day",
                              Worst_Daily_Return = "Worst Trading Day")
```

```r
  )

ggplot(q5_plot, aes(x = Asset, y = Daily_Return, fill = Trading_Day_Type)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("Best Trading Day" = "#2E45B8",
                               "Worst Trading Day" = "#C91D42")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Best and Worst Single-Day Returns",
       subtitle = "Largest one-day gains and losses by asset, 2020 to 2025",
       x = NULL, y = "Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

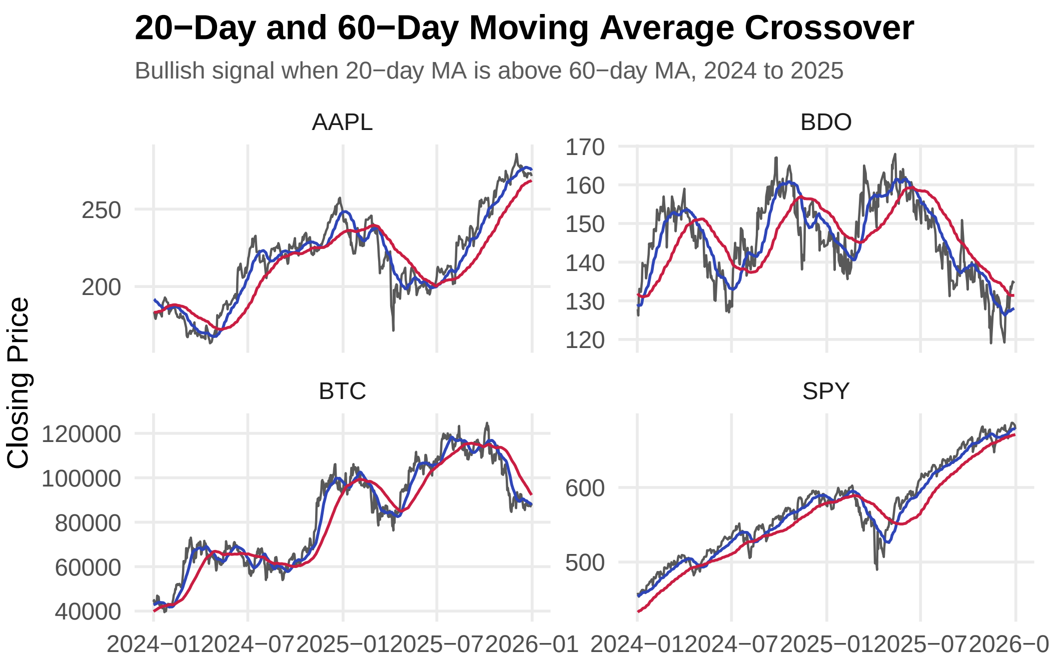



```r
ggsave("../reports/figures/fig_q5.pdf", width = 7, height = 4, device =
  "pdf")
```

Figure 6: 20-Day and 60-Day Moving Average Crossover

Multi-panel faceted line chart showing closing prices with 20-day and 60-day moving averages from 2024 to 2025.

```r
q6_plot <- q6_ma |>
  filter(Date >= as.Date("2024-01-01"))

ggplot(q6_plot, aes(x = Date)) +
  geom_line(aes(y = Close), linewidth = 0.4, color = "#595959") +
  geom_line(aes(y = MA_20), linewidth = 0.5, color = "#2E45B8") +
  geom_line(aes(y = MA_60), linewidth = 0.5, color = "#C91D42") +
  facet_wrap(~ Asset, scales = "free_y") +
  labs(title = "20-Day and 60-Day Moving Average Crossover",
       subtitle = "Bullish signal when 20-day MA is above 60-day MA, 2024 to
        ↪ 2025",
       x = NULL, y = "Closing Price") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```

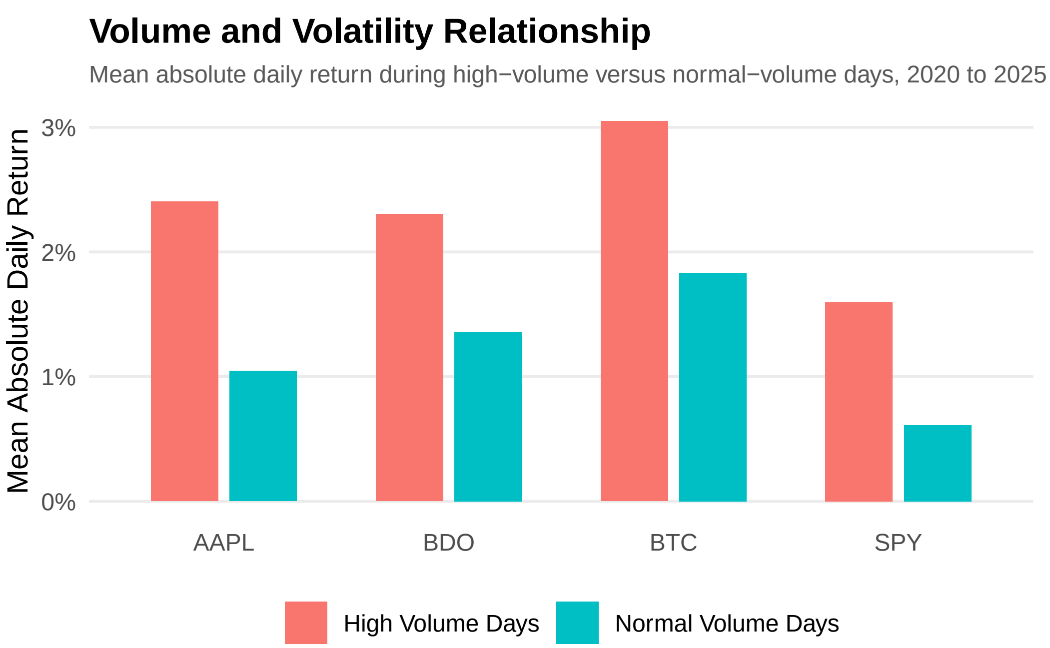



```r
ggsave("../reports/figures/fig_q6.pdf", width = 7, height = 4.5, device =
  ↪ "pdf")
```

Figure 7: Volume and Volatility Relationship

Grouped bar chart comparing mean absolute daily return during high-volume versus normal-volume days.

```r
ggplot(q7, aes(x = Asset, y = Mean_Absolute_Return, fill = Volume_Regime)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Volume and Volatility Relationship",
    subtitle = "Mean absolute daily return during high-volume versus
     ↪  normal-volume days, 2020 to 2025",
    x = NULL,
    y = "Mean Absolute Daily Return",
    fill = NULL
  ) +
  theme_minimal(base_size = 11) +
  theme(
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    axis.ticks = element_blank(),
    plot.title = element_text(face = "bold", size = 13),
    plot.subtitle = element_text(size = 9, color = "#595959"),
    legend.position = "bottom"
  )
```

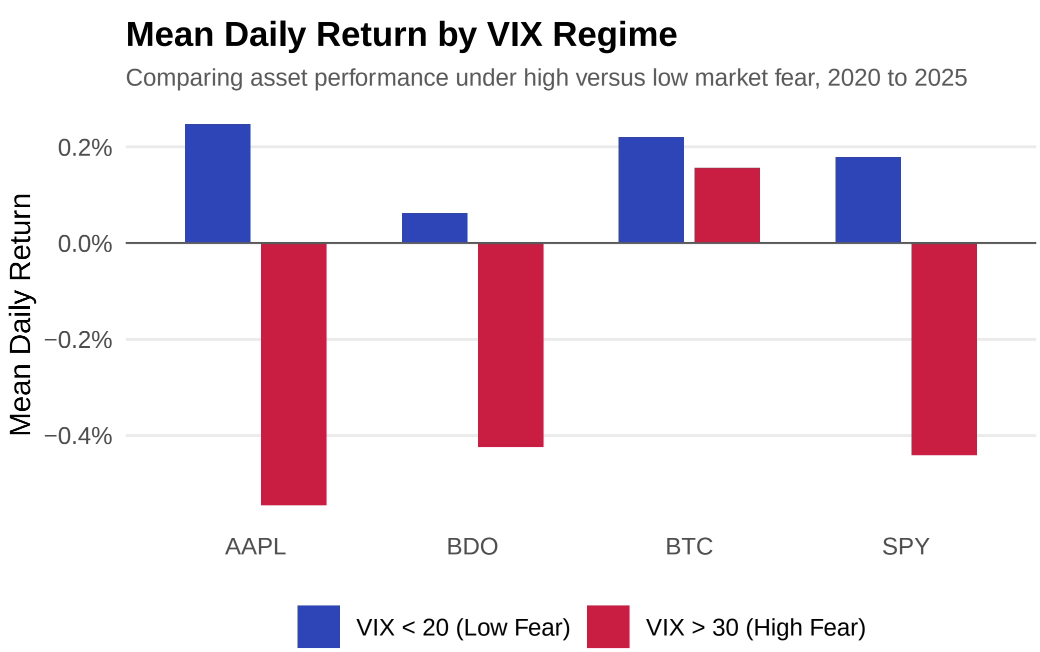



```r
ggsave("../reports/figures/fig_q7.pdf", width = 7, height = 4, device =
 ↪  "pdf")
```

Figure 8: Mean Daily Return by VIX Regime

Grouped bar chart comparing asset returns during high-fear (VIX > 30) versus low-fear (VIX < 20) regimes.

```
ggplot(q8_plot, aes(x = Asset, y = Mean_Daily_Return, fill = VIX_Regime)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("VIX > 30 (High Fear)" = "#C91D42",
                               "VIX < 20 (Low Fear)" = "#2E45B8")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Mean Daily Return by VIX Regime",
       subtitle = "Comparing asset performance under high versus low market
        ↪ fear, 2020 to 2025",
       x = NULL, y = "Mean Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

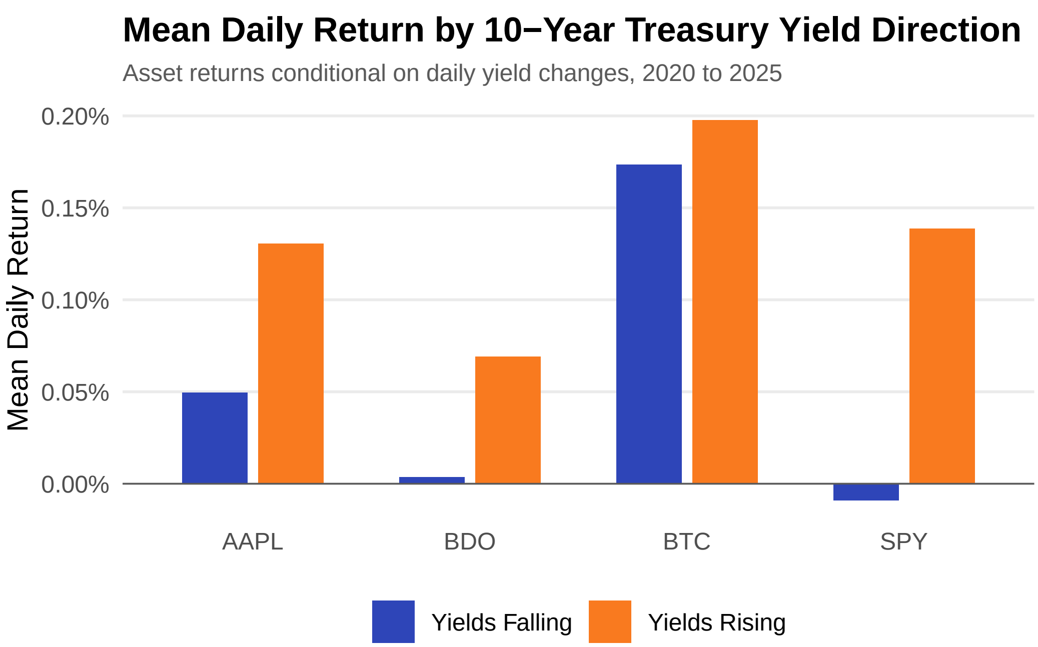



```
ggsave("../reports/figures/fig_q8.pdf", width = 7, height = 4, device =
  ↪ "pdf")
```

Figure 9: Mean Daily Return by 10-Year Treasury Yield Direction

Grouped bar chart showing asset returns conditional on whether daily yields are rising or falling.

```
ggplot(q9_plot, aes(x = Asset, y = Mean_Daily_Return, fill =
  Yield_Direction)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("Yields Rising" = "#F97A1F",
                               "Yields Falling" = "#2E45B8")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Mean Daily Return by 10-Year Treasury Yield Direction",
       subtitle = "Asset returns conditional on daily yield changes, 2020 to
         2025",
       x = NULL, y = "Mean Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```

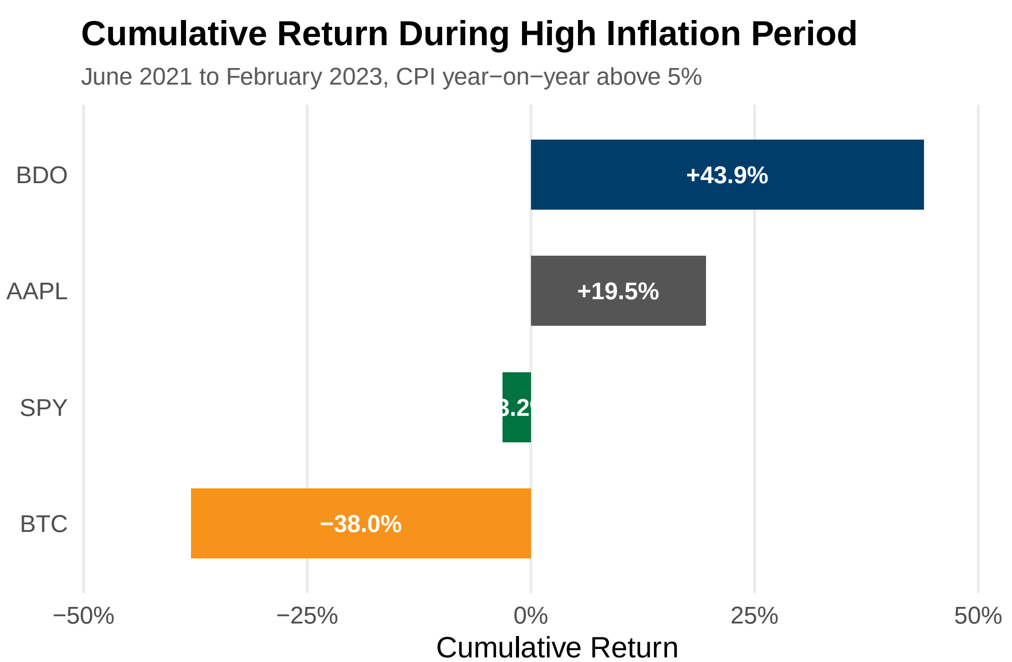



```
ggsave("../reports/figures/fig_q9.pdf", width = 7, height = 4, device =
  "pdf")
```

Figure 10: Cumulative Return During High Inflation Period

Horizontal bar chart showing total cumulative returns during the period when CPI year-on-year exceeded 5%.

```r
ggplot(q10_plot, aes(x = Total_Cumulative_Return, y = Asset, fill = Asset))
↪  +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(x = Total_Cumulative_Return / 2,
                label = sprintf("%+.1f%%", Total_Cumulative_Return * 100)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format(),
                     expand = expansion(mult = c(0.15, 0.15))) +
  labs(title = "Cumulative Return During High Inflation Period",
       subtitle = "June 2021 to February 2023, CPI year-on-year above 5%",
       x = "Cumulative Return", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```r
ggsave("../reports/figures/fig_q10.pdf", width = 7, height = 3.5, device =
↪  "pdf")
```

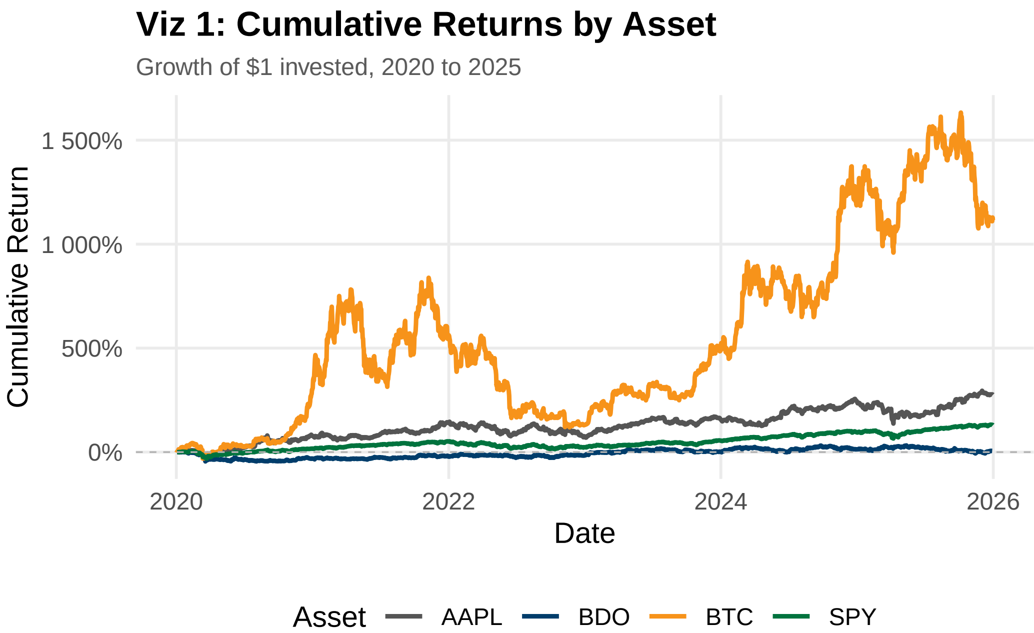
Appendix A18 - Exploratory Data Visualization (SEBALLOS, Josiah Dweyn Panganiban)

Contains 8 exploratory figures covering cumulative returns, price trends, return distributions, rolling volatility, drawdown analysis, risk-return profiles, and portfolio risk contribution. These complement the SQL-driven charts in A17.

Viz Figure 1: Cumulative Returns by Asset

Tracks the growth of USD 1 invested in each asset from 2020 to 2025, indexed to 100% at the start.

```
fig1_data <- sql_long |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1) |>
  ungroup()

ggplot(fig1_data, aes(x = Date, y = Cumulative_Return, color = Asset)) +
  geom_line(linewidth = 0.8) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "#B3B3B3",
   ↪  linewidth = 0.3) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Viz 1: Cumulative Returns by Asset",
       subtitle = "Growth of $1 invested, 2020 to 2025",
       x = "Date", y = "Cumulative Return", color = "Asset") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```



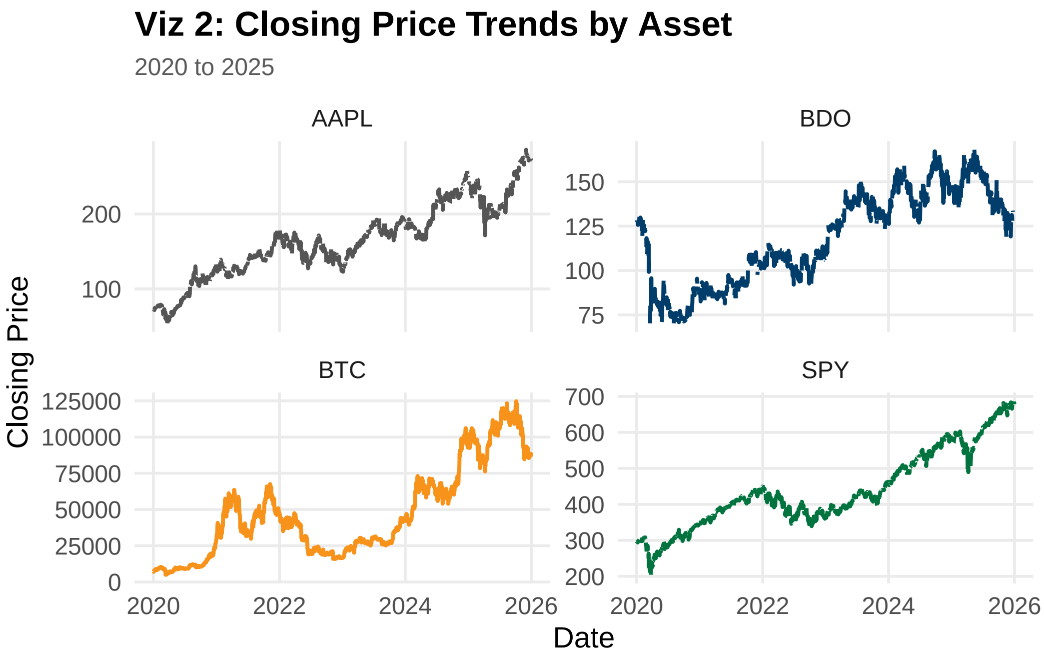

```
ggsave("../reports/fig_viz1.pdf", width = 7, height = 4, device = "pdf")
```

Viz 1: Cumulative Returns by Asset tracks the growth of USD 1 invested in each asset from 2020 to 2025. BTC shows the steepest and most volatile climb, reaching cumulative returns above 1,000% with sharp cyclical peaks and drawdowns. AAPL and SPY both trend steadily upward, with AAPL consistently outperforming the SPY benchmark. BDO remains close to the 0% line for most of the period, indicating limited net nominal gain relative to the other three assets.

Viz Figure 2: Closing Price Trends by Asset

Plots nominal closing prices on individual scales to compare price-level dynamics across assets of different magnitudes.

```
ggplot(sql_long, aes(x = Date, y = Close, color = Asset)) +
  geom_line(linewidth = 0.7, show.legend = FALSE, na.rm = TRUE) +
  facet_wrap(~ Asset, scales = "free_y", ncol = 2) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  labs(title = "Viz 2: Closing Price Trends by Asset",
       subtitle = "2020 to 2025",
       x = "Date", y = "Closing Price") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```r
ggsave("../reports/fig_viz2.pdf", width = 7, height = 5, device = "pdf")
```

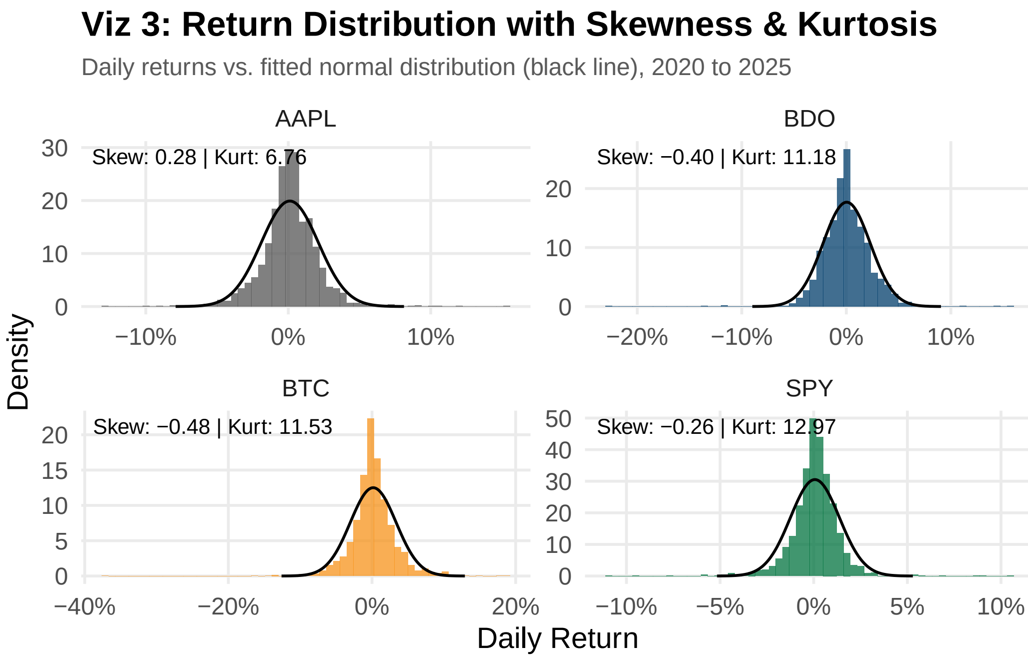
Viz 2: Closing Price Trends by Asset plots nominal closing prices rather than normalized returns, revealing that BDO experienced meaningful price swings on its own scale despite appearing flat in Viz 1. Each asset uses its own y-axis since raw prices are not directly comparable across magnitudes.

Viz Figure 3: Return Distribution with Skewness and Kurtosis

Overlays each asset's daily return histogram against a fitted normal curve to assess distribution shape and tail risk.

```r
fig3_data <- sql_long |> filter(!is.na(Daily_Return))

calc_skew <- function(x) {
  n <- length(x); m <- mean(x)
  (sum((x - m)^3) / n) / (sd(x)^3)
}
calc_kurt <- function(x) {
  n <- length(x); m <- mean(x)
  (sum((x - m)^4) / n) / (sd(x)^2)^2 - 3
}

fig3_stats <- fig3_data |>
  group_by(Asset) |>
  summarise(
    Mean = mean(Daily_Return), SD = sd(Daily_Return),
    Skewness = calc_skew(Daily_Return), Kurtosis = calc_kurt(Daily_Return),
```

```
    .groups = "drop"
  ) |>
  mutate(label = sprintf("Skew: %.2f | Kurt: %.2f", Skewness, Kurtosis))

normal_curves <- fig3_stats |>
  rowwise() |>
  reframe(
    Asset = Asset,
    x = seq(Mean - 4 * SD, Mean + 4 * SD, length.out = 200),
    y = dnorm(x, mean = Mean, sd = SD)
  )

ggplot(fig3_data, aes(x = Daily_Return)) +
  geom_histogram(aes(y = after_stat(density), fill = Asset), bins = 60,
                alpha = 0.75, show.legend = FALSE) +
  geom_line(data = normal_curves, aes(x = x, y = y), color = "black",
   ↪  linewidth = 0.5) +
  geom_text(data = fig3_stats, aes(x = -Inf, y = Inf, label = label),
          hjust = -0.05, vjust = 1.5, size = 2.8) +
  facet_wrap(~ Asset, scales = "free", ncol = 2) +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                              BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format()) +
  labs(title = "Viz 3: Return Distribution with Skewness & Kurtosis",
      subtitle = "Daily returns vs. fitted normal distribution (black
       ↪  line), 2020 to 2025",
      x = "Daily Return", y = "Density") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```r
ggsave("../reports/fig_viz3.pdf", width = 7, height = 5, device = "pdf")
```

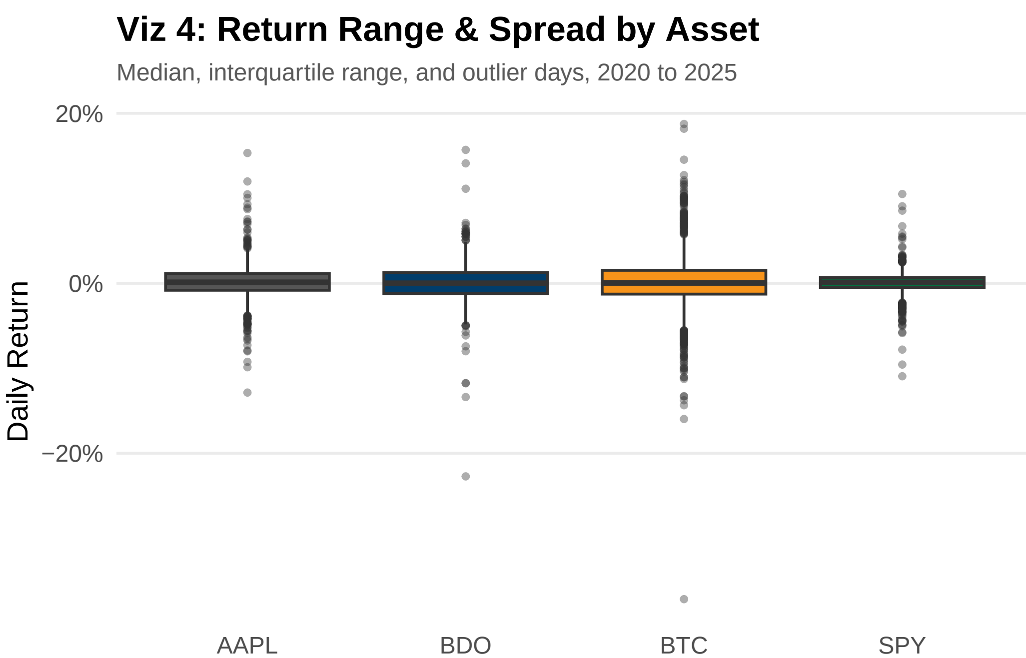
Viz 3: Return Distribution with Skewness and Kurtosis overlays each asset's daily return distribution against a fitted normal curve. AAPL is the only asset with positive skewness (0.28), meaning its rare extreme days lean toward the upside. BTC and BDO both show negative skewness (-0.48 and -0.40 respectively), indicating tail risk weighted toward sudden losses. SPY records the highest kurtosis of all four assets (12.97) despite a comparatively mild skew (-0.26), meaning its returns are usually tightly clustered near zero but still capable of extreme shock days. All four kurtosis values exceed 3, the benchmark for a normal distribution, confirming none of these assets follow a standard bell curve.

Viz Figure 4: Return Range and Spread by Asset

Boxplot showing median, interquartile range, and outlier days for each asset.

```r
ggplot(fig3_data, aes(x = Asset, y = Daily_Return, fill = Asset)) +
  geom_boxplot(show.legend = FALSE, outlier.size = 0.8, outlier.alpha = 0.4)
   ↪ +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Viz 4: Return Range & Spread by Asset",
       subtitle = "Median, interquartile range, and outlier days, 2020 to
        ↪ 2025",
       x = NULL, y = "Daily Return") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
```

```
      axis.ticks = element_blank(),
      plot.title = element_text(face = "bold", size = 13),
      plot.subtitle = element_text(size = 9, color = "#595959"))
```




```
ggsave("../reports/fig_viz4.pdf", width = 7, height = 4, device = "pdf")
```

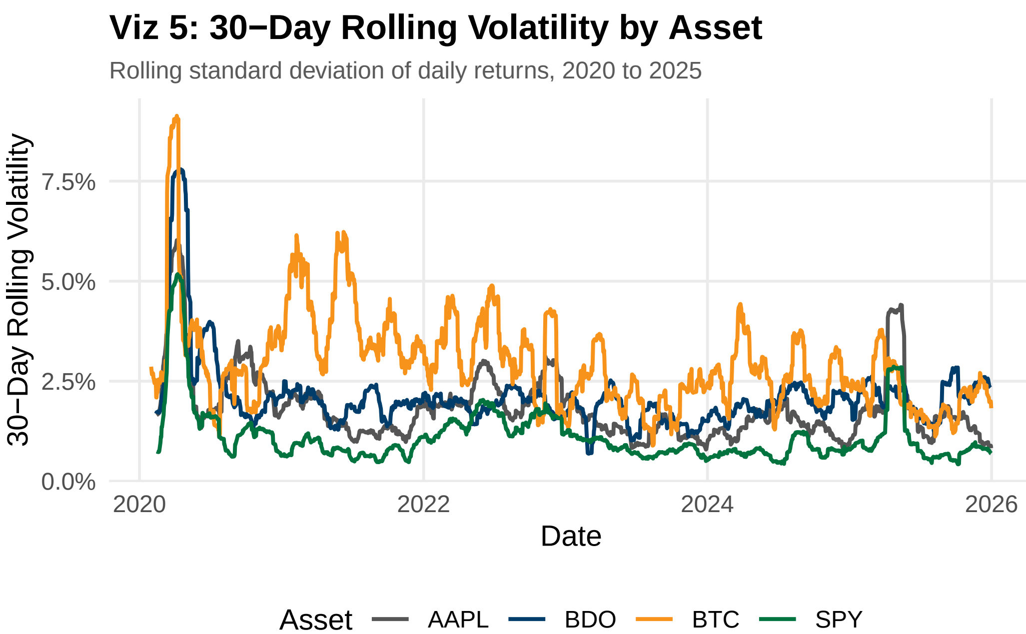
Viz 4: Return Range and Spread by Asset shows each asset's typical daily return range and outlier days. BTC has the widest box and most extreme outliers in both directions, consistent with its high kurtosis in Viz 3. SPY has the narrowest, most compressed box, reflecting diversification across 500 companies. AAPL and BDO fall in between, with BDO showing a visible skew toward downside outlier days.

Viz Figure 5: 30-Day Rolling Volatility by Asset

Tracks rolling 30-day standard deviation of daily returns to visualize volatility evolution over time.

```
rolling_sd <- function(x, n = 30) {
  sapply(seq_along(x), function(i) {
    if (i < n) return(NA)
    sd(x[(i - n + 1):i], na.rm = TRUE)
  })
}

fig5_data <- sql_long |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
```

```r
  mutate(Rolling_Vol = rolling_sd(Daily_Return, 30)) |>
  ungroup() |>
  filter(!is.na(Rolling_Vol))

ggplot(fig5_data, aes(x = Date, y = Rolling_Vol, color = Asset)) +
  geom_line(linewidth = 0.7) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Viz 5: 30-Day Rolling Volatility by Asset",
       subtitle = "Rolling standard deviation of daily returns, 2020 to
        ↪  2025",
       x = "Date", y = "30-Day Rolling Volatility", color = "Asset") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```



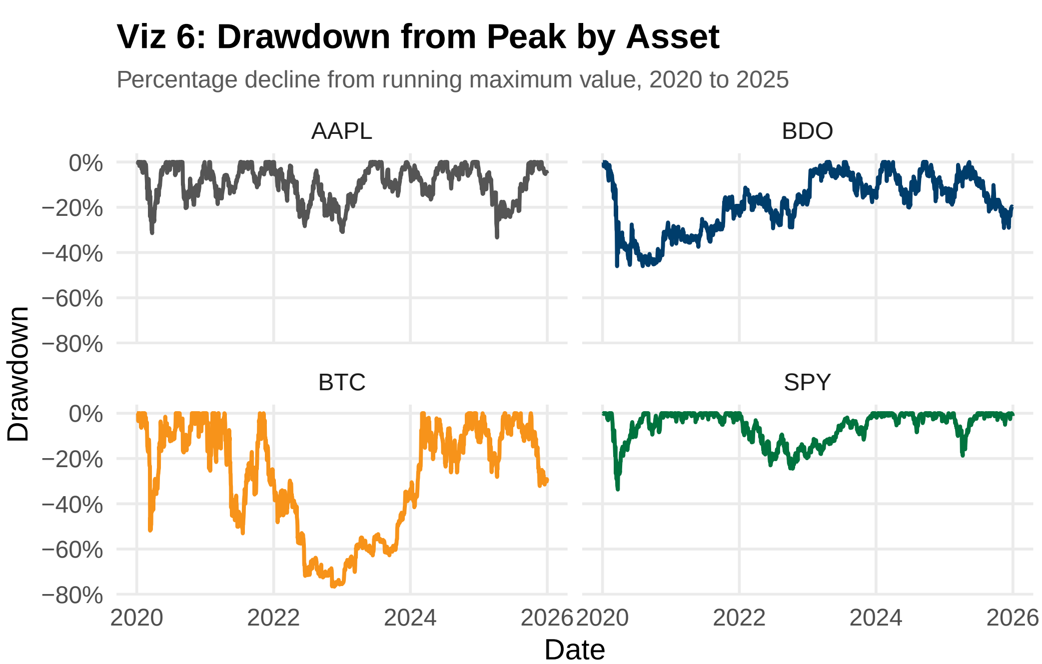

```r
ggsave("../reports/fig_viz5.pdf", width = 7, height = 4, device = "pdf")
```

Viz 5: 30-Day Rolling Volatility by Asset shows a sharp synchronized volatility spike across all assets in early 2020 from the COVID-19 market shock. BTC consistently maintains the highest rolling volatility baseline with visible cyclical waves from its bull and bear cycles. SPY maintains the lowest and most stable volatility floor. AAPL and BDO show occasional idiosyncratic spikes tied to specific events rather than broad macro shocks.

Viz Figure 6: Drawdown from Peak by Asset

Computes percentage decline from each asset's running historical high to visualize recovery patterns.

```
fig6_data <- fig1_data |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Running_Max = cummax(1 + Cumulative_Return),
         Drawdown = (1 + Cumulative_Return) / Running_Max - 1) |>
  ungroup()

ggplot(fig6_data, aes(x = Date, y = Drawdown, color = Asset)) +
  geom_line(linewidth = 0.7) +
  facet_wrap(~ Asset, ncol = 2) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Viz 6: Drawdown from Peak by Asset",
       subtitle = "Percentage decline from running maximum value, 2020 to
        ↪  2025",
       x = "Date", y = "Drawdown") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        legend.position = "none",
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```



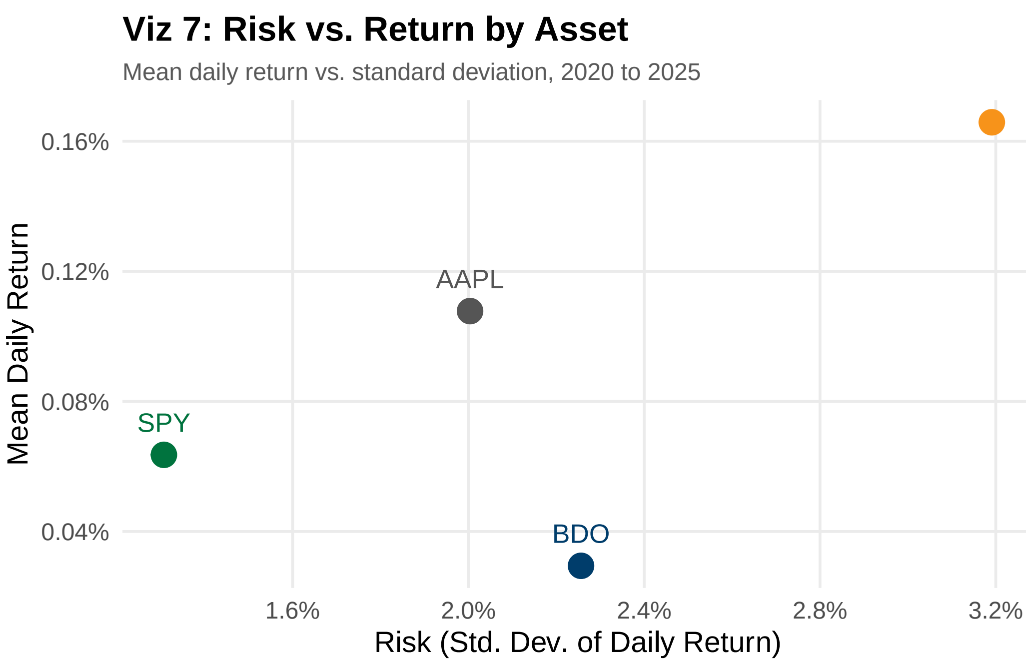
```
ggsave("../reports/fig_viz6.pdf", width = 7, height = 5, device = "pdf")
```

Viz 6: Drawdown from Peak by Asset shows each asset's percentage decline from its running historical high. BTC shows the deepest drawdowns falling well below -60% and takes longest to recover to a prior peak. SPY shows the shallowest fastest-recovering drawdowns consistent with its diversified benchmark role. BDO shows a prolonged period in negative territory indicating slower capital recovery through a downturn.

Viz Figure 7: Risk vs. Return by Asset

Scatter plot comparing mean daily return against standard deviation across all four assets.

```
fig7_data <- sql_long |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  summarise(Mean_Return = mean(Daily_Return), SD_Return = sd(Daily_Return),
   ↪  .groups = "drop")

ggplot(fig7_data, aes(x = SD_Return, y = Mean_Return, color = Asset)) +
  geom_point(size = 4) +
  geom_text(aes(label = Asset), vjust = -1.2, size = 3.5, show.legend =
   ↪  FALSE) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format()) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Viz 7: Risk vs. Return by Asset",
       subtitle = "Mean daily return vs. standard deviation, 2020 to 2025",
       x = "Risk (Std. Dev. of Daily Return)", y = "Mean Daily Return") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        legend.position = "none",
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```
ggsave("../reports/fig_viz7.pdf", width = 7, height = 5, device = "pdf")
```

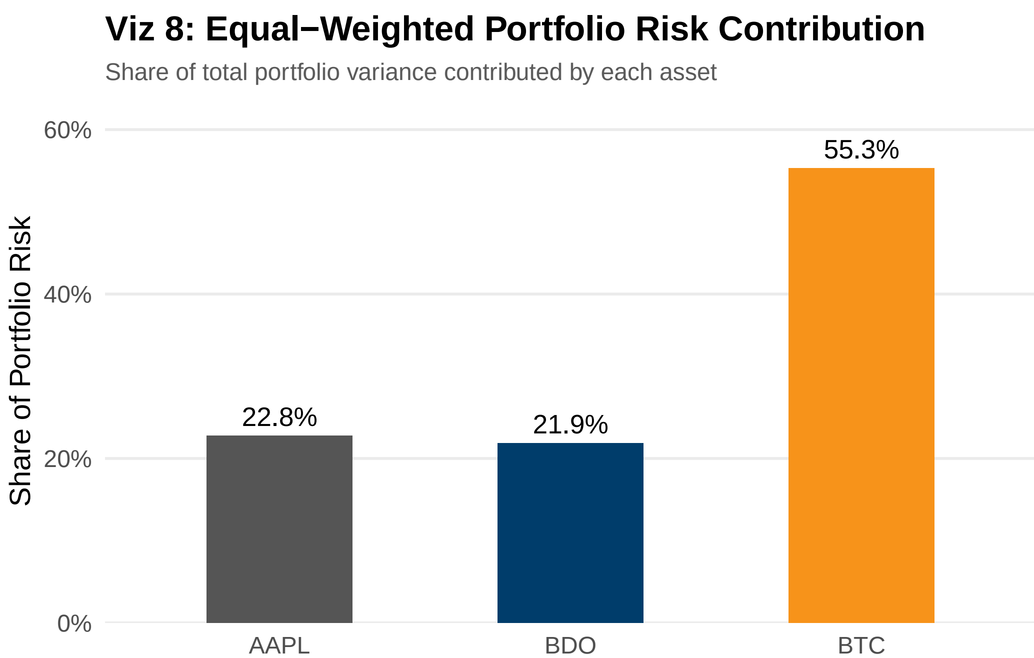
Viz 7: Risk vs. Return by Asset plots each asset's mean daily return against standard deviation. BTC delivers the highest mean return (0.166%) but also highest risk (3.19% daily SD). AAPL offers a more efficient middle ground with 0.108% mean return at only 2.00% volatility. SPY sits at the lowest-risk end (1.31% SD) with a modest 0.0636% mean return. BDO is the weakest performer on a risk-adjusted basis, carrying similar risk to AAPL (2.26% SD) while delivering the lowest mean return (0.0295%).

Viz Figure 8: Equal-Weighted Portfolio Risk Contribution

Decomposes total portfolio variance into each asset's contribution to show that equal dollar weights do not produce equal risk.

```
returns_matrix <- sql_long |>
  filter(Asset %in% c("AAPL", "BDO", "BTC"), !is.na(Daily_Return)) |>
  select(Date, Asset, Daily_Return) |>
  pivot_wider(names_from = Asset, values_from = Daily_Return) |>
  select(AAPL, BDO, BTC) |>
  na.omit()

cov_matrix <- cov(returns_matrix)
weights <- c(AAPL = 1/3, BDO = 1/3, BTC = 1/3)
port_var <- as.numeric(t(weights) %*% cov_matrix %*% weights)
marginal_contrib <- cov_matrix %*% weights
risk_contrib <- as.numeric(weights * marginal_contrib / port_var)

fig8_data <- data.frame(Asset = names(weights), Risk_Contribution =
↪  risk_contrib)
```

```r
ggplot(fig8_data, aes(x = Asset, y = Risk_Contribution, fill = Asset)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = scales::percent(Risk_Contribution, accuracy = 0.1)),
            vjust = -0.5, size = 3.5) +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B", BTC =
   ↪  "#F7931A")) +
  scale_y_continuous(labels = scales::percent_format(), expand =
   ↪  expansion(mult = c(0, 0.15))) +
  labs(title = "Viz 8: Equal-Weighted Portfolio Risk Contribution",
       subtitle = "Share of total portfolio variance contributed by each
        ↪  asset",
       x = NULL, y = "Share of Portfolio Risk") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```r
ggsave("../reports/fig_viz8.pdf", width = 7, height = 4, device = "pdf")
```

Viz 8: Equal-Weighted Portfolio Risk Contribution decomposes total portfolio variance under an equal one-third allocation. BTC alone contributes 55.3% of total portfolio variance despite representing only one-third of invested capital. AAPL and BDO contribute 22.8% and 21.9% respectively, roughly proportional to their capital weights. Equal-dollar allocation does not produce equal-risk allocation, supporting a reduced BTC weight to lower portfolio risk without proportional loss in expected return.

Sec 7: Descriptive Statistics – GALEDO, Enrique Lorenzo Hermoso

Mean, median, variance, std dev, min, max, range, skewness, kurtosis.

```
# Sec 7: Descriptive Statistics -- compute mean, median, variance, skewness,
↪  kurtosis -- GALEDO, Enrique Lorenzo Hermoso
```

Sec 8: Data Visualization – SEBALLOS, Josiah Dweyn Panganiban

At least 8 professional-quality figures.

```
# Sec 8: Data Visualization -- create 8 professional figures with ggplot2 --
↪  SEBALLOS, Josiah Dweyn Panganiban
```

Sec 9: Exploratory Analysis – SEECHUNG, Camille Castro

Compare returns, volatility, inflationary periods, high-VIX periods, interest-rate environments.

```
# Sec 9: Exploratory Financial Analysis -- compare returns, volatility,
↪  inflation, VIX, interest rates -- SEECHUNG, Camille Castro
```

Sec 10: Investment Recommendation – SEECHUNG, Camille Castro

Allocate USD 1,000,000.

```
# Sec 10: Investment Recommendation -- USD 1,000,000 allocation among AAPL,
↪  BDO, BTC, cash -- SEECHUNG, Camille Castro
```

Sec 11: Executive Dashboard – CRUZ, Ricardo Miguel Iñigo

```
# Sec 11: Executive Dashboard -- key stats, charts, SQL findings,
↪  recommendation summary -- CRUZ, Ricardo Miguel Iñigo
```

B. Appendix B - Additional Tables and Figures

The following tables and figures present the full sequence of data extraction, cleaning, integration, and quantitative analysis (Q1-Q10) in chronological order.

Table 15: Data Acquisition Summary

Dataset	Source	Original Observations	Missing Values	Duplicates
AAPL	Yahoo Finance	2512	0	0
BDO	Investing.com	2441	0	0
BTC	Yahoo Finance	3652	0	0
SPY	Yahoo Finance	2514	0	0
CPI	FRED	952	0	0
DGS10	FRED	16105	0	0
FEDFUNDS	FRED	863	0	0
VIX	FRED	9216	0	0

Table 16: Data Acquisition and Cleaning Summary

Dataset	Original_Obs	After_Cleaning	After_Joins
AAPL	2512	1508	NA
BDO	2441	1468	NA
BTC	3652	2192	NA
SPY	2514	1508	NA
CPI	952	71	NA
DGS10	16105	1500	NA
FEDFUNDS	863	72	NA
VIX	9216	1535	NA
Assets (wide)	NA	2192	NA
Final Dataset	NA	NA	2192

Table 17: Join Comparison

Join Type	Rows	Description
left_join	2192	Preserves all asset dates; macro NA on non-trading days
inner_join	46	Only dates common to all eight datasets
full_join	2192	Union including macro-only periods
anti_join	2121	Asset dates missing CPI, reflecting weekend and holiday gaps

Table 18: Observation Reduction Across Processing Stages

Dataset	Original	After Cleaning	Final
AAPL	2512	1508	—
BDO	2441	1468	—
BTC	3652	2192	—
SPY	2514	1508	—
CPI	952	71	—
DGS10	16 105	1500	—
FEDFUNDS	863	72	—
VIX	9216	1535	—
Final Dataset	—	—	2192

Table 19: Q1 Data Output

Asset	Mean_Daily_Return	SD_Daily_Return	Risk_Adjusted_Return
AAPL	0.0011	0.0200	0.0538
BTC	0.0017	0.0319	0.0520
SPY	0.0006	0.0131	0.0486
BDO	0.0003	0.0226	0.0131

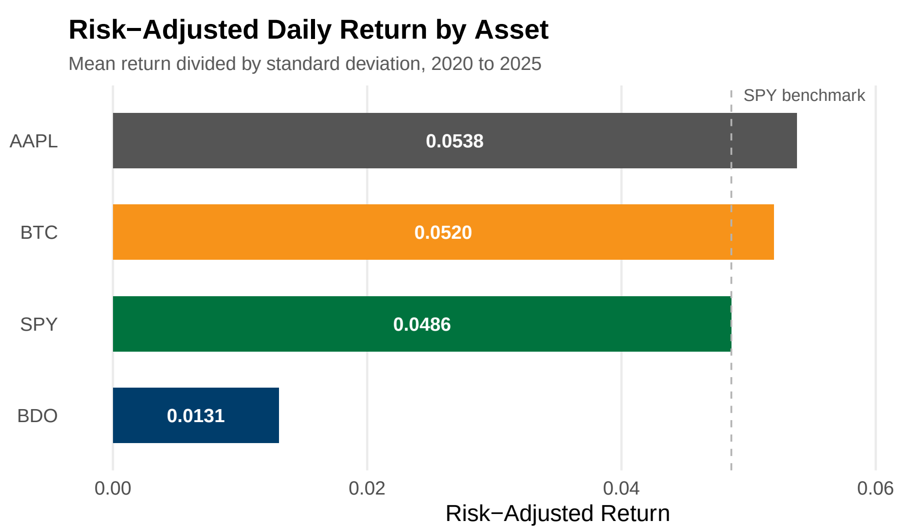


Figure 19: Risk-adjusted daily return by asset. Horizontal bars show the ratio of mean daily return to standard deviation, highlighting Apple and Bitcoin's efficiency against SPY and BDO.

Table 20: Q2 Summary Data Output

Asset	Start_Date	End_Date	Total_Cumulative_Return	Observations
AAPL	2020-02-03	2020-03-31	-0.1764	41
BDO	2020-02-03	2020-03-31	-0.3006	40
BTC	2020-02-01	2020-03-31	-0.3114	60
SPY	2020-02-03	2020-03-31	-0.1941	41

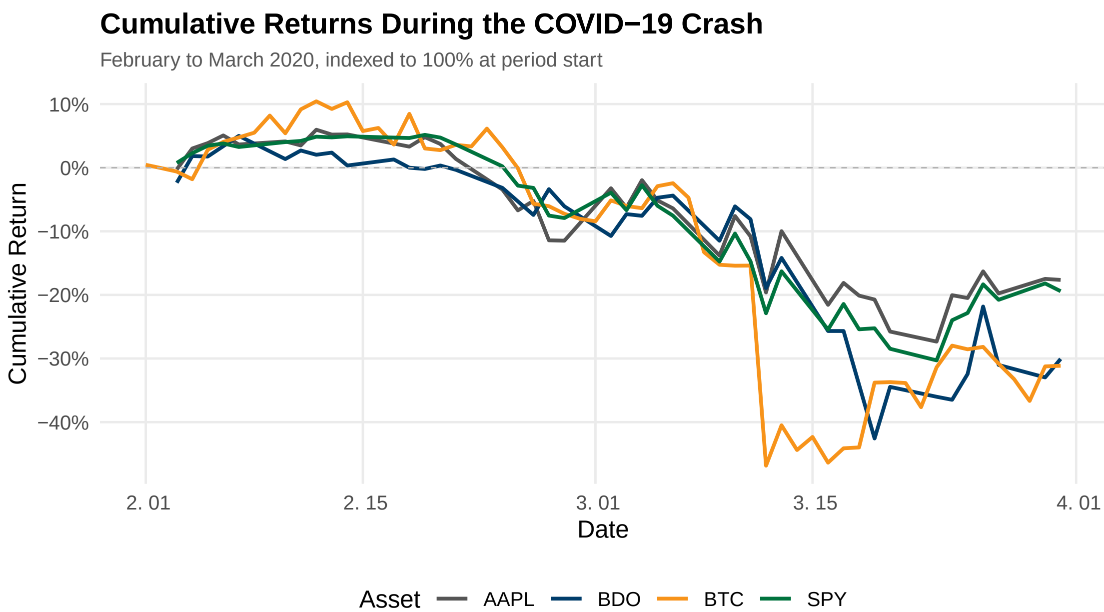


Figure 20: Cumulative returns during the COVID-19 crash, February to March 2020. Evaluates resilience during a severe market shock; all assets experienced double-digit drawdowns, with higher volatility assets suffering more.

Table 21: Q3 Correlation Matrix Output

	AAPL	BDO	BTC	SPY
AAPL	1.0000	0.1044	0.2836	0.7839
BDO	0.1044	1.0000	0.0507	0.1235
BTC	0.2836	0.0507	1.0000	0.3825
SPY	0.7839	0.1235	0.3825	1.0000

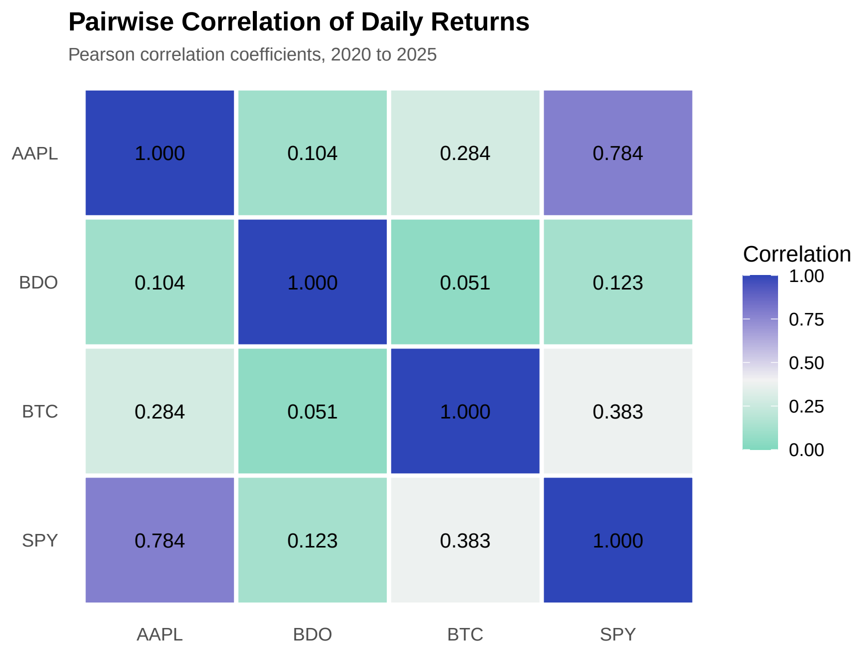


Figure 21: Pairwise Pearson correlation coefficients of daily returns. Values closer to 1 indicate stronger positive co-movement. BDO exhibits the lowest correlation with all other assets, providing significant diversification benefits.

Table 22: Q4 Data Output

Asset	Start_Date	End_Date	Start_Close	End_Close	Total_Return	Annualized_Return	Mean_Daily_Return	SD_Daily_Return	Obs
BTC	2020-01-02	2025-12-31	6985.47	87508.83	11.5273	0.5244	0.0017	0.0319	2191
AAPL	2020-01-03	2025-12-31	71.63	271.36	2.7884	0.2489	0.0011	0.0200	1507
SPY	2020-01-03	2025-12-31	293.88	678.32	1.3082	0.1498	0.0006	0.0131	1507
BDO	2020-01-03	2025-12-29	128.17	134.60	0.0502	0.0082	0.0003	0.0226	1467

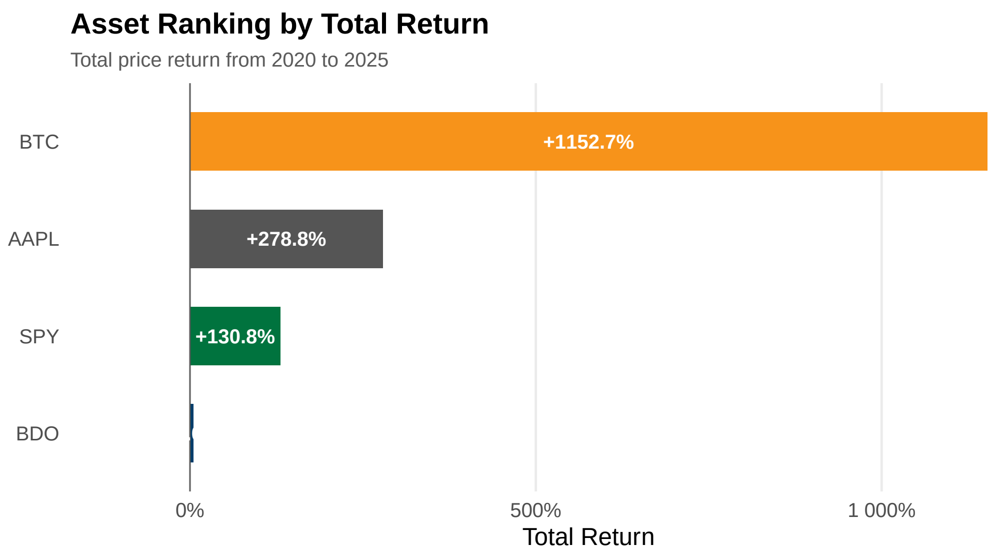


Figure 22: Asset ranking by total price return from 2020 to 2025. Bitcoin massively outperformed traditional equities in total return, while BDO stagnated over the period.

Table 23: Q5 Data Output

Asset	Best_Date	Best_Daily_Return	Worst_Date	Worst_Daily_Return	Return_Range	Obs
BTC	2021-02-08	0.1875	2020-03-12	-0.3717	0.5592	2191
BDO	2020-03-26	0.1570	2020-03-19	-0.2272	0.3842	1467
AAPL	2025-04-09	0.1533	2020-03-16	-0.1286	0.2819	1507
SPY	2025-04-09	0.1050	2020-03-16	-0.1094	0.2144	1507

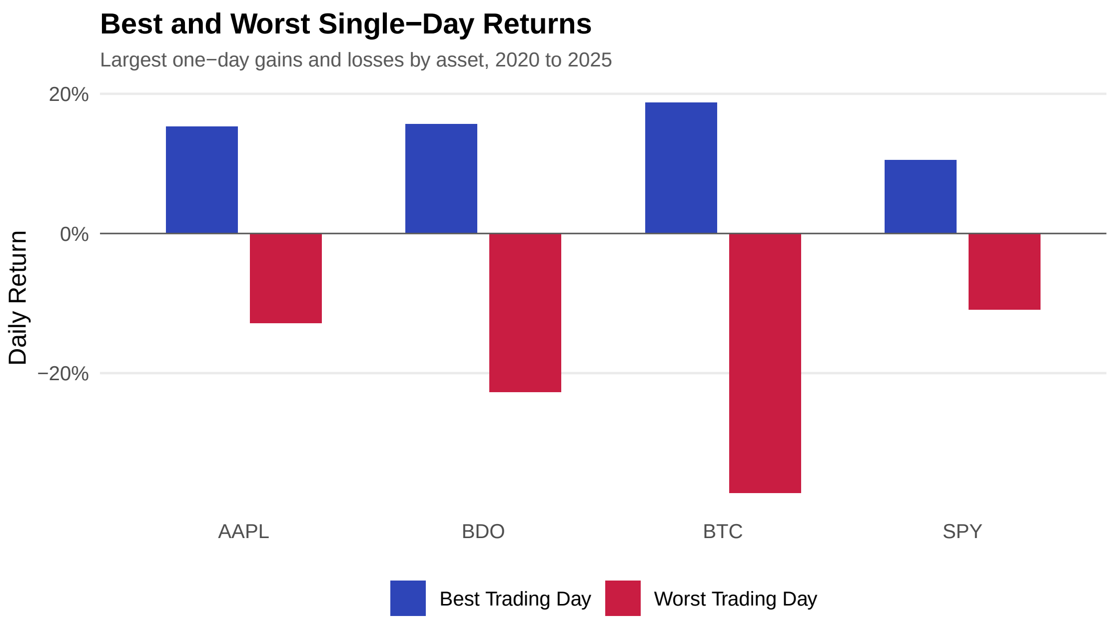


Figure 23: Best and worst single-day returns by asset, 2020 to 2025. Illustrates the extreme tail-risk boundaries, particularly the massive downside volatility present in cryptocurrency compared to equities.

Table 24: Q6 Data Output

Asset	Signal	Mean_Daily_Return	SD_Daily_Return	Obs
AAPL	Bullish	0.0015	0.0168	892
AAPL	Bearish	0.0008	0.0212	557
BDO	Bullish	0.0006	0.0195	667
BDO	Bearish	0.0003	0.0211	742
BTC	Bullish	0.0028	0.0300	1182
BTC	Bearish	0.0002	0.0343	951
SPY	Bearish	0.0013	0.0162	361
SPY	Bullish	0.0006	0.0092	1088

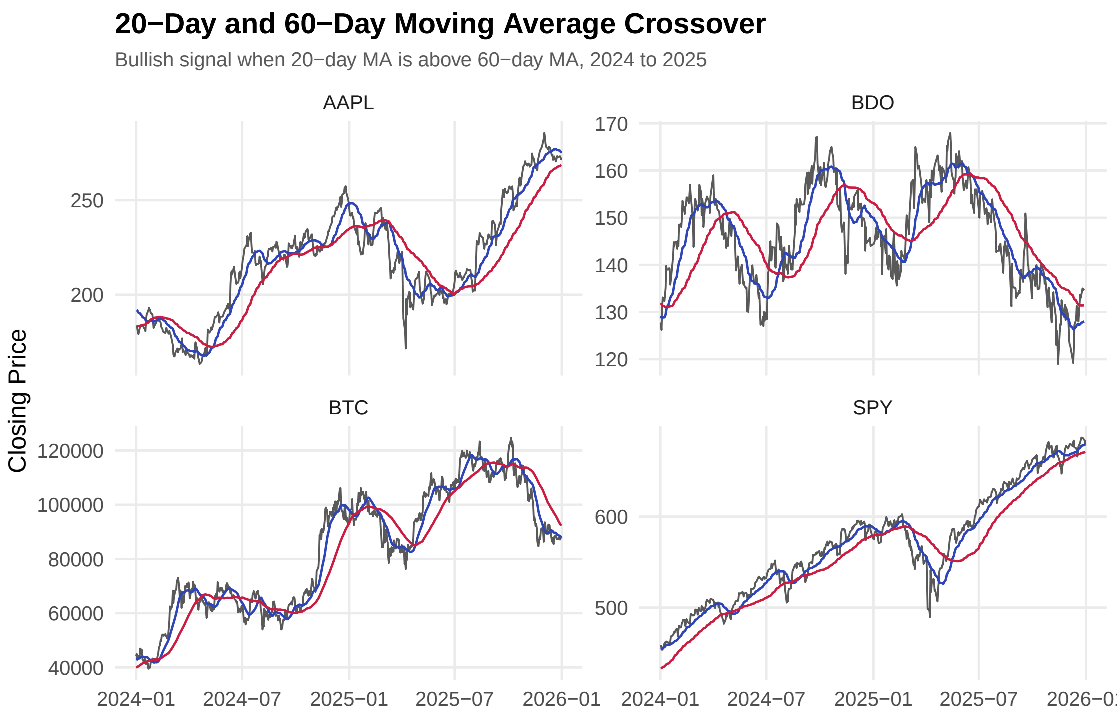


Figure 24: 20-day and 60-day moving average crossovers, 2024 to 2025. Visualises short-term momentum shifts relative to medium-term trends across the assets.

Table 25: Q7 Data Output

Asset	Volume_Regime	Obs	Mean_Daily_Return	Mean_Abs_Return	Volatility	Avg_Volume
AAPL	High Volume Days	377	0.0015	0.0240	0.0323	152332404
AAPL	Normal Volume Days	1130	0.0009	0.0105	0.0137	61873902
BDO	High Volume Days	367	-0.0001	0.0230	0.0322	6581172
BDO	Normal Volume Days	1100	0.0004	0.0136	0.0183	2551276
BTC	High Volume Days	548	0.0017	0.0305	0.0450	65153421094
BTC	Normal Volume Days	1643	0.0016	0.0183	0.0261	26930318305
SPY	High Volume Days	377	-0.0029	0.0160	0.0220	129389767
SPY	Normal Volume Days	1130	0.0018	0.0061	0.0078	63711998

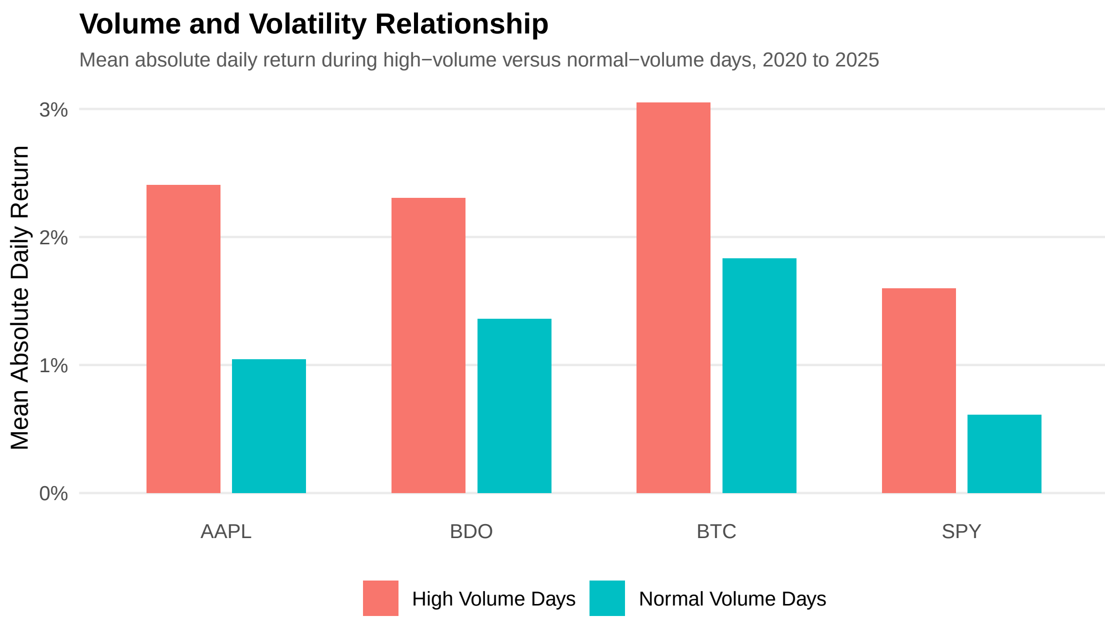


Figure 25: Mean absolute daily return during high-volume versus normal-volume days, 2020 to 2025. Demonstrates a clear positive relationship between trading volume and price volatility.

Table 26: Q8 Data Output

Asset	VIX_Regime	Mean_Daily_Return	SD_Daily_Return	Obs
AAPL	High Volatility (VIX $>$ 30)	-0.0054	0.0389	148
AAPL	Low Volatility (VIX $<$ 20)	0.0025	0.0134	859
AAPL	Moderate (20 $\leq$ VIX $\leq$ 30)	0.0007	0.0208	528
BDO	High Volatility (VIX $>$ 30)	-0.0042	0.0411	148
BDO	Low Volatility (VIX $<$ 20)	0.0006	0.0185	859
BDO	Moderate (20 $\leq$ VIX $\leq$ 30)	0.0011	0.0213	528
BTC	High Volatility (VIX $>$ 30)	0.0016	0.0550	148
BTC	Low Volatility (VIX $<$ 20)	0.0022	0.0288	859
BTC	Moderate (20 $\leq$ VIX $\leq$ 30)	0.0020	0.0380	528
SPY	High Volatility (VIX $>$ 30)	-0.0044	0.0303	148
SPY	Low Volatility (VIX $<$ 20)	0.0018	0.0069	859
SPY	Moderate (20 $\leq$ VIX $\leq$ 30)	0.0002	0.0123	528

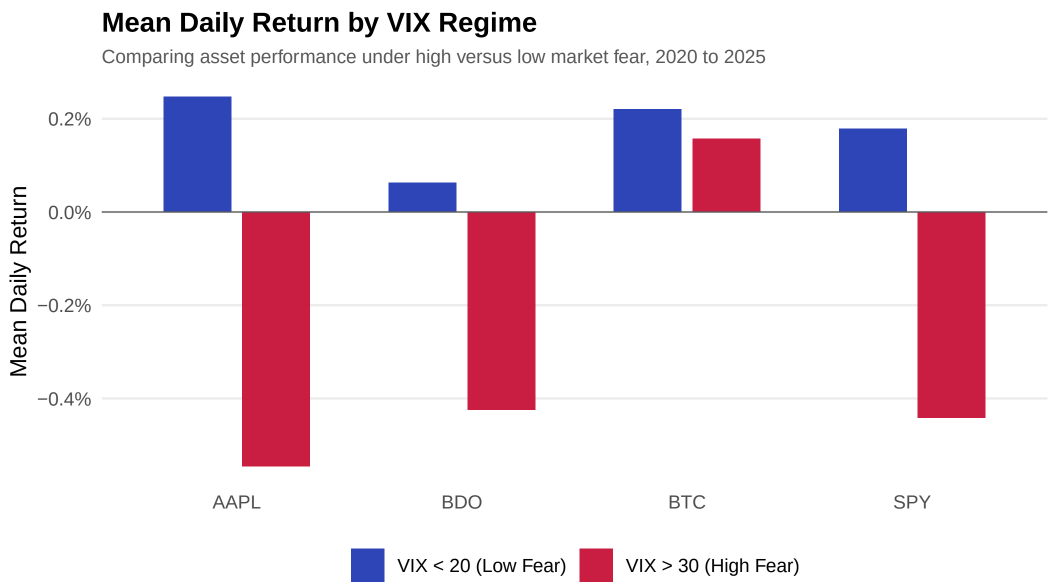


Figure 26: Mean daily return by VIX regime. Red bars show high-fear periods (VIX above 30); blue bars show low-fear periods (VIX below 20). Bitcoin uniquely displayed positive expected returns during market turmoil.

Table 27: Q9 Data Output

Asset	Yield_Direction	Mean_Daily_Return	SD_Daily_Return	Obs
AAPL	No Change	0.0019	0.0195	103
AAPL	Yields Falling	0.0005	0.0200	684
AAPL	Yields Rising	0.0013	0.0201	712
BDO	No Change	-0.0020	0.0208	103
BDO	Yields Falling	0.0000	0.0253	684
BDO	Yields Rising	0.0007	0.0201	712
BTC	No Change	0.0028	0.0372	103
BTC	Yields Falling	0.0017	0.0343	684
BTC	Yields Rising	0.0020	0.0363	712
SPY	No Change	-0.0001	0.0101	103
SPY	Yields Falling	-0.0001	0.0130	684
SPY	Yields Rising	0.0014	0.0135	712

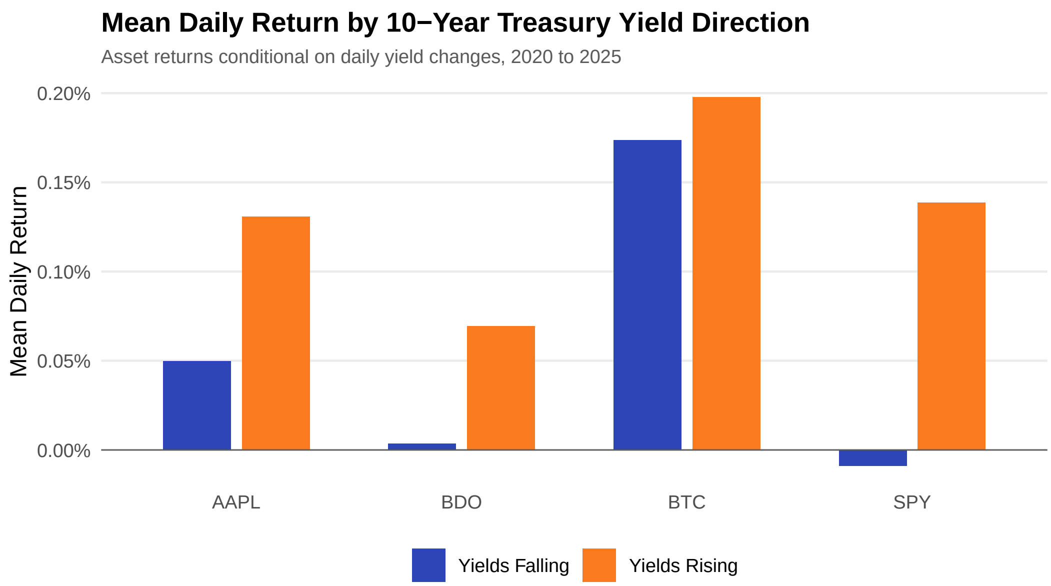


Figure 27: Mean daily return by 10-year Treasury yield direction. Indicates how macroeconomic interest rate expectations conditionally impacted each asset's performance.

Table 28: Q10 Data Output

Asset	Start_Date	End_Date	Start_CPI_YoY	End_CPI_YoY	Total_Cumul_Return	Mean_Daily_Return	Obs
BDO	2021-06-01	2023-02-28	0.053	0.0596	0.4391	0.0010	435
AAPL	2021-06-01	2023-02-28	0.053	0.0596	0.1952	0.0006	440
SPY	2021-06-01	2023-02-28	0.053	0.0596	-0.0317	0.0000	440
BTC	2021-06-01	2023-02-28	0.053	0.0596	-0.3800	-0.0002	638

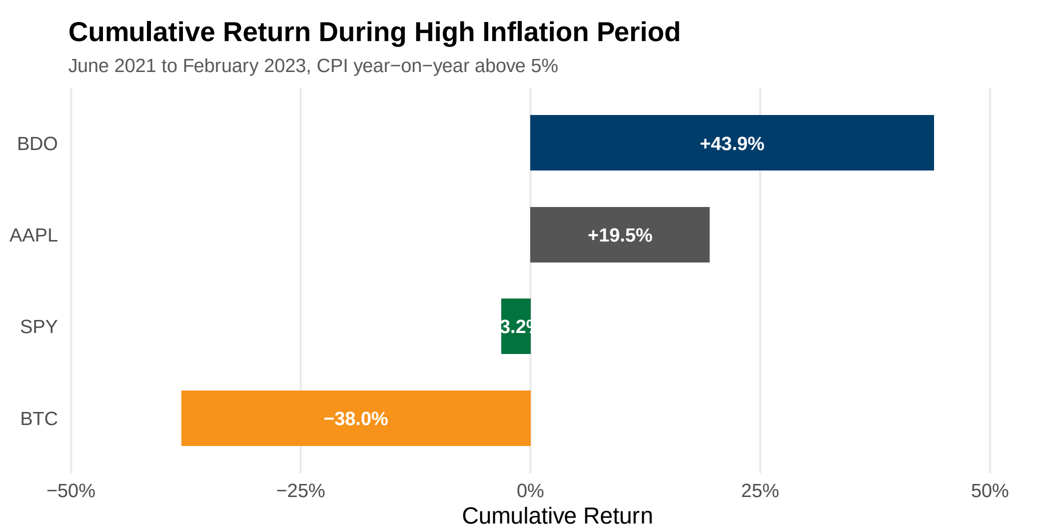


Figure 28: Cumulative return during the high-inflation period, June 2021 to February 2023. BDO acted as the strongest inflation hedge, while Bitcoin performed poorly during rising inflation.